

Faculty of Science
Dept. of Mathematics



Notes on Differentiation

Mat 103

Prepared by

Department of Mathematics

For

First Level Students

2021

Contents

1 Functions **1**

1.1 Basic Definitions and Examples . . . 1
1.2 Graphs of Functions . . . 8
1.3 Combining Functions . . . 12
1.3.1 Algebra of Functions . . . 12
1.3.2 Shifting a Graph of a Function . . . 14
1.3.3 The composite functions . . . 16
1.4 Inverse Functions . . . 19
1.5 Classifying Functions . . . 22
1.5.1 Increasing and Decreasing Functions . . . 22
1.5.2 Even Functions and Odd Functions: Symmetry . . . 22
1.5.3 Linear and Quadratic Functions . . . 24
1.5.4 Trigonometric Functions . . . 29
1.5.5 Exponential and Logarithmic Functions . . . 34
1.5.6 Hyperbolic Functions . . . 36
1.5.7 Inverse Of Hyperbolic Functions . . . 37
1.6 Exercises . . . 38

2 Limits and continuity **41**

2.1 limits of Function values . . . 41
2.1.1 The limit laws . . . 42
2.1.2 Eliminating Common Factors from Zero Denominators . . . 43
2.1.3 The Sandwich Theorem . . . 44
2.2 The Precise Definition of a limit . . . 46
2.3 One-sided Limits . . . 50
2.3.1 Limits involving $\frac{\sin\theta}{\theta}$ . . . 51
2.4 Continuity . . . 53
2.4.1 Intermediate value property . . . 59
2.5 Limits involving infinity; Asymptotes of graphs . . . 59
2.5.1 Limits at Infinity of Rational Functions . . . 60
2.5.2 Horizontal Asymptotes . . . 61

2.5.3 Oblique Asymptotes . . . 63
2.6 Infinite Limits . . . 64
2.6.1 Vertical Asymptotes . . . 67
2.7 Exercises . . . 71

3 Differentiation **75**
3.1 Definition of the Derivative . . . 75
3.2 Differentiation Rules . . . 78
3.2.1 Powers, Multiples, Sums, and Differences . . . 78
3.2.2 Products and Quotients . . . 80
3.3 Derivatives of Trigonometric Functions . . . 82
3.4 The Chain Rule . . . 85
3.5 Derivatives of Inverse Trigonometric Functions . . . 89
3.6 Derivative of the Natural Exponential Function . . . 91
3.7 Derivative Of Logarithmic Functions . . . 92
3.8 Derivatives of Hyperbolic Functions . . . 94
3.9 Derivatives Of Inverse Hyperbolic Functions . . . 95
3.10 Implicit Differentiation . . . 97
3.11 Exercises . . . 98

4 Applications of Derivatives **101**
4.1 Tangent and Normal Lines . . . 101
4.2 Maclaurin and Taylor series . . . 102
4.3 Mean Value Theorem . . . 106
4.4 Extreme Values of Functions . . . 107
4.5 Concavity and Points of Inflection: . . . 110
4.6 L'Hospital's Rule . . . 112
4.7 EXERCISES . . . 115

References **117**

Chapter 1

Functions

1.1 Basic Definitions and Examples

We begin with a short discussion of real numbers. This gives us the opportunity to recall some basic properties and standard notation.

Definition 1.1. ***A real number*** *is a number represented by a decimal or "decimal expansion."*

There are three types of decimal expansions: finite, repeating, and infinite but nonrepeating. For example,

$$\frac{3}{8} = 0.375, \quad \frac{1}{7} = 0.142857142857\ldots = 0.\overline{142857}, \ \pi = 3.141592653589793\ldots$$

The number $\frac{3}{8}$ is represented by a finite decimal, whereas $\frac{1}{7}$ is represented by a repeating or periodic decimal. The bar over 142857 indicates that this sequence repeats indefinitely. The decimal expansion of π is infinite but nonrepeating.

The set of all real numbers is denoted by a boldface $\mathbf{R}$. When there is no risk of confusion, we refer to a real number simply as a number. We also use the standard symbol $\in$ for the phrase "belongs to." Thus, $a \in \mathbf{R}$ reads " a belongs to $\mathbf{R}$ ".

The set of integers is commonly denoted by the letter $\mathbf{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$. A whole number is a nonnegative integer-that is, one of the numbers $0, 1, 2, \ldots$

Definition 1.2. *A real number is called* ***rational*** *if it can be represented by a fraction p/q, where p and q are integers with $q \neq 0$.*

The set of rational numbers is denoted $\mathbf{Q}$ (for "quotient"). Numbers that are not rational, such as π and $\sqrt{2}$, are called irrational.

We can tell whether a number is rational from its decimal expansion: Rational numbers have finite or repeating decimal expansions, and irrational numbers have infinite, non- repeating decimal expansions. Furthermore, the decimal expansion of a number is

unique, apart from the following exception: Every finite decimal is equal to an infinite decimal in which the digit 9 repeats. For example,

$$1 = 0.999\ldots, \quad \frac{3}{8} = 0.375 = 0.374999\ldots, \quad \frac{47}{20} = 2.35 = 2.34999\ldots$$

We visualize real numbers as points on a line (Figure 1.43). For this reason, real numbers are often referred to as points. The point corresponding to 0 is called the origin.

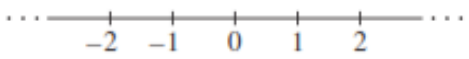


FIGURE 1.1: The set of real numbers represented as a line.

Definition 1.3. *The* ***absolute value*** *of a real number a, denoted $|a|$, is defined by (Figure 1.2)*

$$|a| = \textit{ distance from the origin } = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

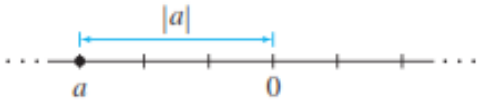


FIGURE 1.2: $|a|$ is the distance from a to the origin.

For example, $|1.2| = 1.2$ and $|-8.35| = 8.35$. The absolute value satisfies

$$|a| = |-a|, \quad |ab| = |a||b|$$

The distance between two real numbers a and b is $|b-a|$, which is the length of the line segment joining a and b (Figure 1.3). Beware that $|a+b|$ is not equal to $|a|+|b|$ unless

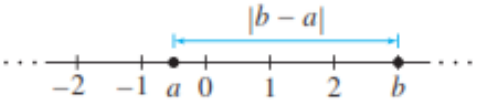


FIGURE 1.3: The distance from a to b is $|b-a|$.

a and b have the same sign or at least one of a and b is zero. If they have opposite signs, cancellation occurs in the sum $a+b$, and $|a+b| < |a|+|b|$.

For example, $|2+5| = |2|+|5|$ but $|-2+5| = 3$, which is less than $|-2|+|5| = 7$. In any case, $|a+b|$ is never larger than $|a|+|b|$ and this gives us the simple but important **triangle inequality:**

$$|a+b| \leq |a| + |b|$$

We use standard notation for intervals. Given real numbers $a < b$, there are four intervals with endpoints a and b (Figure 1.4). They all have length $b-a$ but differ according to which endpoints are included.

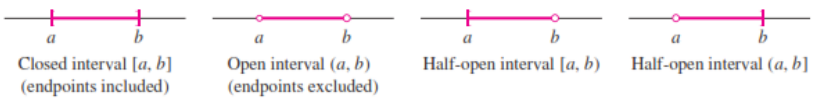


FIGURE 1.4: The four intervals with endpoints a and b.

Definition 1.4. *The* ***closed interval*** $[a,b]$ *is the set of all real numbers* x *such that* $a \leq x \leq b$ *:*

$$[a,b] = \{x \in \mathbf{R} : a \leq x \leq b\}$$

We usually write this more simply as $\{x : a \leq x \leq b\}$, it being understood that x belongs to $\mathbf{R}$. The open and half-open intervals are the sets

$$\underbrace{(a,b) = \{x : a < x < b\}}_{\text{Open interval (endpoints excluded)}}, \underbrace{[a,b) = \{x : a \leq x < b\}}_{\text{Half-open interval}}, \underbrace{(a,b] = \{x : a < x \leq b\}}_{\text{Half-open interval}}$$

The infinite interval $(-\infty, \infty)$ is the entire real line $\mathbf{R}$. A half-infinite interval is closed if it contains its finite endpoint and is open otherwise (Figure 1.5):

$$[a,\infty) = \{x : a \leq x\}, \quad (-\infty, b] = \{x : x \leq b\}$$

Open and closed intervals may be described by inequalities. For example, the interval



FIGURE 1.5: Closed half-infinite intervals.

$(-r, r)$ is described by the inequality $|x| < r$ (Figure 1.6):

$$|x| < r \quad \Leftrightarrow \quad -r < x < r \quad \Leftrightarrow \quad x \in (-r, r)$$

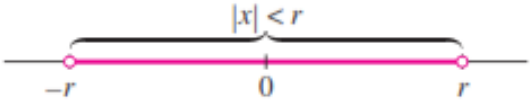


FIGURE 1.6: $|x| < r$.

More generally, for an interval symmetric about the value c (Figure 1.7),

$$|x - c| < r \quad \Leftrightarrow \quad c - r < x < c + r \quad \Leftrightarrow \quad x \in (c - r, c + r)$$

Closed intervals are similar, with $<$ replaced by $\leq$. We refer to r as the radius and to



FIGURE 1.7: $|x - c| < r$.

c as the midpoint or center. The intervals (a, b) and $[a, b]$ have midpoint $c = \frac{1}{2}(a + b)$ and radius $r = \frac{1}{2}(b - a)$ (Figure 1.7).

Example 1.1.1. Describe $[7, 13]$ using inequalities.
Solution: The midpoint of the interval $[7, 13]$ is $c = \frac{1}{2}(7 + 13) = 10$ and its radius is $r = \frac{1}{2}(13 - 7) = 3$ (Figure 1.8). Therefore,

$$[7, 13] = \{x \in \mathbf{R} : |x - 10| \leq 3\}$$

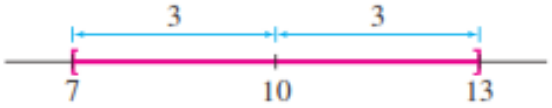


FIGURE 1.8: The interval $[7, 13]$ is described by $|x - 10| \leq 3$.

Example 1.1.2. Describe the set

$$S = \left\{ x : \left| \frac{1}{2}x - 3 \right| > 4 \right\}$$

in terms of intervals.

Solution: It is easier to consider the opposite inequality $\left|\frac{1}{2}x - 3\right| \leq 4$.

$$\begin{aligned} \left|\frac{1}{2}x - 3\right| \leq 4 \quad \Leftrightarrow \quad & -4 \leq \frac{1}{2}x - 3 \leq 4 \\ & -1 \leq \frac{1}{2}x \leq 7 \quad \text{(add 3)} \\ & -2 \leq x \leq 14 \quad \text{(multiply by 2)} \end{aligned}$$

Thus, $\left|\frac{1}{2}x - 3\right| \leq 4$ is satisfied when x belongs to $[-2, 14]$. The set S is the complement, consisting of all numbers x not in $[-2, 14]$. We can describe S as the union of two intervals: $S = (-\infty, -2) \cup (14, \infty)$ (Figure 1.9).



FIGURE 1.9: The set $S = \left\{x : \left|\frac{1}{2}x - 3\right| > 4\right\}$.

We now review some definitions and notation concerning functions.

Definition 1.5. *A function f from a set D to a set Y is a rule that assigns, to each element x in D, a unique element $y = f(x)$ in Y. We write*

$$f : D \to Y$$

The set **D** , called the **domain** of f, is the set of "allowable inputs." For $x \in D$, $f(x)$ is called the value of f at x, and the set Y is the co-domain of f. (Figure 1.10).

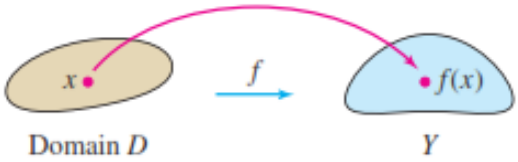


FIGURE 1.10: A function assigns an element $f(x)$ in Y to each $x \in D$.

The range R of f is the subset of Y consisting of all values $f(x)$:

$$R = \{y \in Y : f(x) = y \text{ for some } x \in D\}$$

Informally, we think of f as a "machine" that produces an output y for every input x in the domain D (Figure 1.11). The domain and range of a function can be any sets

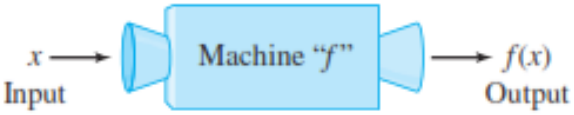


FIGURE 1.11: Think of f as a "machine" that takes the input x and produces the output $f(x)$.

of objects, but often in calculus they are sets of real numbers interpreted as points of a coordinate line. Often a function is given by a formula that describes how to calculate the output value from the input variable.

For instance, the equation $A = pr^2$ is a rule that calculates the area A of a circle from its radius r (so r, interpreted as a length, can only be positive in this formula). When we define a function $y = f(x)$ with a formula and the domain is not stated explicitly or restricted by context, the domain is assumed to be the largest set of real x-values for which the formula gives real y-values, which is called the **natural domain**. If we want to restrict the domain in some way, we must say so. The domain of $y = x^2$ is the entire set of real numbers. To restrict the domain of the function to, say, positive values of x, we would write "$y = x^2, x > 0$."

When the range of a function is a set of real numbers, the function is said to be **real-valued**. The domains and ranges of most real-valued functions of a real variable we consider are intervals or combinations of intervals. The intervals may be open, closed, or half open, and may be finite or infinite. Sometimes the range of a function is not easy to find.

A function f is like a machine that produces an output value $f(x)$ in its range whenever we feed it an input value x from its domain (Figure 1.11). The function keys on a calculator give an example of a function as a machine. For instance, the $\sqrt{x}$ key on a calculator gives an output value (the square root) whenever you enter a nonnegative number x and press the $\sqrt{x}$ key. A function can also be pictured as an arrow diagram (Figure 1.2). Each arrow associates an element of the domain D with a unique or single element in the set Y. In Figure 1.12, the arrows indicate that $f(a)$ is associated with a, $f(x)$ is associated with x, and so on. Notice that a function can have the same value at two different input elements in the domain (as occurs with $f(a)$ in Figure 1.2), but each input element x is assigned a single output value $f(x)$.

Example 1.1.3. Let's verify the natural domains and associated ranges of some simple functions.

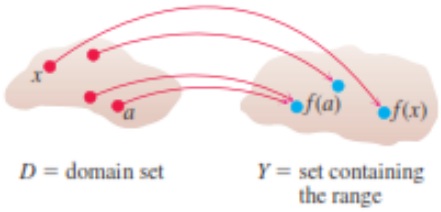


FIGURE 1.12

Solution: The domains in each case are the values of x for which the formula makes sense.

Function	Domain ($\boldsymbol{x}$)	Range ($\boldsymbol{y}$)
$y = x^2$	$(-\infty, \infty)$	$[0, \infty)$
$y = 1/x$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$
$y = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$y = \sqrt{4 - x}$	$(-\infty, 4]$	$[0, \infty)$
$y = \sqrt{1 - x^2}$	$[-1, 1]$	$[0, 1]$

The formula $y = x^2$ gives a real y-value for any real number x, so the domain is $(-\infty, \infty)$. The range of $y = x^2$ is $[0, \infty)$ **because** the square of any real number is nonnegative and every nonnegative number y is the square of its own square root, $y = (\sqrt{y})^2$ for $y \geq 0$.

The formula $y = 1/x$ gives a real y-value for every x except $x = 0$. For consistency in the rules of arithmetic, we cannot divide any number by zero. The range of $y = 1/x$, the set of reciprocals of all nonzero real numbers, is the set of all nonzero real numbers, since $y = 1/(1/y)$. That is, for $y \neq 0$ the number $x = 1/y$ is the input assigned to the output value y.

The formula $y = \sqrt{x}$ gives a real y-value only if $x \geq 0$. The range of $y = \sqrt{x}$ is $[0, \infty)$ because every nonnegative number is some number's square root (namely, it is the square root of its own square).

In $y = \sqrt{4 - x}$, the quantity $4 - x$ cannot be negative. That is, $4 - x \geq 0$, or $x \leq 4$. The formula gives real y-values for all $x \leq 4$. The range of $\sqrt{4 - x}$ is $[0, \infty)$, the set of all nonnegative numbers.

The formula $y = \sqrt{1 - x^2}$ gives a real y-value for every x in the closed interval from -1 to 1 . Outside this domain, $1 - x^2$ is negative and its square root is not a real number. The values of $1 - x^2$ vary from 0 to 1 on the given domain, and the square roots of these values do the same. The range of $\sqrt{1 - x^2}$ is $[0, 1]$.

Example 1.1.4. 1. Find the range of the function $f : [-1, 8] \to Y$ where $f(x) = 3x + 4$.

2. Find the domain of the function $f(x) = \frac{3}{x^2-2x}$

3. Find the domain of the function $g(x) = \sqrt{x^2 - 2x - 8}$

Solution:

1.
$$\begin{aligned}\text{Im}(f) &= \{f(x) : -1 \leq x \leq 6\} \\ &= \{f(x) : -2 \leq 2x \leq 12\} \\ &= \{5 \leq f(x) = 2x + 3 \leq 19\} \\ &= [5, 19]\end{aligned}$$

2. Note that $f(x)$ is a real number if and only if the denominator $x^2 - 2x \neq 0$ So,

$$\begin{aligned}D(f) &= \{x \in R; x^2 - 2x \neq 0\} \\ &= \{x \in R : x(x-2) \neq 0\} \\ D(f) &= \{x \in R; x \neq 0, x \neq 2\} \\ &= (-\infty, 0) \cup (0, 2) \cup (2, \infty) \\ &= R \backslash \{0, 2\}\end{aligned}$$

3. Note that $f(x)$ is a real number if and only if the radicand $x^2 - 2x - 8$ is non-negative. Thus, $f(x)$ exists if and only if $x^2 - 2x - 8 \geq 0$ or , equivalently, $(x-4)(x+2) \geq 0$. By solving this inequality, we obtain $x \geq 4$ or $x \leq -2$ Hence, $D(f) = (-\infty, -2] \cup [4, \infty)$

Example 1.1.5. Let
$$f(x) = \frac{\sqrt{4+x}}{1-x}.$$

(a) Find the domain of f.

(b) Find $f(5), f(-2), f(-a)$, and $-f(a)$.

Solution:
(a) Note that $f(x)$ exists if and only if $4+x \geq 0$ and $1-x \neq 0$ or, equivalently, $x \geq -4$ and $x \neq -1$ Hence, $D(f) = [-4, 1) \cup (1, \infty)$.
(b) Left as an exercise.

1.2 Graphs of Functions

If f is a function with domain D, its graph consists of the points in the Cartesian plane whose coordinates are the input-output pairs for f. In set notation, the graph is

$$\{(x, f(x)) \mid x \in D\}$$

The graph of the function $f(x) = x + 2$ is the set of points with coordinates (x, y) for which $y = x + 2$. Its graph is the straight line sketched in Figure 1.13. The graph of

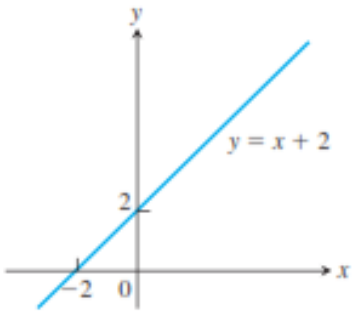


FIGURE 1.13

a function f is a useful picture of its behavior. If (x, y) is a point on the graph, then $y = f(x)$ is the height of the graph above (or below) the point x. The height may be positive or negative, depending on the sign of $f(x)$ (Figure 1.14).

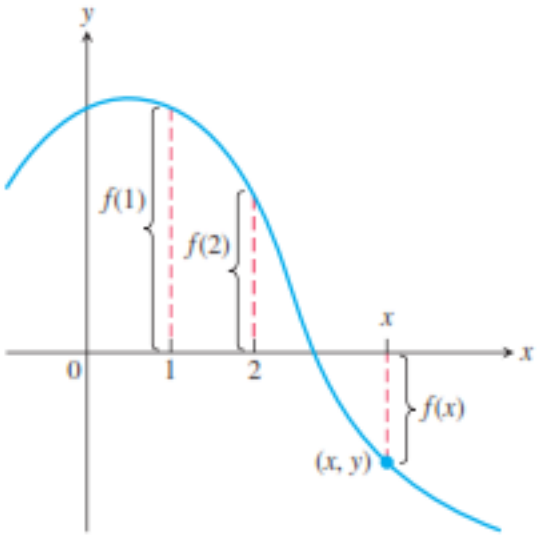


FIGURE 1.14

Example 1.2.1. Graph the function $y = x^2$ over the interval $[-2, 2]$.

Solution: Make a table of xy-pairs that satisfy the equation $y = x^2$. Plot the points (x, y) whose coordinates appear in the table, and draw a smooth curve (labeled with its equation) through the plotted points (see Figure 1.15).

x	$y = x^2$
-2	4
-1	1
0	0
1	1
$\frac{3}{2}$	$\frac{9}{4}$
2	4

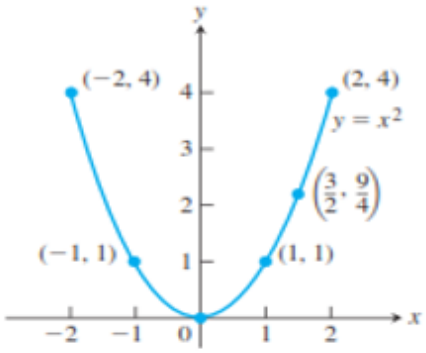


FIGURE 1.15

Vertical Line Test for a Function

Not every curve in the coordinate plane can be the graph of a function. A function f can have only one value $f(x)$ for each x in its domain, so no *vertical* line can intersect the graph of a function more than once. If a is in the domain of the function f, then the vertical line $x = a$ will intersect the graph of f at the single point $(a, f(a))$.

A circle cannot be the graph of a function, since some vertical lines intersect the circle, twice. The circle graphed in Figure 1.17a, however, does contain the graphs of functions of x, such as the upper semicircle defined by the function $f(x) = \sqrt{1 - x^2}$ and the lower semicircle defined by the function $g(x) = -\sqrt{1 - x^2}$ (Figures 1.17 b and 1.17c).

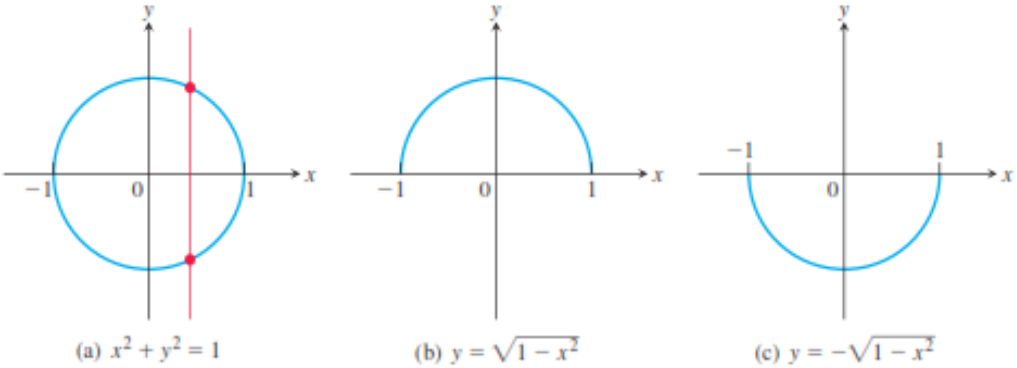


FIGURE 1.16

Picewise-Defined Functions

Sometimes a function is described in pieces by using different formulas on different parts of its domain. One example is the **absolute value function**

$$|x| = \begin{cases} x, & x \geq 0 \quad \text{First formula} \\ -x, & x < 0, \quad \text{Second formula} \end{cases}$$

The right-hand side of the equation means that the function equals x if $x \geq 0$, and equals $-x$ if $x < 0$. Piecewise-defined functions often arise when real-world data are modeled. Here are some other examples.

Example 1.2.2. The function

$$f(x) = \begin{cases} -x, & x < 0 \quad \text{First formula} \\ x^2, & 0 \leq x \leq 1 \quad \text{Second formula} \\ 1, & x > 1 \quad \text{Third formula} \end{cases}$$

is defined on the entire real line but has values given by different formulas, depending on the position of x. The values of f are given by $y = -x$ when $x < 0, y = x^2$ when $0 \leq x \leq 1$, and $y = 1$ when $x > 1$. The function, however, is just one function whose domain is the entire set of real numbers (Figure 1.17).

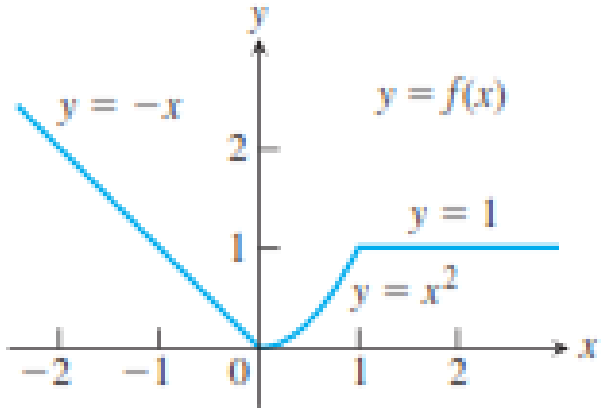


FIGURE 1.17

Example 1.2.3. The function whose value at any number x is the greatest integer less than or equal to x is called the greatest integer function or the integer floor function.

It is denoted $\lfloor x \rfloor$. Figure 1.17 shows the graph. Observe that

$$\begin{array}{llll} \lfloor 2.4 \rfloor = 2, & \lfloor 1.9 \rfloor = 1, & \lfloor 0 \rfloor = 0, & \lfloor -1.2 \rfloor = -2 \\ \lfloor 2 \rfloor = 2, & \lfloor 0.2 \rfloor = 0, & \lfloor -0.3 \rfloor = -1, & \lfloor -2 \rfloor = -2. \end{array}$$

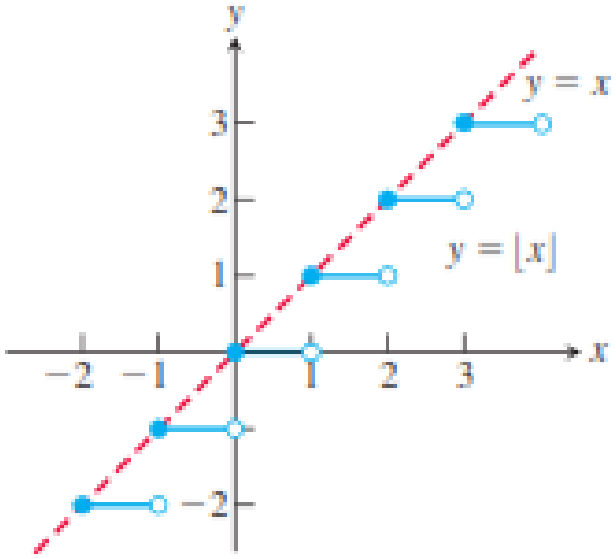


FIGURE 1.18

1.3 Combining Functions

In this section we look at the main ways functions are combined or transformed to form new functions. We often complicated functions from simpler functions by combining them in various ways, using arithmetic operations and composition.

Like numbers, functions can be added, subtracted, multiplied, and divided (except where the denominator is zero) to produce new functions.

1.3.1 Algebra of Functions

Definition 1.6. *If f and g are functions, then for every x that belongs to the domains of both f and g (that is, for $x \in D(f) \cap D(g)$), we define functions $f+g$, $f-g$, and fg by the formulas*

$$(f+g)(x) = f(x) + g(x)$$
$$(f-g)(x) = f(x) - g(x)$$
$$(fg)(x) = f(x)g(x)$$

Notice that the + sign on the left-hand side of the first equation represents the operation of addition of functions, whereas the + on the right-hand side of the equation means addition of the real numbers $f(x)$ and $g(x)$.

Definition 1.7. *At any point of $D(f) \cap D(g)$ at which $g(x) \neq 0$, we can also define the function f/g. by the formula*

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \quad (\text{ where } g(x) \neq 0)$$

Definition 1.8. *Functions can also be multiplied by constants: If c is a real number, then the function cf is defined for all x in the domain of f by*

$$(cf)(x) = cf(x)$$

Example 1.3.1. The functions defined by the formulas

$$f(x) = \sqrt{x} \quad \text{and} \quad g(x) = \sqrt{1-x}$$

have domains $D(f) = [0, \infty)$ and $D(g) = (-\infty, 1]$. The points common to these domains are the points

$$[0, \infty) \cap (-\infty, 1] = [0, 1]$$

Example 1.3.2. Let

$$f(x) = \sqrt{9-x^2} \quad \text{and} \quad g(x) = \sqrt{x^2-1}$$

Solution: Note that, the domain of f is the closed interval $[-3, 3]$ and the domain of g is $(-\infty, -1] \cup [1, \infty)$.

Consequently, the domain of $f+g, f-g, fg$ is $[-3, -1]U[1, 3]$, but the domain of f/g is $[-3, -1) \cup (1, 3]$. So,

$$(f+g)(x) = f(x) + g(x) = \sqrt{9-x^2} + \sqrt{x^2-1}$$

$$(f-g)(x) = f(x) - g(x) = \sqrt{9-x^2} - \sqrt{x^2-1}$$

$$(fg)(x) = f(x)g(x) = \sqrt{9-x^2}\sqrt{x^2-1}$$

$$(f/g)(x) = \frac{f(x)}{g(x)} = \frac{\sqrt{9-x^2}}{\sqrt{x^2-1}}$$

The following table summarizes the formulas and domains for the various algebraic combinations of the two functions. We also write $f \cdot g$ for the product function fg.

Function	Formula	Domain
$f+g$	$(f+g)(x) = \sqrt{x} + \sqrt{1-x}$	$[0,1] = D(f) \cap D(g)$
$f-g$	$(f-g)(x) = \sqrt{x} - \sqrt{1-x}$	$[0,1]$
$g-f$	$(g-f)(x) = \sqrt{1-x} - \sqrt{x}$	$[0,1]$
$f \cdot g$	$(f \cdot g)(x) = f(x)g(x) = \sqrt{x(1-x)}$	$[0,1]$
f/g	$\frac{f}{g}(x) = \frac{f(x)}{g(x)} = \sqrt{\frac{x}{1-x}}$	$[0,1)(x = 1$ excluded$)$
g/f	$\frac{g}{f}(x) = \frac{g(x)}{f(x)} = \sqrt{\frac{1-x}{x}}$	$(0,1](x = 0$ excluded$)$

1.3.2 Shifting a Graph of a Function

A common way to obtain a new function from an existing one is by adding a constant to each output of the existing function, or to its input variable. The graph of the new function is the graph of the original function shifted vertically or horizontally, as follows.
Shift Formulas:

$$\begin{cases} VerticalShifts, \quad y = f(x) + k & Shifts the graph of f up k units if k>0 \\ & Shifts it down |k| \quad units if k<0 \\ horizontalShifts, \quad y = f(x+h) & Shifts the graph of f left h units if h>0 \\ & Shifts it right |h| units if h<0. \end{cases}$$

Example 1.3.3. (a) Adding 1 to the right-hand side of the formula $y = x^2$ to get $y = x^2 + 1$ shifts the graph up 1 unit (Figure 1.19).
(b) Adding -2 to the right-hand side of the formula $y = x^2$ to get $y = x^2 - 2$ shifts the graph down 2 units (Figure 1.19).
(c) Adding 3 to x in $y = x^2$ to get $y = (x+3)^2$ shifts the graph 3 units to the left, while adding -2 shifts the graph 2 units to the right (Figure 1.20).
(d) Adding -2 to x in $y = |x|$, and then adding -1 to the result, gives $y = |x-2| - 1$ and shifts the graph 2 units to the right and 1 unit down (Figure 1.21).

Scaling and Reflecting a Graph of a Function

To scale the graph of a function $y = f(x)$ is to stretch or compress it, vertically or horizontally. This is accomplished by multiplying the function f, or the independent variable x, by an appropriate constant c. Reflections across the coordinate axes are special cases where $c = -1$.

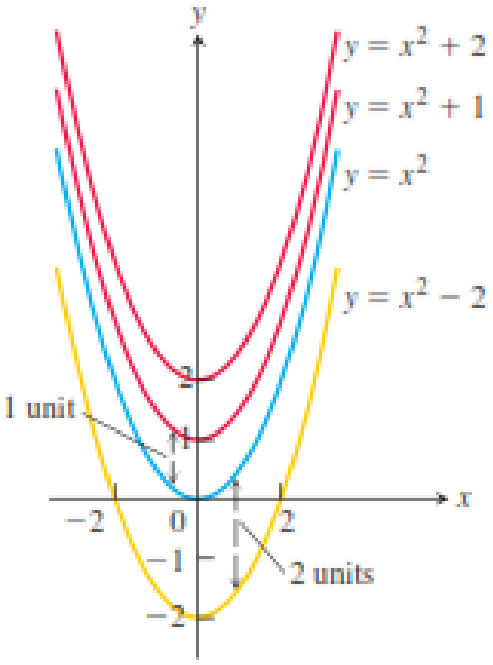


FIGURE 1.19

Vertical and Horizontal Scaling and Reflecting Formulas
For $c > 1$, the graph is scaled:

$y = cf(x)$ Stretches the graph of f vertically by a factor of c
$y = \frac{1}{c}f(x)$ Compresses the graph of f vertically by a factor of c
$y = f(cx)$ Compresses the graph of f horizontally by a factor of c.
$y = f(x/c)$ Stretches the graph of f horizontally by a factor of c

For $c = -1$, the graph is reflected:

$y = -f(x)$ **Reflects the graph of** f across the x-axis.
$y = \frac{1}{c}f(x)$ Compresses the graph of f vertically by a factor of c
$y = f(-x)$ Reflects the graph of f across the y-axis.

Example 1.3.4. Here we scale and reflect the graph of $y = \sqrt{x}$.
(a) Vertical: Multiplying the right-hand side of $y = \sqrt{x}$ by 3 to get $y = 3\sqrt{x}$ stretches the graph vertically by a factor of 3 , whereas multiplying by $1/3$ compresses the graph by a factor of 3 (Figure 1.22).
(b) Horizontal: The graph of $y = \sqrt{3x}$ is a horizontal compression of the graph of

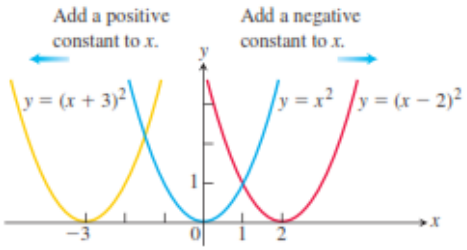


FIGURE 1.20

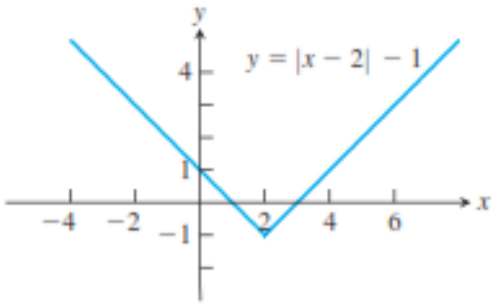


FIGURE 1.21

$y = \sqrt{x}$ by a factor of 3, and $y = \sqrt{x/3}$ is a horizontal stretching by a factor of 3 (Figure 1.23).

Note that $y = \sqrt{3x} = \sqrt{3}\sqrt{x}$ so a horizontal compression may correspond to a vertical stretching by a different scaling factor. Likewise, a horizontal stretching may correspond to a vertical compression by a different scaling factor.

(c) Reflection: The graph of $y = -\sqrt{x}$ is a reflection of $y = \sqrt{x}$ across the x-axis, and $y = \sqrt{-x}$ is a reflection across the y-axis (Figure 1.24).

1.3.3 The composite functions

Composition is another important way of constructing new functions.

Definition 1.9. *The composition of f and g is the function $f \circ g$ ("f composed with g") is defined by $(f \circ g)(x) = f(g(x))$. Moreover, the domain of $f \circ g$ is the set of values of x in the domain of g such that $g(x)$ lies in the domain of f.*

The definition implies that fog can be formed when the range of g lies in the domain of f. To find $(fog)(x)$, first find $g(x)$ and second find $f(g(x))$.)Figure 1.25) pictures fog as a machine diagram, and (Figure 1.25) shows the composite as an arrow diagram.

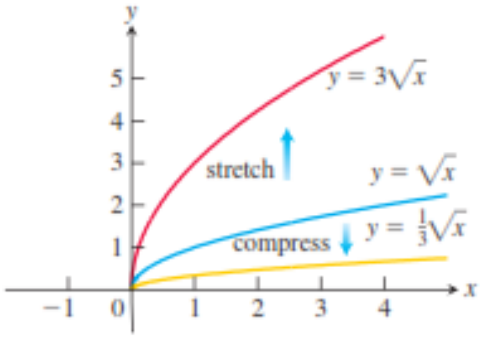


FIGURE 1.22

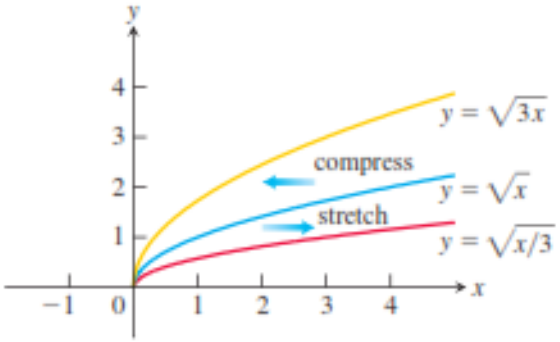


FIGURE 1.23

To evaluate the composite function gof (when defined), we find (x) first and then $g((x))$. The domain of gof is the set of numbers x in the domain of such that $f(x)$ lies in the domain of$g$. The functions fog and gof are usually quite different.

Example 1.3.5. If $f(x) = \sqrt{x}$ and $g(x) = x + 1$, find

(a) $(f \circ g)(x)$ (c) $(f \circ f)(x)$

(b) $(g \circ f)(x)$ (d) $(g \circ g)(x)$

Solution:

(a) $(f \circ g)(x) = f(g(x)) = \sqrt{g(x)} = \sqrt{x+1}$ and its domain is $[-1, \infty)$.

(b) $(g \circ f)(x) = g(f(x)) = f(x) + 1 = \sqrt{x} + 1$ and its domain is $[0, \infty)$.

(c) $(f \circ f)(x) = f(f(x)) = \sqrt{f(x)} = \sqrt{\sqrt{x}} = x^{1/4}$ and its domain is $[0, \infty)$

(d) $(g \circ g)(x) = g(g(x)) = g(x) + 1 = (x + 1) + 1 = x + 2$ and its domain is $(-\infty, \infty)$.

To see why the domain of $f \circ g$ is $[-1, \infty)$, notice that $g(x) = x + 1$ is defined for all real x but belongs to the domain of f only if $x + 1 \geq 0$, that is to say, when $x \geq -1$.

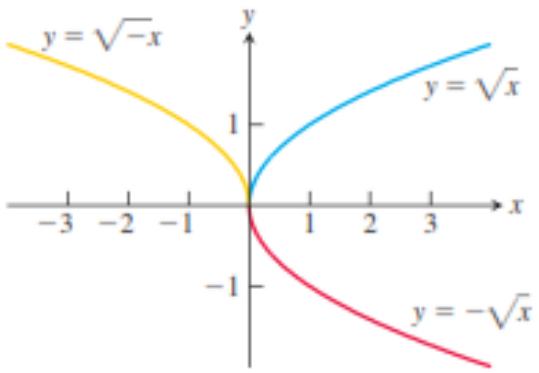


FIGURE 1.24

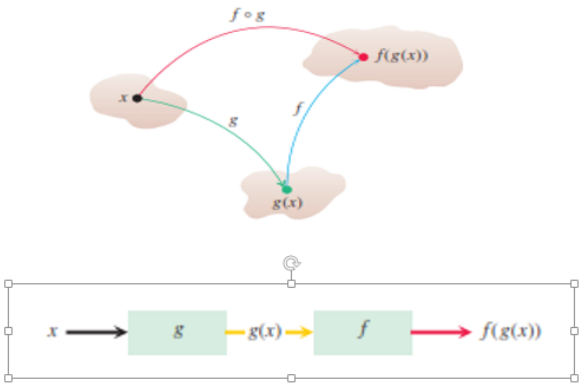


FIGURE 1.25

Notice that if $f(x) = x^2$ and $g(x) = \sqrt{x}$, then $(f \circ g)(x) = (\sqrt{x})^2 = x$. However, the domain of $f \circ g$ is $[0, \infty)$, not $(-\infty, \infty)$, since $\sqrt{x}$ requires $x \geq 0$.

Example 1.3.6. Compute the composite functions $f \circ g$ and $g \circ f$ and discuss their domains, where

$$f(x) = \sqrt{x}, \quad g(x) = 1 - x$$

Solution: We have

$$(f \circ g)(x) = f(g(x)) = f(1 - x) = \sqrt{1 - x}$$

The square root $\sqrt{1 - x}$ is defined if $1 - x \geq 0$ or $x \leq 1$, so the domain of $f \circ g$ is $\{x : x \leq 1\}$. On the other hand,

$$(g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = 1 - \sqrt{x}$$

The domain of $g \circ f$ is $\{x : x \geq 0\}$.

Example 1.3.7. If $f(x) = \frac{x}{x+2}, g(x) = \frac{x-1}{x}$, Find

$$(f \circ g)(x), (g \circ f)(x), D(f \circ g), \text{and} D(g \circ f).$$

Solution: Note that $D(f) = R - \{2\}$, and $D(g) = R - \{0\}$

$$\begin{aligned}
(f \circ g)(x) &= f(g(x)) = f\left(\frac{x-1}{x}\right) \\
&= \frac{\frac{x-1}{x}}{\frac{x-1}{x} + 2} = \frac{x-1}{3x-1} \\
(g \circ f)(x) &= g(f(x)) = g\left(\frac{x}{x+2}\right) \\
&= \frac{\frac{x}{x+2} - 1}{\frac{x}{x+2}} = \frac{-2}{x} \\
D(f \circ g)(x) &= \{x : x \in D(g), g(x) \in D(f)\} \\
&= \left\{x : x \neq 0, \frac{x-1}{x} \neq -2\right\} \\
&= \left\{x : x \neq 0, x \neq \frac{1}{3}\right\} \\
&= R - \left\{0, \frac{1}{3}\right\} \\
D(g \circ f)(x) &= \{x : x \in D(f), f(x) \in D(g)\} \\
&= \left\{x : x \neq -2, \frac{x}{x+2} \neq 0\right\} \\
&= \{x : x \neq -2, x \neq 0\} \\
&= R - \{-2, 0\}
\end{aligned}$$

Exercise: If $f(x) = x^2 - 9$ and $g(x) = \sqrt{x}$, find $(f \circ g)(x), (g \circ f)(x), D(f \circ g)$, and $D(g \circ f)$.

1.4 Inverse Functions

One-to-One Functions A function is a rule that assigns a value from its range to each element in its domain. Some functions assign the same range value to more than one element in the domain. The function $f(x) = x^2$ assigns the same value, 1, to both of the numbers -1 and $+1$; the sines of $\pi/3$ and $2\pi/3$ are both $\sqrt{3}/2$. Other functions assume each value in their range no more than once. The square roots and cubes of different numbers are always different. A function that has distinct values at distinct

elements in its domain is called one-to-one. These functions take on any one value in their range exactly once.

Definition 1.10. *A function $f(x)$ is one-to-one on a domain D if $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$ in D.*

Since each output of a one-to-one function comes from just one input, the effect of the function can be inverted to send an output back to the input from which it came.

Definition 1.11. *Suppose that f is a one-to-one function on a domain D with range R. The inverse function f^{-1} is defined by*

$$f^{-1}(b) = a \quad \text{if} \quad f(a) = b$$

The domain of f^{-1} is R and the range of f^{-1} is D.

The symbol f^{-1} for the inverse of f is read " f inverse." The "-1" in f^{-1} is not an exponent; $f^{-1}(x)$ does not mean $1/f(x)$. Notice that the domains and ranges of f and f^{-1} are interchanged.

If we apply f to send an input x to the output $f(x)$ and follow by applying f^{-1} to $f(x)$, we get right back to x, just where we started. Similarly, if we take some number y in the range of f, apply f^{-1} to it, and then apply f to the resulting value $f^{-1}(y)$, we get back the value y with which we began. Composing a function and its inverse has the same effect as doing nothing.

$$\begin{aligned} (f^{-1} \circ f)(x) &= x, \quad \text{for all } x \text{ in the domain of } f \\ (f \circ f^{-1})(y) &= y, \quad \text{for all } y \text{ in the domain of } f^{-1} \text{ (or range of } f) \end{aligned}$$

Only a one-to-one function can have an inverse. The reason is that if $f(x_1) = y$ and $f(x_2) = y$ for two distinct inputs x_1 and x_2, then there is no way to assign a value to $f^{-1}(y)$ that satisfies both $f^{-1}(f(x_1)) = x_1$ and $f^{-1}(f(x_2)) = x_2$

A function that is increasing on an interval satisfies the inequality $f(x_2) > f(x_1)$ when $x_2 > x_1$, so it is one-to-one and has an inverse. Decreasing functions also have an inverse. Functions that are neither increasing nor decreasing may still be one-to-one and have an inverse, as with the function $f(x) = 1/x$ for $x \neq 0$ and $f(0) = 0$, defined on $(-\infty, \infty)$ and passing the horizontal line test.

Finding Inverses

The process of passing from f to f^{-1} can be summarized as a two-step procedure.

1. Solve the equation $y = f(x)$ for x. This gives a formula $x = f^{-1}(y)$ where x is expressed as a function of y.

2. Interchange x and y, obtaining a formula $y = f^{-1}(x)$ where f^{-1} is expressed in the conventional format with x as the independent variable and y as the dependent variable.

Example 1.4.1. Find the inverse of $y = \frac{1}{2}x + 1$, expressed as a function of x.

Solution: 1. Solve for x in terms of y: $y = \frac{1}{2}x+1$ The graph is a straight line satisfying the horizontal line test (Fig. 1.60).

$$2y = x + 2$$
$$x = 2y - 2$$

2. Interchange x and y : $y = 2x - 2$ The inverse of the function $f(x) = (1/2)x + 1$ is the function $f^{-1}(x) = 2x - 2$. (See Figure 1.60.) To check, we verify that both composites give the identity function:

$$f^{-1}(f(x)) = 2\left(\frac{1}{2}x + 1\right) - 2 = x + 2 - 2 = x$$
$$f\left(f^{-1}(x)\right) = \frac{1}{2}(2x - 2) + 1 = x - 1 + 1 = x.$$

Example 1.4.2. Find the inverse of the function $y = x^2, x \geq 0$, expressed as a function of x.

Solution: For $x \geq 0$, the graph satisfies the horizontal line test, so the function is one-to-one and has an inverse. To find the inverse, we first solve for x in terms of y:

$$y = x^2$$
$$\sqrt{y} = \sqrt{x^2} = |x| = x \quad |x| = x \text{ because } x \geq 0.$$

We then interchange x and y, obtaining

$$y = \sqrt{x}$$

The inverse of the function $y = x^2, x \geq 0$, is the function $y = \sqrt{x}$. Notice that the function $y = x^2, x \geq 0$, with domain restricted to the nonnegative real numbers, is one-to-one and has an inverse. On the other hand, the function $y = x^2$, with no domain restrictions, is not one-to-one and therefore has no inverse.

1.5 Classifying Functions

1.5.1 Increasing and Decreasing Functions

If the graph of a function climbs or rises as you move from left to right, we say that the function is increasing. If the graph descends or falls as you move from left to right, the function is decreasing.

1. If $f(x_2) > f(x_1)$ whenever $x_1 < x_2$, then f is said to be **increasing** on I.
2. If $f(x_2) < f(x_1)$ whenever $x_1 < x_2$, then f is said to be **decreasing** on I.

It is important to realize that the definitions of increasing and decreasing functions must be satisfied for every pair of points x_1 and x_2 in I with $x_1 < x_2$. Because we use the inequality $<$ to compare the function values, instead of $\leq$, it is sometimes said that f is strictly increasing or decreasing on I. The interval I may be finite (also called bounded) or infinite (unbounded) and by definition never consists of a single point .

Example 1.5.1. The function graphed in Figure 1.17 is decreasing on $(-\infty, 0]$ and increasing on $[0, 1]$. The function is neither increasing nor decreasing on the interval $[1, \infty)$ because of the strict inequalities used to compare the function values in the definitions.

1.5.2 Even Functions and Odd Functions: Symmetry

The graphs of even and odd functions have characteristic symmetry properties.

A function $y = f(x)$ is an

- **Even function of** x if $f(-x) = f(x)$.
- **Odd function of** x if $f(-x) = -f(x)$.

for every x in the function's domain.

The names even and odd come from powers of x. If y is an even power of x, as in $y = x^2$ or $y = x^4$, it is an even function of x because $(-x)^2 = x^2$ and $(-x)^4 = x^4$. If y is an odd power of x, as in $y = x$ or $y = x^3$, it is an odd function of x because $(-x)^1 = -x$ and $(-x)^3 = -x^3$

The graph of an even function is **symmetric about the y-axis**. Since $f(-x) = f(x)$, a point (x, y) lies on the graph if and only if the point $(-x, y)$ lies on the graph (Figure 1.26 a). A reflection across the y-axis leaves the graph unchanged.

The graph of an odd function is **symmetric about the origin**. Since $f(-x) = -f(x)$, a point (x, y) lies on the graph if and only if the point $(-x, -y)$ lies on the graph (Figure 1.26 b). Equivalently, a graph is symmetric about the origin if a rotation of

180° about the origin leaves the graph unchanged. Notice that the definitions imply that both x and $-x$ must be in the domain of f.

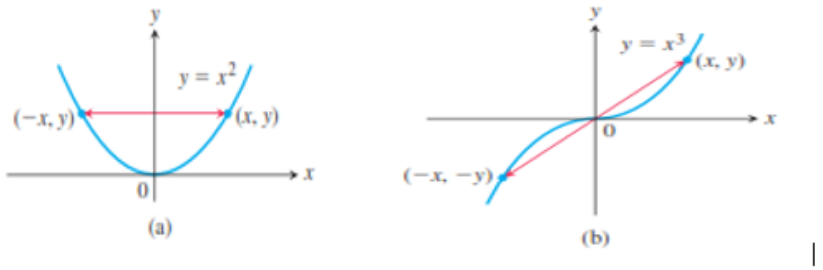


FIGURE 1.26

Example 1.5.2. Here are several functions illustrating the definition.

1. $f(x) = x^2$ Even function: $(-x)^2 = x^2$ for all x; symmetry about y-axis.
2. $f(x) = x^2 + 1$ Even function: $(-x)^2 + 1 = x^2 + 1$ for all x; symmetry about y-axis.
3. $f(x) = x$ Odd function: $(-x) = -x$ for all x; symmetry about the origin.
4. $f(x) = x + 1$ Not odd: $f(-x) = -x + 1$, but $-f(x) = -x - 1$. The two are not equal. Not even: $(-x) + 1 \neq x + 1$ for all $x \neq 0$

Example 1.5.3. Determine whether the function is even, odd, or neither.

(a) $f(x) = x^4$

(b) $g(x) = x^{-1}$

(c) $h(x) = x^2 + x$

Solution:

(a) $f(-x) = (-x)^4 = x^4$. Thus, $f(x) = f(-x)$, and f is even.

(b) $g(-x) = (-x)^{-1} = -x^{-1}$. Thus, $g(-x) = -g(x)$, and g is odd.

(c) $h(-x) = (-x)^2 + (-x) = x^2 - x$. We see that $h(-x)$ is not equal to $h(x)$ or to $-h(x) = -x^2 - x$. Therefore, h is neither even nor odd.

1.5.3 Linear and Quadratic Functions

Linear functions are the simplest of all functions, and their graphs (lines) are the simplest of all curves. However, linear functions and lines play an enormously important role in calculus. For this reason, you should be thoroughly familiar with the basic properties of linear functions and the different ways of writing an equation of a line.

Let's recall that a **linear function** is a function of the form

$$f(x) = mx + b \quad (m \quad \text{and} \quad b \quad \text{constants})$$

The graph of f is a line of slope m, and since $f(0) = b$, the graph intersects the y-axis at the point $(0, b)$ (Figure 1). The number b is called the y-intercept.

The slope-intercept form of the line with slope m and y-intercept b is given by

$$y = mx + b$$

We use the symbols Δx and Δy to denote the change (or increment) in x and $y = f(x)$ over an interval $[x_1, x_2]$ (Figure 1.27):

$$\Delta x = x_2 - x_1, \quad \Delta y = y_2 - y_1 = f(x_2) - f(x_1)$$

The slope m of a line is equal to the ratio

$$m = \frac{\Delta y}{\Delta x} = \frac{\text{vertical change}}{\text{horizontal change}} = \frac{\text{rise}}{\text{run}}$$

This follows from the formula $y = mx + b$:

$$\frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{(mx_2 + b) - (mx_1 + b)}{x_2 - x_1} = \frac{m(x_2 - x_1)}{x_2 - x_1} = m$$

The slope m measures the rate of change of y with respect to x. In fact, by writing

$$\Delta y = m\Delta x$$

we see that a 1 -unit increase in x (i.e., $\Delta x = 1$) produces an m-unit change Δy in y. For example, if $m = 5$, then y increases by 5 units per unit increase in x. The rate-of-change interpretation of the slope is fundamental in calculus.

Graphically, the slope m measures the steepness of the line $y = mx + b$. (Figure 1.28 A). shows lines through a point of varying slope m.

Note the following properties:

Steepness: The larger the absolute value $|m|$, the steeper the line.

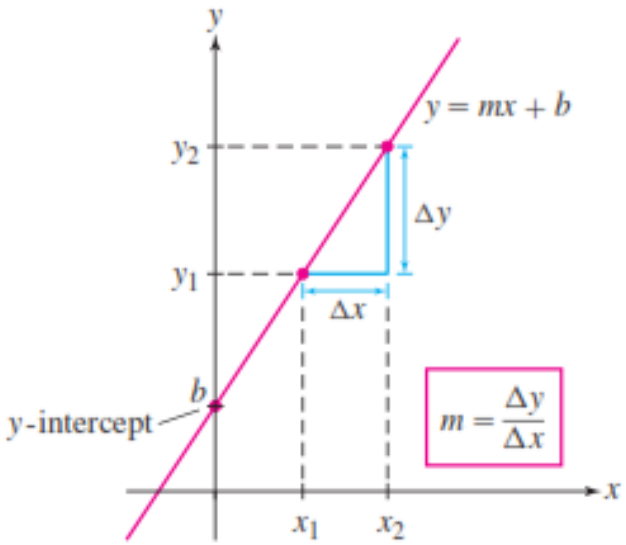


FIGURE 1.27: The slope m is the ratio "rise over run."

Positive slope: If $m > 0$, the line slants upward from left to right.

Negative slope: If $m < 0$, the line slants downward from left to right: $f(x) = mx+b$ is increasing if $m > 0$ and decreasing if $m < 0$.

The horizontal line $y = b$ has slope $m = 0$ (Figure 1.28 B).

A **vertical line** has equation $x = c$, where c is a constant. The slope of a vertical line is undefined. It is not possible to write the equation of a vertical line in slope-intercept form $y = mx + b$. A vertical line is not the graph of a function (Figure 1.28 B).

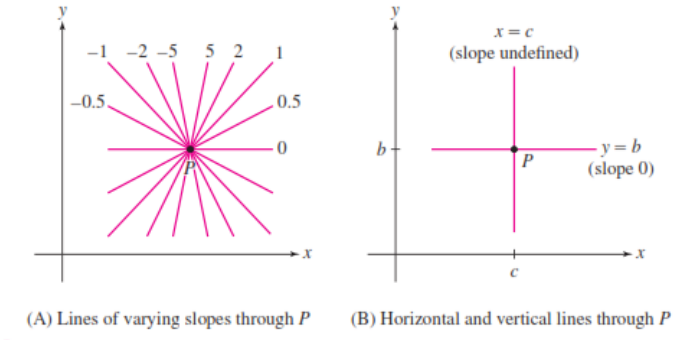


FIGURE 1.28: The slope m is the ratio "rise over run."

Next, we recall the relation between the slopes of **parallel** and **perpendicular** lines (Figure 1.29):

- Lines of slopes m_1 and m_2 are parallel if and only if $m_1 = m_2$.
- Lines of slopes m_1 and m_2 are perpendicular if and only if

$$m_1 = -\frac{1}{m_2} \quad (\text{ or } m_1 m_2 = -1)$$

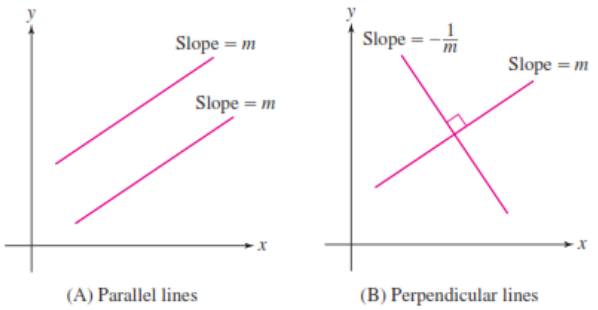


FIGURE 1.29: The slope m is the ratio "rise over run."

As mentioned above, it is important to be familiar with the standard ways of writing the equation of a line. The general linear equation is

$$ax + by = c \qquad (1)$$

where a and b are not both zero. For $b = 0$, we obtain the vertical line $ax = c$. When $b \neq 0$, we can rewrite Eq. (1) in slope-intercept form. For example, $-6x + 2y = 3$ can be rewritten as $y = 3x + \frac{3}{2}$.

Two other forms we will use frequently are the point-slope and point-point forms. Given a point $P = (a, b)$ and a slope m, the equation of the line through P with slope m is $y - b = m(x - a)$. Similarly, the line through two distinct points $P = (a_1, b_1)$ and $Q = (a_2, b_2)$ has slope (Figure 7)

$$m = \frac{b_2 - b_1}{a_2 - a_1}$$

Therefore, we can write its equation as $y - b_1 = m\,(x - a_1)$.

Additional Equations for Lines:

1. Point-slope form of the line through $P = (a, b)$ with slope m :

$$y - b = m(x - a)$$

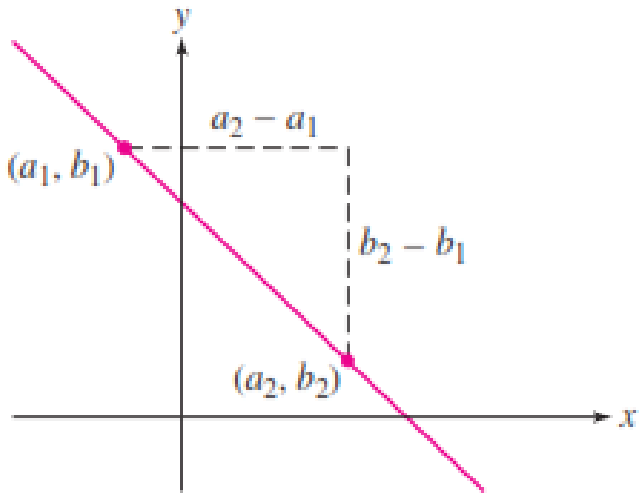


FIGURE 1.30

2. Point-point form of the line through

$$P = (a_1, b_1) \text{ and} Q = (a_2, b_2) : y - b_1 = m\,(x - a_1)\,, \text{ where } m = \frac{b_2 - b_1}{a_2 - a_1}$$

Example 1.5.4. Find the equation of the line through $(9, 2)$ with slope $-\frac{2}{3}$.
Solution In point-slope form:

$$y - 2 = -\frac{2}{3}(x - 9)$$

In slope-intercept form:

$$y = -\frac{2}{3}(x - 9) + 2 \quad \text{or} \quad y = -\frac{2}{3}x + 8.$$

Example 1.5.5. Find the equation of the line through $(2, 1)$ and $(9, 5)$.
Solution The line has slope

$$m = \frac{(5 - 1)}{(9 - 2)} = \frac{4}{7}$$

Because $(2, 1)$ lies on the line, its equation in point-slope form is

$$y - 1 = -\frac{4}{7}(x - 2).$$

A quadratic function: is a function defined by a quadratic polynomial

$$f(x) = ax^2 + bx + c \quad (a, b, c, \text{ constants with } a \neq 0)$$

The graph of f is a parabola (Figure 1.31). The parabola opens upward if the leading coefficient a is positive and downward if a is negative. The discriminant of $f(x)$ is the quantity

$$D = b^2 - 4ac$$

The roots of f are given by the quadratic formula :

$$\text{Roots of } f = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-b \pm \sqrt{D}}{2a}$$

The sign of D determines whether or not f has real roots (Figure 1.31). If $D > 0$, then f has two real roots, and if $D = 0$, it has one real root (a "double root"). If $D < 0$, then $\sqrt{D}$ is imaginary and f has no real roots.

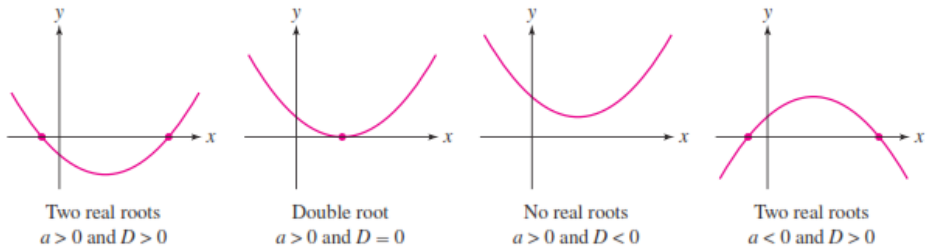


FIGURE 1.31

When f has two real roots r_1 and r_2, then $f(x)$ factors as

$$f(x) = a\,(x - r_1)\,(x - r_2)$$

For example, $f(x) = 2x^2 - 3x + 1$ has discriminant $D = b^2 - 4ac = 9 - 8 = 1 > 0$ and by the quadratic formula, its roots are $(3 \pm 1)/4$ or 1 and $\frac{1}{2}$. Therefore,

$$f(x) = 2x^2 - 3x + 1 = 2(x - 1)\left(x - \frac{1}{2}\right)$$

The technique of **completing the square** consists of writing a quadratic polynomial as a multiple of a square plus a constant. Then

$$x^2 + bx + c = x^2 + bx + \left(\frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$$

If there is a constant a multiplying the x^2 term, we factor that out first, as demonstrated in the following example.

Example 1.5.6. Completing the Square Complete the square for the quadratic polynomial $f(x) = 4x^2 - 12x + 3$

Solution: First factor out the leading coefficient:

$$4x^2 - 12x + 3 = 4\left(x^2 - 3x + \frac{3}{4}\right)$$

Then complete the square for the term $x^2 - 3x$:

$$x^2 - 3x = x^2 - 3x + \left(\frac{3}{2}\right)^2 - \left(\frac{3}{2}\right)^2 = \left(x - \frac{3}{2}\right)^2 - \frac{9}{4}$$

Therefore,

$$4x^2 - 12x + 3 = 4\left(\left(x - \frac{3}{2}\right)^2 - \frac{9}{4} + \frac{3}{4}\right) = 4\left(x - \frac{3}{2}\right)^2 - 6.$$

1.5.4 Trigonometric Functions

The Six Basic Trigonometric Functions: You are probably familiar with defining the trigonometric functions of an acute angle in terms of the sides of a right triangle (Figure 1.32).

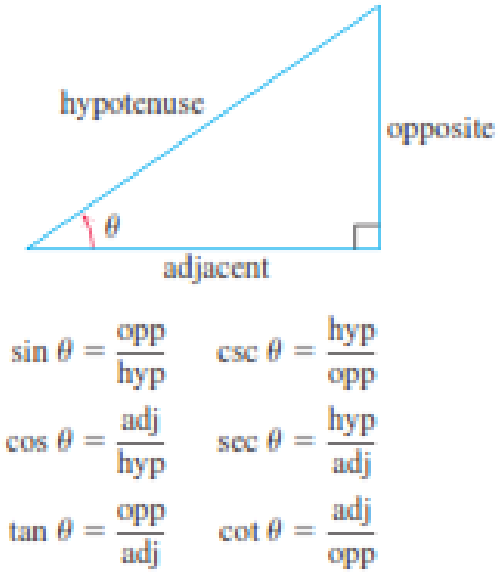


FIGURE 1.32: Trigonometric ratios of an acute angle.

We extend this definition to obtuse and negative angles by first placing the angle in standard position in a circle of radius r. We then define the trigonometric functions in terms of the coordinates of the point P(x, y) where the angle's terminal ray intersects the circle (Figure 1.32).

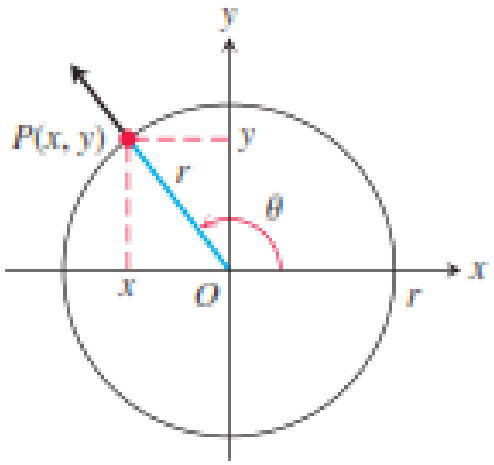


FIGURE 1.33

$$\begin{aligned} \text{sine: } \sin\theta &= \frac{y}{r} & \text{cosecant: } \csc\theta &= \frac{r}{y} \\ \text{cosine: } \cos\theta &= \frac{x}{r} & \text{secant: } \sec\theta &= \frac{r}{x} \\ \text{tangent: } \tan\theta &= \frac{y}{x} & \text{cotangent: } \cot\theta &= \frac{x}{y}. \end{aligned}$$

These extended definitions agree with the right-triangle definitions when the angle is acute. Notice also that whenever the quotients are defined,

$$\begin{aligned} \tan\theta &= \frac{\sin\theta}{\cos\theta} & \cot\theta &= \frac{1}{\tan\theta} \\ \sec\theta &= \frac{1}{\cos\theta} & \csc\theta &= \frac{1}{\sin\theta}. \end{aligned}$$

As you can see, $\tan\theta$ and $\sec\theta$ are not defined if $x = \cos\theta = 0$. This means they are not defined if θ is $\pm\pi/2, \pm 3\pi/2, \ldots$. Similarly, $\cot\theta$ and $\csc\theta$ are not defined for values of θ for which $y = 0$, namely $\theta = 0, \pm\pi, \pm 2\pi, \ldots$.

The exact values of these trigonometric ratios for some angles can be read from the triangles in Figure 1.41. For instance,

$$\begin{array}{lll} \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}} & \sin\frac{\pi}{6} = \frac{1}{2} & \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2} \\ \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}} & \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2} & \cos\frac{\pi}{3} = \frac{1}{2} \\ \tan\frac{\pi}{4} = 1 & \tan\frac{\pi}{6} = \frac{1}{\sqrt{3}} & \tan\frac{\pi}{3} = \sqrt{3} \end{array}$$

The CAST rule (Figure 1.42) is useful for remembering when the basic trigonometric functions are positive or negative. For instance, from the triangle in (Figure **??**), we

see that

$$\sin\frac{2\pi}{3}=\frac{\sqrt{3}}{2},\quad \cos\frac{2\pi}{3}=-\frac{1}{2},\quad \tan\frac{2\pi}{3}=-\sqrt{3}$$

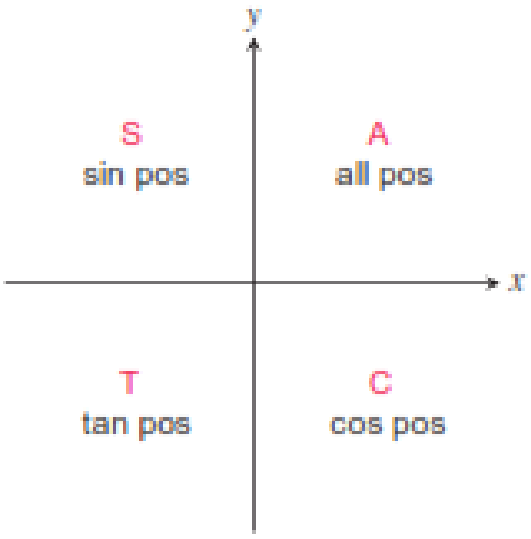


FIGURE 1.34: The CAST rule.

Periodicity and Graphs of the Trigonometric Functions

When an angle of measure θ and an angle of measure $\theta+2\pi$ are in standard position, their terminal rays coincide. The two angles therefore have the same trigonometric function values: $\sin(\theta+2\pi)=\sin\theta, \tan(\theta+2\pi)=\tan\theta$, and so on. Similarly, $\cos(\theta-2\pi)=\cos\theta, \sin(\theta-2\pi)=\sin\theta$, and so on. We describe this repeating behavior by saying that the six basic trigonometric functions are periodic.

Definition 1.12. *A function $f(x)$ is **periodic** if there is a positive number p such that $f(x+p)=f(x)$ for every value of x. The smallest such value of p is the period of f.*

When we graph trigonometric functions in the coordinate plane, we usually denote the independent variable by x instead of θ. (Figure 1.37) shows that the tangent and cotangent functions have period $p=\pi$, and the other four functions have period 2π. Also, the symmetries in these graphs reveal that the cosine and secant functions are even and the other four functions are odd (although this does not prove those results).

Example 1.5.7. Find the values of the six trigonometric functions at $x=4\pi/3$.

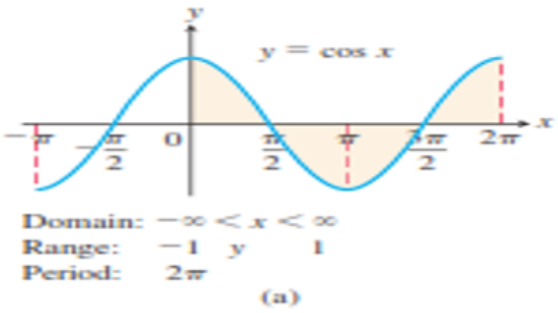


FIGURE 1.35

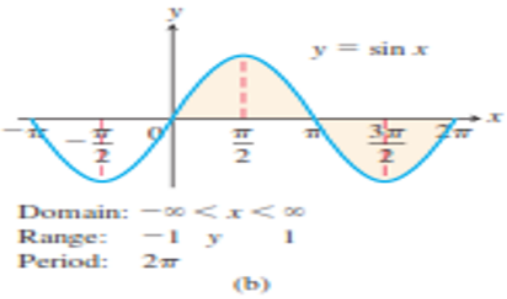


FIGURE 1.36

Solution: The point P on the unit circle corresponding to the angle $x = 4\pi/3$ lies opposite the point with angle $\pi/3$ (Figure 1.37).

$$\sin\frac{4\pi}{3} = -\sin\frac{\pi}{3} = -\frac{\sqrt{3}}{2}, \quad \cos\frac{4\pi}{3} = -\cos\frac{\pi}{3} = -\frac{1}{2}$$

The remaining values are

$$\tan\frac{4\pi}{3} = \frac{\sin 4\pi/3}{\cos 4\pi/3} = \frac{-\sqrt{3}/2}{-1/2} = \sqrt{3}, \quad \cot\frac{4\pi}{3} = \frac{\cos 4\pi/3}{\sin 4\pi/3} = \frac{\sqrt{3}}{3}$$
$$\sec\frac{4\pi}{3} = \frac{1}{\cos 4\pi/3} = \frac{1}{-1/2} = -2, \quad \csc\frac{4\pi}{3} = \frac{1}{\sin 4\pi/3} = \frac{-2\sqrt{3}}{3}$$

Example 1.5.8. Find the angles x such that $\sec x = 2$

Solution: Because $\sec x = 1/\cos x$, we must solve $\cos x = \frac{1}{2}$. From (Figure 1.37). we see that $x = \pi/3$ and $x = -\pi/3$ are solutions. We may add any integer multiple of 2π, so the general solution is $x = \pm\pi/3 + 2\pi k$ for any integer k.

Trigonometric Identities

A key feature of trigonometric functions is that they satisfy a large number of identities. First and foremost, sine and cosine satisfy a fundamental identity, which is

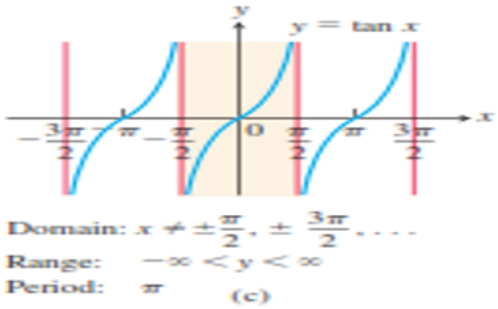


FIGURE 1.37

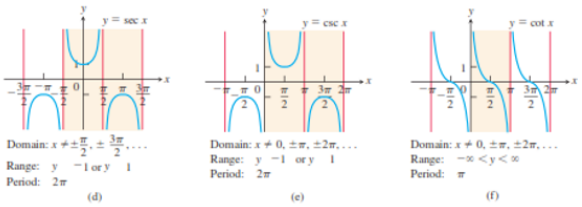


FIGURE 1.38

equivalent to the **Pythagorean Theorem:**

$$\sin^2 x + \cos^2 x = 1 \qquad (1)$$

Equivalent versions are obtained by dividing Eq. (1) by $\cos^2 x$ or $\sin^2 x$:

$$\tan^2 x + 1 = \sec^2 x, \quad 1 + \cot^2 x = \csc^2 x$$

Example 1.5.9. Suppose that $\cos\theta = \frac{2}{5}$. Calculate $\tan\theta$ in the following two cases:

(a) $0 < \theta < \frac{\pi}{2}$ and

(b) $\pi < \theta < 2\pi$

Solution: First, using the identity $\cos^2\theta + \sin^2\theta = 1$, we obtain

$$\sin\theta = \pm\sqrt{1 - \cos^2\theta} = \pm\sqrt{1 - \frac{4}{25}} = \pm\frac{\sqrt{21}}{5}$$

(a) If $0 < \theta < \frac{\pi}{2}$, then $\sin\theta$ is positive and we take the positive square root:

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\sqrt{21}/5}{2/5} = \frac{\sqrt{21}}{2}.$$

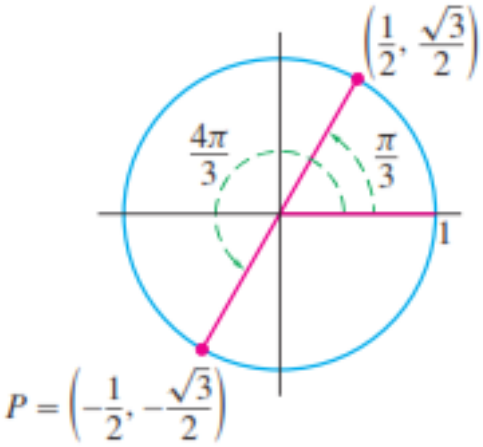


FIGURE 1.39

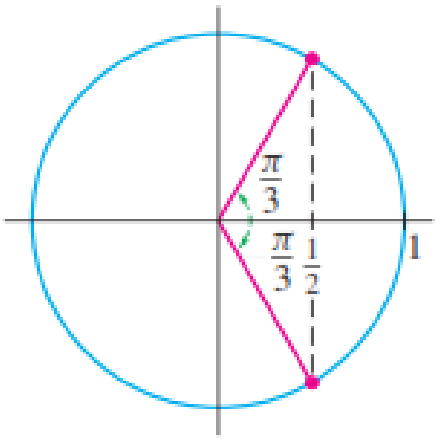


FIGURE 1.40

To visualize this computation, draw a right triangle with angle θ such that $\cos\theta = \frac{2}{5}$. The opposite side then has length $\sqrt{21} = \sqrt{5^2 - 2^2}$ by the Pythagorean Theorem.
(b) If $\pi < \theta < 2\pi$, then $\sin\theta$ is negative and $\tan\theta = -\frac{\sqrt{21}}{2}$.

1.5.5 Exponential and Logarithmic Functions

Definition 1.13. *Functions of the form $f(x) = a^T$, where the base $a > 0$ is a positive constant and $a \neq 1$, are called **exponential** functions.*

All exponential functions have domain $(-\infty, \infty)$ and range $(0, \infty)$, so an exponential function never assumes the value 0. The graphs of some exponential functions are shown in (Figure 1.41) . The exponential functions has the following properties

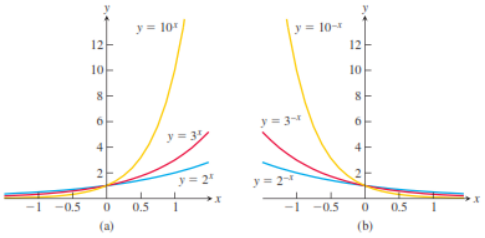


FIGURE 1.41: Graphs of exponential functions.

1. $a^0 = 1$.
2. $a^{m+n} = a^m a^n$.
3. $a^{m-n} = a^m/a^n$.
4. $(a^m)^n = a^{mn}$.

When a is the natural base i.e, $a = e \cong 2.718$ then $f(x) = e^x$ is called the natural exponential function.

Definition 1.14. *These are the functions* $f(x) = \log_a x$, *where the base* $a \neq 1$ *is a positive constant. They are the inverse functions of the exponential functions.*

(Figure 1.42)shows the graphs of four logarithmic functions with various bases. In each case the domain is $(0, \infty)$ and the range is $(-\infty, \infty)$.

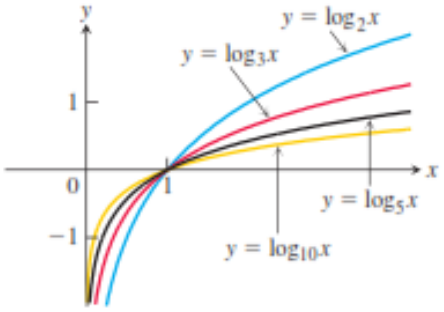


FIGURE 1.42: Graphs of four logarithmic functions.

For all real numbers a, m, n, p, x, and y, where $x > 0$, a > 0, a $\neq 1$.

1. If $a^x = a^y$, then $x = y$.
2. $\log_a x = y$ if and only if $x = a^y$.
3. $\log_{10} x = \log x$
4. $\log_e x = \ln x$.
5. $\log_a a^x = x, \quad \ln e^x = x$.
6. $a^{\log_a x} = x, \quad e^{\ln x} = x$.
7. $\log_a mn = \log_a m + \log_a n$.
8. $\log_a \dfrac{m}{n} = \log_a m - \log_a n$.

9. $\log_a m^p = p\,(\log_a m)$.
10. $\log_a x = \dfrac{\log x}{\log a} = \dfrac{\ln x}{\ln a}, a \neq 1$.
11. If $\log_a x = \log_a y$, then $x = y$.

1.5.6 Hyperbolic Functions

The hyperbolic functions are certain special combinations of e^x and e^{-x} that play a role in engineering and physics . The hyperbolic sine and cosine, often called "cinch" and "cosh," are defined as follows:

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}.$$

As the terminology suggests, there are similarities between the hyperbolic and trigonometric functions. Here are some examples:

- **Parity:** The trigonometric functions and their hyperbolic analogs have the same parity. Thus, $f(x) = \sin x$ and $f(x) = \sinh x$ are both odd, and $f(x) = \cos x$ and $f(x) = \cosh x$ are both even (Figure 2):

$$\sinh(-x) = -\sinh x, \quad \cosh(-x) = \cosh x.$$

- **Identities:** The basic trigonometric identity $\sin^2 x + \cos^2 x = 1$ has a hyperbolic analog:

$$\cosh^2 x - \sinh^2 x = 1.$$

  The addition formulas satisfied by $\sin x$ and $\cos x$ also have hyperbolic analogs:

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y,$$
$$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y.$$

- **Hyperbola instead of the circle:** Because of the identity $\cosh^2 t - \sinh^2 t = 1$, the point $(\cosh t, \sinh t)$ lies on the hyperbola $x^2 - y^2 = 1$, just as $(\cos t, \sin t)$ lies on the unit circle $x^2 + y^2 = 1$.

- **Other hyperbolic functions:** The hyperbolic tangent, cotangent, secant, and cosecant functions are defined like their trigonometric counterparts:

$$\tanh x = \tfrac{\sinh x}{\cosh x} = \tfrac{e^x - e^{-x}}{e^x + e^{-x}}, \quad \operatorname{sech} x = \tfrac{1}{\cosh x} = \tfrac{2}{e^x + e^{-x}}$$
$$\coth x = \tfrac{\cosh x}{\sinh x} = \tfrac{e^x + e^{-x}}{e^x - e^{-x}}, \quad \operatorname{csch} x = \tfrac{1}{\sinh x} = \tfrac{2}{e^x - e^{-x}}.$$

1.5.7 Inverse Of Hyperbolic Functions

Just as the inverse trigonometric functions are useful in certain integrations, the inverse hyperbolic functions are useful with others. Because the hyperbolic functions are defined in terms of exponential functions, their inverses can be expressed in terms of logarithms as shown in the following . It is often more convenient to refer to $sinh^{-1}x$ than to $\ln(x+\sqrt{x^2+1})$, especially when one is working on theory and does not need to compute actual values.

Logarithmic definitions of Inverse Hyperbolic Functions

1. $\cosh^{-1} x = \ln\left(x + \sqrt{x^2-1}\right); \quad x \geq 1,$
2. $\tanh^{-1} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right); \quad |x| < 1,$
3. $\operatorname{sech}^{-1} x = \ln\left(\frac{1+\sqrt{1-x^2}}{x}\right); \quad 0 < x \leq 1,$
4. $\sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right),$
5. $\coth^{-1} x = \dfrac{1}{2}\ln\left(\dfrac{x+1}{x-1}\right); \quad |x| > 1,$
6. $\operatorname{csch}^{-1} x = \ln\left(\frac{1}{x} + \dfrac{\sqrt{1+x^2}}{|x|}\right); \quad x \neq 0.$

Example 1.5.10. Prove that $\sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right)$.

Soultion: Let's consider the inverses of the hyperbolic functions. We begin with the function $f(x) = sinhx$.

$$\begin{aligned} y &= \frac{e^x - e^{-x}}{2} \\ 2y &= e^x - e^{-x} \\ 2ye^x &= e^{2x} - 1 \end{aligned}$$

Then

$$(e^x)^2 - 2ye^x - 1 = 0$$

Hence

$$e^x = \frac{2y \pm \sqrt{4y^2+4}}{2} = y \pm \sqrt{y^2+1} = y + \sqrt{y^2+1}$$

That is, $x = \ln\left(y + \sqrt{y^2+1}\right)$. Finally, interchange the variable to find that

$$\sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right).$$

In a similar manner we can derive the inverses of the other hyperbolic functions.

In such a situation, the logarithmic representation is useful. The reader is not encouraged to memorize these, but rather know they exist and know how to use them when needed.

Function	Domain	Range	Function	Domain	Range
$\cosh x$	$[0,\infty)$	$[1,\infty)$	$\cosh^{-1} x$	$[1,\infty)$	$[0,\infty)$
$\sinh x$	$(-\infty,\infty)$	$(-\infty,\infty)$	$\sinh^{-1} x$	$(-\infty,\infty)$	$(-\infty,\infty)$
$\tanh x$	$(-\infty,\infty)$	$(-1,1)$	$\tanh^{-1} x$	$(-1,1)$	$(-\infty,\infty)$
$\operatorname{sech} x$	$[0,\infty)$	$(0,1]$	$\operatorname{sech}^{-1} x$	$(0,1]$	$[0,\infty)$
$\operatorname{csch} x$	$(-\infty,0)\cup(0,\infty)$	$(-\infty,0)\cup(0,\infty)$	$\operatorname{csch}^{-1} x$	$(-\infty,0)\cup(0,\infty)$	$(-\infty,0)\cup(0,\infty)$
$\coth x$	$(-\infty,0)\cup(0,\infty)$	$(-\infty,-1)\cup(1,\infty)$	$\coth^{-1} x$	$(-\infty,-1)\cup(1,\infty)$	$(-\infty,0)\cup(0,\infty)$

FIGURE 1.43

1.6 Exercises

1. Determine whether the function is even, odd, or neither.

   (a) $y = x^2 + 1$

   (b) $y = x^5 - x^3 - x$

   (c) $y = 1 - \cos x$

   (d) $y = \sec x \tan x$

   (e) $y = \frac{x^4+1}{x^3-2x}$

   (f) $y = x - \sin x$

   (g) $y = x + \cos x$

   (h) $y = x \cos x$

   (i) $f(x) = e^{-x^2}$

   (j) $f(x) = 1 + \sin^{-1}(-x)$

   (k) $f(x) = |e^x|$.

   (l) $f(x) = e^{\ln|x|+1}$

2. Suppose that f and g are both odd functions defined on the entire real line. Which of the following (where defined) are even? odd?

   a. fg, b. f^3, c. $f(\sin x)$, d. $g(\sec x)$.

3. If $f(a-x) = f(a+x)$, show that $g(x) = f(x+a)$ is an even function.

4. Find the (a) domain and (b) range.

   (a) $y = |x| - 2$

   (b) $y = -2 + \sqrt{1-x}$

   (c) $y = \sqrt{16 - x^2}$

   (d) $y = 3^{2-x} + 1$

   (e) $y = 2e^{-x} - 3$

   (f) $y = \tan(2x - \pi)$

   (g) $y = 2\sin(3x + \pi) - 1$

   (h) $y = x^{2/5}$

(i) $y = \ln(x-3)+1$

(j) $y = -1 + \sqrt[3]{2-x}$

(k) $y = \begin{cases} \sqrt{-x}, & -4 \le x \le 0 \\ \sqrt{x}, & 0 < x \le 4 \end{cases}$

(l) $y = \begin{cases} -x-2, & -2 \le x \le -1 \\ x, & -1 < x \le 1 \\ -x+2, & 1 < x \le 2 \end{cases}$

(m) $f(x) = 1 + e^{-\sin x}$

(n) $g(x) = e^x + \ln\sqrt{x}$

(o) $f(x) = e^{1/x^2}$

(p) $g(x) = \ln|4-x^2|$

(q) $h(x) = \sin^{-1}\left(\frac{x}{3}\right)$

(r) $f(x) = \cos^{-1}(\sqrt{x}-1)$

(s) $h(x) = \ln(\cos^{-1} x)$

(t) $f(x) = \sqrt{\pi - \sin^{-1} x}$

5. State whether each function is increasing, decreasing, or neither.

   (a) Volume of a sphere as a function of its radius

   (b) Greatest integer function

   (c) Height above Earth's sea level as a function of atmospheric pressure (assumed nonzero)

   (d) Kinetic energy as a function of a particle's velocity.

6. Find the largest interval on which the given function is increasing.

   (a) $f(x) = |x-2|+1$

   (b) $f(x) = (x+1)^4$

   (c) $g(x) = (3x-1)^{1/3}$

   (d) $R(x) = \sqrt{2x-1}$

7. Find

   a. $(f \circ g)(-1)$. b. $(g \circ f)(2)$. c. $(f \circ f)(x)$. d. $(g \circ g)(x)$.

   (a) $f(x) = \frac{1}{x}, \quad g(x) = \frac{1}{\sqrt{x+2}}$

   (b) $f(x) = 2-x, \quad g(x) = \sqrt[3]{x+1}$

8. (a) write formulas for $f \circ g$ and $g \circ f$ and find the (b) domain and (c) range of each.

   (a) $f(x) = 2-x^2, \quad g(x) = \sqrt{x+2}$

   (b) $f(x) = \sqrt{x}, \quad g(x) = \sqrt{1-x}$

Chapter 2

Limits and continuity

2.1 limits of Function values

Frequently when studying a function $y = f(x)$, we find ourselves interested in the function's behavior near a particular point c, but not at c. This might be the case, for instance, if c is an irrational number, like π or $\sqrt{2}$, whose values can only be approximated by "close" rational numbers at which we actually evaluate the function instead. Another situation occurs when trying to evaluate a function at c leads to division by zero, which is undefined. We encountered this last circumstance when seeking the instantaneous rate of change in y by considering the quotient function $\Delta y/h$ for h closer and closer to zero. Here's a specific example in which we explore numerically how a function behaves near a particular point at which we cannot directly evaluate the function.

Example 2.1.1. *How does the function*

$$f(x) = \frac{x^2 - 1}{x - 1}$$

behave near $x = 1$ *?*

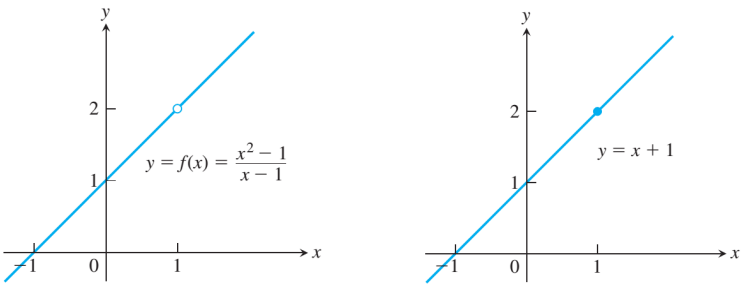


Figure 2.1

Solution: The given formula defines f for all real numbers x except $x = 1$ (we cannot divide by zero). For any $x \neq 1$, we can simplify the formula by factoring the numerator

and canceling common factors:

$$f(x) = \frac{(x-1)(x+1)}{x-1} = x+1 \quad \text{for} \quad x \neq 1$$

The graph of f is the line $y = x+1$ with the point $(1,2)$ removed. This removed point is shown as a "hole" in Figure 2.1. Even though $f(1)$ is not defined, it is clear that we can make the value of $f(x)$ as close as we want to 2 by choosing x close enough to 1 (Table 2.1).

x	$f(x) = \frac{x^2-1}{x-1}$
0.9	1.9
1.1	2.1
0.99	1.99
1.01	2.01
0.999	1.999
1.001	2.001
0.999999	1.999999
1.000001	2.000001

TABLE 2.1 As x gets closer to 1, $f(x)$ gets closer to 2 .

Generalizing the idea illustrated in Example 1, suppose $f(x)$ is defined on an open interval about c, except possibly at c itself. If $f(x)$ is arbitrarily close to the number L (as close to L as we like) for all x sufficiently close to c, we say that f approaches the limit L as x approaches c, and write

$$\lim_{x \to c} f(x) = L$$

which is read "the limit of $f(x)$ as x approaches c is L. " For instance, in Example 1 we would say that $f(x)$ approaches the limit 2 as x approaches 1, and write

$$\lim_{x \to 1} f(x) = 2, \quad \text{or} \quad \lim_{x \to 1} \frac{x^2-1}{x-1} = 2$$

Essentially, the definition says that the values of $f(x)$ are close to the number L whenever x is close to c (on either side of c).

2.1.1 The limit laws

To calculate limits of functions that are arithmetic combinations of functions having known limits, we can use several fundamental rules.

Theorem 2.1 (Limits Laws). *If L, M, c, and k are real numbers and $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

1. *Sum Rule:* $\lim_{x\to c}(f(x) + g(x)) = L + M$.
2. *Difference Rule:* $\lim_{x\to c}(f(x) - g(x)) = L - M$.
3. *Constant Multiple Rule:* $\lim_{x\to c}(k \cdot f(x)) = k \cdot L$.
4. *Product Rule:* $\lim_{x\to c}(f(x) \cdot g(x)) = L \cdot M$.
5. *Quotient Rule:* $\lim_{x\to c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$.
6. *Power Rule:* $\lim_{x\to c}[f(x)]^n = L^n, n$ *a positive integer*
7. *Root Rule:* $\lim_{x\to c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, n$ *a positive integer (If n is even, we assume that $\lim_{x\to c} f(x) = L > 0$..)*

Moreover, If $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$, then

$$\lim_{x\to c} P(x) = P(c) = a_n c^n + a_{n-1}c^{n-1} + \cdots + a_0.$$

If $P(x)$ and $Q(x)$ are polynomials and $Q(c) \neq 0$, then

$$\lim_{x\to c} \frac{P(x)}{Q(x)} = \frac{P(c)}{Q(c)}.$$

Example 2.1.2.

$$\lim_{x\to -1} \frac{x^3 + 4x^2 - 3}{x^2 + 5} = \frac{(-1)^3 + 4(-1)^2 - 3}{(-1)^2 + 5} = \frac{0}{6} = 0.$$

2.1.2 Eliminating Common Factors from Zero Denominators

In case of rational functions, the last case of Theorem 2.1 applies only if the denominator of the rational function is not zero at the limit point c. If the denominator is zero, canceling common factors in the numerator and denominator may reduce the fraction to one whose denominator is no longer zero at c. If this happens, we can find the limit by substitution in the simplified fraction.

Example 2.1.3. *Evaluate*

$$\lim_{x\to 1} \frac{x^2 + x - 2}{x^2 - x}$$

Solution: We cannot substitute $x = 1$ because it makes the denominator zero. We test the numerator to see if it, too, is zero at $x = 1$. It is, so it has a factor of $(x - 1)$ in

common with the denominator. Canceling this common factor gives a simpler fraction with the same values as the original for $x \neq 1$

$$\frac{x^2+x-2}{x^2-x} = \frac{(x-1)(x+2)}{x(x-1)} = \frac{x+2}{x}, \quad \text{if } x \neq 1$$

Using the simpler fraction, we find the limit of these values as $x \to 1$ by Theorem 2.1 :

$$\lim_{x\to 1} \frac{x^2+x-2}{x^2-x} = \lim_{x\to 1} \frac{x+2}{x} = \frac{1+2}{1} = 3$$

See Figure 2.2.

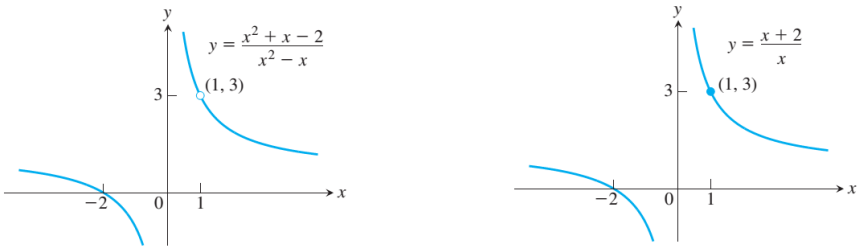


Figure 2.2

2.1.3 The Sandwich Theorem

The following theorem enables us to calculate a variety of limits. It is called the Sandwich Theorem because it refers to a function whose values are sandwiched between the values of two other functions g and h that have the same limit L at a point c. Being trapped between the values of two functions that approach L, the values of must also approach L (Figure 2.3).

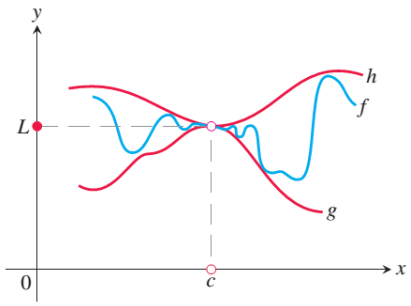


Figure 2.3

Theorem 2.2 (The Sandwich Theorem). *Suppose that $g(x) \leq f(x) \leq h(x)$ for all x in some open interval containing c, except possibly at $x = c$ itself. Suppose also that*

$$\lim_{x\to c} g(x) = \lim_{x\to c} h(x) = L$$

Then $\lim_{x\to c} f(x) = L$.

The Sandwich Theorem is also called the Squeeze Theorem or the Pinching Theorem.

Example 2.1.4. *Given that*

$$1 - \frac{x^2}{4} \le u(x) \le 1 + \frac{x^2}{2} \quad \text{for all } x \neq 0$$

find $\lim_{x\to 0} u(x)$, *no matter how complicated* u *is.*

Solution: Since

$$\lim_{x\to 0}\left(1 - \left(x^2/4\right)\right) = 1 \quad \text{and} \quad \lim_{x\to 0}\left(1 + \left(x^2/2\right)\right) = 1$$

the Sandwich Theorem implies that $\lim_{x\to 0} u(x) = 1$ (Figure 2.4).

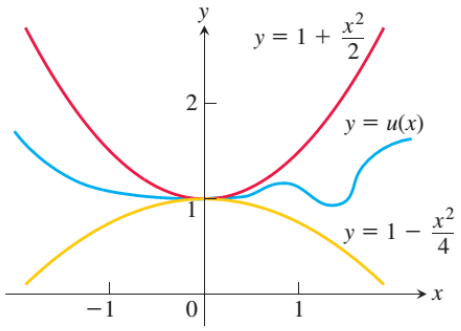


Figure 2.4

Example 2.1.5. *Using the Sandwich Theorem:*

(a) If $\sqrt{5 - 2x^2} \le f(x) \le \sqrt{5 - x^2}$ *for* $-1 \le x \le 1$, *find* $\lim_{x\to 0} f(x)$.

(b) If $2 - x^2 \le g(x) \le 2\cos x$ *for all* x, *find* $\lim_{x\to 0} g(x)$.

Solution: (a) Since

$$\lim_{x\to 0} \sqrt{5 - 2x^2} = \sqrt{5 - 2(0)^2} = \sqrt{5}$$

and

$$\lim_{x\to 0} \sqrt{5 - x^2} = \sqrt{5 - (0)^2} = \sqrt{5},$$

then by the sandwich theorem, $\lim_{x\to 0} f(x) = \sqrt{5}$.

(b) Since

$$\lim_{x\to 0}\left(2 - x^2\right) = 2 - 0 = 2$$

and

$$\lim_{x\to 0} 2\cos x = 2(1) = 2,$$

then by the sandwich theorem, $\lim_{x\to 0} g(x) = 2$.

2.2 The Precise Definition of a limit

We now turn our attention to the precise definition of a limit. We replace vague phrases like "gets arbitrarily close to" in the informal definition with specific conditions that can be applied to any particular example. With a precise definition, we can avoid misunderstandings, prove the limit properties given in the preceding section, and establish many important limits.

To show that the limit of (x) as $x \longrightarrow c$ equals the number L, we need to show that the gap between (x) and L can be made "as small as we choose" if x is kept "close enough" to c. Let us see what this would require if we specified the size of the gap between (x) and L.

Definition 2.3. *Let $f(x)$ be defined on an open interval about c, except possibly at c itself. We say that the limit of $f(x)$ as x approaches c is the number L, and write*

$$\lim_{x \to c} f(x) = L$$

if, for every number $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that for all x,

$$0 < |x - c| < \delta \quad \Rightarrow \quad |f(x) - L| < \epsilon.$$

See Figure 2.5.

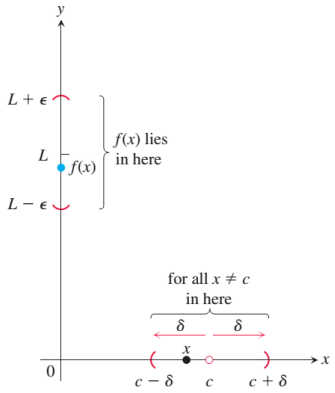


Figure 2.5

Example 2.2.1. *Show that*

$$\lim_{x \to 1}(5x - 3) = 2$$

Solution: Set $c = 1, f(x) = 5x - 3$, and $L = 2$ in the definition of limit. For any given $\epsilon > 0$, we have to find a suitable $\delta > 0$ so that if $x \neq 1$ and x is within distance δ of $c = 1$, that is, whenever

$$0 < |x - 1| < \delta$$

it is true that $f(x)$ is within distance ϵ of $L = 2$, so

$$|f(x) - 2| < \epsilon.$$

We find δ by working backward from the ϵ-inequality:

$$\begin{aligned} |(5x - 3) - 2| = |5x - 5| &< \epsilon \\ 5|x - 1| &< \epsilon \\ |x - 1| &< \epsilon/5 \end{aligned}$$

Thus, we can take $\delta = \epsilon/5$ (Figure 2.6). If $0 < |x - 1| < \delta = \epsilon/5$, then

$$|(5x - 3) - 2| = |5x - 5| = 5|x - 1| < 5(\epsilon/5) = \epsilon$$

which proves that $\lim_{x\to 1}(5x - 3) = 2$ The value of $\delta = \epsilon/5$ is not the only value that will make $0 < |x - 1| < \delta$ imply $|5x - 5| < \epsilon$. Any smaller positive δ will do as well. The definition does not ask for a "best" positive δ, just one that will work.

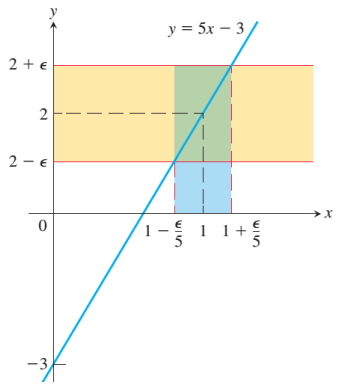


Figure 2.6

Example 2.2.2. *For the limit* $\lim_{x\to 5}\sqrt{x-1} = 2$, *find a* $\delta > 0$ *that works for* $\epsilon = 1$ *That is, find a* $\delta > 0$ *such that for all* x

$$0 < |x - 5| < \delta \quad \Rightarrow \quad |\sqrt{x-1} - 2| < 1$$

Solution: We organize the search into two steps:

1. Solve the inequality $|\sqrt{x-1}-2|<1$ to find an interval containing $x=5$ on which the inequality holds for all $x \neq 5$.

$$\begin{aligned} |\sqrt{x-1}-2| &< 1 \\ -1 < \sqrt{x-1}-2 &< 1 \\ 1 < \sqrt{x-1} &< 3 \\ 1 < x-1 &< 9 \\ 2 < x &< 10 \end{aligned}$$

The inequality holds for all x in the open interval $(2,10)$, so it holds for all $x \neq 5$ in this interval as well.

2. Find a value of $\delta > 0$ to place the centered interval $5-\delta < x < 5+\delta$ (centered at $x=5$) inside the interval $(2,10)$. The distance from 5 to the nearer endpoint of $(2,10)$ is 3. If we take $\delta = 3$ or any smaller positive number, then the inequality $0<|x-5|<\delta$ will automatically place x between 2 and 10 to make $|\sqrt{x-1}-2|<1$:

$$0<|x-5|<3 \quad \Rightarrow \quad |\sqrt{x-1}-2|<1.$$

Example 2.2.3. *Prove the limit statements in the following:*

(a) $\lim_{x\to 4}(9-x)=5$

(b) $\lim_{x\to 3}(3x-7)=2$

(c) $\lim_{x\to 9}\sqrt{x-5}=2$

(d) $\lim_{x\to 0}\sqrt{4-x}=2$

(e) $\lim_{x\to 1} f(x)=1$ *if* $f(x)=\begin{cases} x^2, & x\neq 1 \\ 2, & x=1 \end{cases}$

Solution:

(a) The first step is

$$\begin{aligned} |(9-x)-5|<\epsilon &\Rightarrow -\epsilon < 4-x < \epsilon \\ &\Rightarrow -\epsilon-4 < -x < \epsilon-4 \\ &\Rightarrow \epsilon+4 > x > 4-\epsilon \\ &\Rightarrow 4-\epsilon < x < 4+\epsilon. \end{aligned}$$

The second step is

$$\begin{aligned} |x-4|<\delta &\Rightarrow -\delta < x-4 < \delta \\ &\Rightarrow -\delta+4 < x < \delta+4 \end{aligned}$$

Then $-\delta + 4 = -\epsilon + 4 \Rightarrow \delta = \epsilon$, or $\delta + 4 = \epsilon + 4 \Rightarrow \delta = \epsilon$. Thus choose $\delta = \epsilon$.

(b) Similarly, we have

$$\begin{aligned} |(3x-7)-2| < \epsilon &\Rightarrow -\epsilon < 3x - 9 < \epsilon \\ &\Rightarrow 9 - \epsilon < 3x < 9 + \epsilon \\ &\Rightarrow 3 - \frac{\epsilon}{3} < x < 3 + \frac{\epsilon}{3}. \end{aligned}$$

$$\begin{aligned} |x-3| < \delta &\Rightarrow -\delta < x - 3 < \delta \\ &\Rightarrow -\delta + 3 < x < \delta + 3 \end{aligned}$$

Then $-\delta + 3 = 3 - \frac{\epsilon}{3} \Rightarrow \delta = \frac{\epsilon}{3}$, or $\delta + 3 = 3 + \frac{\epsilon}{3} \Rightarrow \delta = \frac{\epsilon}{3}$. Thus choose $\delta = \frac{\epsilon}{3}$.

(c) In one hand

$$\begin{aligned} |\sqrt{x-5}-2| < \epsilon &\Rightarrow -\epsilon < \sqrt{x-5} - 2 < \epsilon \\ &\Rightarrow 2 - \epsilon < \sqrt{x-5} < 2 + \epsilon \\ &\Rightarrow (2-\epsilon)^2 < x - 5 < (2+\epsilon)^2 \\ &\Rightarrow (2-\epsilon)^2 + 5 < x < (2+\epsilon)^2 + 5. \end{aligned}$$

On the other hand

$$\begin{aligned} |x-9| < \delta &\Rightarrow -\delta < x - 9 < \delta \\ &\Rightarrow -\delta + 9 < x < \delta + 9. \end{aligned}$$

Then $-\delta + 9 = \epsilon^2 - 4\epsilon + 9 \Rightarrow \delta = 4\epsilon - \epsilon^2$, or $\delta + 9 = \epsilon^2 + 4\epsilon + 9 \Rightarrow \delta = 4\epsilon + \epsilon^2$. Thus choose the smaller distance, $\delta = 4\epsilon - \epsilon^2$.

(d) We have

$$\begin{aligned} |\sqrt{4-x}-2| < \epsilon &\Rightarrow -\epsilon < \sqrt{4-x} - 2 < \epsilon \\ &\Rightarrow 2 - \epsilon < \sqrt{4-x} < 2 + \epsilon \\ &\Rightarrow (2-\epsilon)^2 < 4 - x < (2+\epsilon)^2 \\ &\Rightarrow -(2+\epsilon)^2 < x - 4 < -(2-\epsilon)^2 \\ &\Rightarrow -(2+\epsilon)^2 + 4 < x < -(2-\epsilon)^2 + 4. \end{aligned}$$

Moreover

$$|x-0| < \delta \Rightarrow -\delta < x < \delta.$$

Then $-\delta = -(2+\epsilon)^2 + 4 = -\epsilon^2 - 4\epsilon \Rightarrow \delta = 4\epsilon + \epsilon^2$, or $\delta = -(2-\epsilon)^2 + 4 = 4\epsilon - \epsilon^2$. Thus choose the smaller distance, $\delta = 4\epsilon - \epsilon^2$

(e) For $x \neq 1$, we have

$$\begin{aligned} |x^2 - 1| < \epsilon &\Rightarrow -\epsilon < x^2 - 1 < \epsilon \\ &\Rightarrow 1 - \epsilon < x^2 < 1 + \epsilon \\ &\Rightarrow \sqrt{1-\epsilon} < |x| < \sqrt{1+\epsilon} \\ &\Rightarrow \sqrt{1-\epsilon} < x < \sqrt{1+\epsilon}. \end{aligned}$$

$$\begin{aligned} |x - 1| < \delta &\Rightarrow -\delta < x - 1 < \delta \\ &\Rightarrow -\delta + 1 < x < \delta + 1 \end{aligned}$$

Then $-\delta + 1 = \sqrt{1-\epsilon} \Rightarrow \delta = 1 - \sqrt{1-\epsilon}$, or $\delta + 1 = \sqrt{1+\epsilon} \Rightarrow \delta = \sqrt{1+\epsilon} - 1$. Choose $\delta = \min\{1 - \sqrt{1-\epsilon}, \sqrt{1+\epsilon} - 1\}$, that is, the smaller of the two distances.

2.3 One-sided Limits

To have a limit L as x approaches c, a function must be defined on both sides of c and its values (x) must approach L as x approaches c from either side. That is, must be defined in some open interval about c, but not necessarily at c. Because of this, ordinary limits are called **two-sided**.

If f fails to have a two-sided limit at c, it may still have a one-sided limit, that is, a limit if the approach is only from one side. If the approach is from the right, the limit is a right-hand limit. From the left, it is a left-hand limit.

The function $f(x) = x/|x|$ (Figure 2.7) has limit 1 as x approaches 0 from the right, and limit -1 as x approaches 0 from the left. Since these one-sided limit values are not the same, there is no single number that $f(x)$ approaches as x approaches 0. So $f(x)$ does not have a (two-sided) limit at 0 .

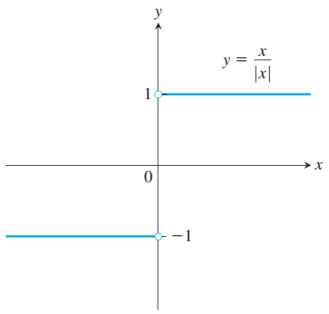


Figure 2.7

Theorem 2.4. *A function $f(x)$ has a limit as x approaches c if and only if it has left-hand and right-hand limits there and these one-sided limits are equal:*

$$\lim_{x\to c} f(x) = L \quad \Leftrightarrow \quad \lim_{x\to c^-} f(x) = L \quad \text{and} \quad \lim_{x\to c^+} f(x) = L.$$

Example 2.3.1. *For the function graphed in Figure 2.8,*

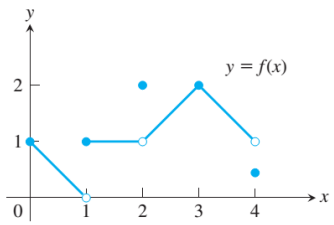


Figure 2.8

First we have $\lim_{x\to 0^+} f(x) = 1$. But $\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0} f(x)$ do not exist. The function is not defined to the left of $x = 0$. And at $x = 1$, $\lim_{x\to 1^-} f(x) = 0$ even though $f(1) = 1$, $\lim_{x\to 1^+} f(x) = 1$. That is, $\lim_{x\to 1} f(x)$ does not exist since the right- and left-hand limits are not equal. At $x = 2$, we have

$$\lim_{x\to 2^-} f(x) = 1, \quad \lim_{x\to 2^+} f(x) = 1, \quad \lim_{x\to 2} f(x) = 1$$

even though $f(2) = 2$. At $x = 3$, $\lim_{x\to 3^-} f(x) = \lim_{x\to 3^+} f(x) = \lim_{x\to 3} f(x) = f(3) = 2$. At $x = 4$, $\lim_{x\to 4^-} f(x) = 1$ even though $f(4) \neq 1$. And $\lim_{x\to 4^+} f(x)$, $\lim_{x\to 4} f(x)$ do not exist. The function is not defined to the right of $x = 4$.

At every other point c in $[0, 4]$, (x) has limit (c).

2.3.1 Limits involving $\frac{\sin\theta}{\theta}$

We can use Sandwich Theorem to prove the following theorem.

Theorem 2.5. *We have*

$$\lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$$

where θ given in radians.

Proof. The plan is to show that the right-hand and left-hand limits are both 1. Then we will know that the two-sided limit is 1 as well. To show that the right-hand limit is 1 , we begin with positive values of θ less than $\pi/2$ (Figure 2.9).

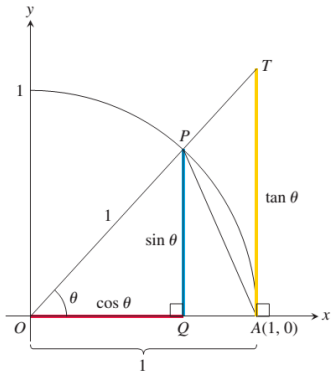


Figure 2.9

Notice that

$$\text{Area } \Delta OAP < \text{ area sector } OAP < \text{ area } \Delta OAT.$$

We can express these areas in terms of θ as follows:

$$\text{Area } \Delta OAP = \frac{1}{2} \text{ base } \times \text{ height } = \frac{1}{2}(1)(\sin\theta) = \frac{1}{2}\sin\theta.$$

$$\text{Area sector } OAP = \frac{1}{2}r^2\theta = \frac{1}{2}(1)^2\theta = \frac{\theta}{2}.$$

$$\text{Area}\Delta OAT = \frac{1}{2} \text{ base} \times \text{ height} = \frac{1}{2}(1)(\tan\theta) = \frac{1}{2}\tan\theta.$$

Thus,

$$\frac{1}{2}\sin\theta < \frac{1}{2}\theta < \frac{1}{2}\tan\theta$$

This last inequality goes the same way if we divide all three terms by the number $(1/2)\sin\theta$, which is positive, since $0 < \theta < \pi/2$:

$$1 < \frac{\theta}{\sin\theta} < \frac{1}{\cos\theta}$$

Taking reciprocals reverses the inequalities:

$$1 > \frac{\sin\theta}{\theta} > \cos\theta$$

Since $\lim_{\theta\to 0^+}\cos\theta = 1$, the Sandwich Theorem gives

$$\lim_{\theta\to 0^+}\frac{\sin\theta}{\theta} = 1$$

To consider the left-hand limit, we recall that $\sin\theta$ and θ are both odd functions. Therefore, $f(\theta) = (\sin\theta)/\theta$ is an even function, with a graph symmetric about the y-axis. This symmetry implies that the left-hand limit at 0 exists and has the same value

as the right-hand limit:

$$\lim_{\theta\to 0^-}\frac{\sin\theta}{\theta}=1=\lim_{\theta\to 0^+}\frac{\sin\theta}{\theta},$$

so $\lim_{\theta\to 0}(\sin\theta)/\theta=1$. □

Solved problems

(a) Show that $\lim_{h\to 0}\frac{\cos h-1}{h}=0$.

Solution: Using the half-angle formula $\cos h=1-2\sin^2(h/2)$, we calculate

$$\begin{aligned} lim_{h\to 0}\frac{\cos h-1}{h} &= \lim_{h\to 0}-\frac{2\sin^2(h/2)}{h}\\ &= -\lim_{\theta\to 0}\frac{\sin\theta}{\theta}\sin\theta\\ &= -(1)(0)=0. \end{aligned}$$

(b) Show that $\lim_{x\to 0}\frac{\sin 2x}{5x}=\frac{2}{5}$.

Solution: We need a $2x$ in the denominator, not a $5x$. We produce it by multiplying numerator and denominator by $\frac{2}{5}$:

$$\begin{aligned} \lim_{x\to 0}\frac{\sin 2x}{5x} &= \lim_{x\to 0}\frac{(2/5)\cdot\sin 2x}{(2/5)\cdot 5x}\\ &= \frac{2}{5}\lim_{x\to 0}\frac{\sin 2x}{2x} \qquad \text{Sine rule applies with } \theta=2x\\ &= \frac{2}{5}(1)=\frac{2}{5}. \end{aligned}$$

(c) Find $\lim_{t\to 0}\frac{\tan t\sec 2t}{3t}$.

Solution: From the definition of $\tan t$ and $\sec 2t$, we have

$$\begin{aligned} \lim_{t\to 0}\frac{\tan t\sec 2t}{3t} &= \lim_{t\to 0}\frac{1}{3}\cdot\frac{1}{t}\cdot\frac{\sin t}{\cos t}\cdot\frac{1}{\cos 2t}\\ &= \frac{1}{3}\lim_{t\to 0}\frac{\sin t}{t}\cdot\frac{1}{\cos t}\cdot\frac{1}{\cos 2t}\\ &= \frac{1}{3}(1)(1)(1)=\frac{1}{3}. \end{aligned}$$

2.4 Continuity

Definition 2.6. *Let c be a real number on the x-axis. The function f is continuous at **c** if*

$$\lim_{x\to c}f(x)=f(c)$$

The function f is right-continuous at $\boldsymbol{c}$ *(or continuous from the right) if*

$$\lim_{x \to c^+} f(x) = f(c)$$

The function f is left-continuous at c (or continuous from the left) if

$$\lim_{x \to c^-} f(x) = f(c).$$

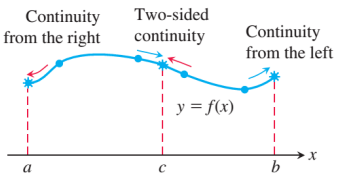


Figure 2.10

It follows that a function f is continuous at an interior point c of its domain if and only if it is both right-continuous and left-continuous at c (Figure 2.10). We say that a function is continuous over a closed interval $[a, b]$ if it is rightcontinuous at a, left-continuous at b, and continuous at all interior points of the interval.

This definition applies to the infinite closed intervals $[a, \infty)$ and $(-\infty, b]$ as well, but only one endpoint is involved. If a function is not continuous at an interior point c of its domain, we say that f is discontinuous at c, and that c is a point of discontinuity of f. Note that a function f can be continuous, right-continuous, or left-continuous only at a point c for which $f(c)$ is defined.

Example 2.4.1. *The function $f(x) = \sqrt{4 - x^2}$ is continuous over its domain $[-2, 2]$. It is right-continuous at $x = -2$, and left-continuous at $x = 2$.*

We summarize continuity at an interior point in the form of a test.

Continuity Test A function $f(x)$ is continuous at a point $x = c$ if and only if it meets the following three conditions.

(1) $f(c)$ exists (c lies in the domain of f).

(2) $\lim_{x \to c} f(x)$ exists (f has a limit as $x \to c$).

(3) $\lim_{x \to c} f(x) = f(c)$ (the limit equals the function value).

For one-sided continuity and continuity at an endpoint of an interval, the limits in parts 2 and 3 of the test should be replaced by the appropriate one-sided limits.

Example 2.4.2. *The function $y = \lfloor x \rfloor$ is graphed in Figure 2.11 It is discontinuous at every integer because the left-hand and right-hand limits are not equal as $x \to n$:*

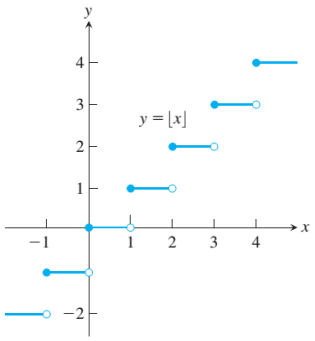


Figure 2.11

$$\lim_{x\to n^-} \lfloor x \rfloor = n-1 \quad \textit{and} \quad \lim_{x\to n^+} \lfloor x \rfloor = n$$

Since $\lfloor n \rfloor = n$, the greatest integer function is right-continuous at every integer n (but not left-continuous).

The greatest integer function is continuous at every real number other than the integers. For example,

$$\lim_{x\to 1.5} \lfloor x \rfloor = 1 = \lfloor 1.5 \rfloor$$

In general, if $n-1 < c < n, n$ an integer, then

$$\lim_{x\to c} \lfloor x \rfloor = n-1 = \lfloor c \rfloor$$

Continuous Functions: Generally, we want to describe the continuity behavior of a function throughout its entire domain, not only at a single point. We know how to do that if the domain is a closed interval. In the same way, we define a continuous function as one that is **continuous** at every point in its domain. This is a property of the function. A function always has a specified domain, so if we change the domain, we change the function, and this may change its continuity property as well. If a function is discontinuous at one or more points of its domain, we say it is a **discontinuous function**.

Algebraic combinations of continuous functions are continuous wherever they are defined.

Theorem 2.7 (Properties of Continuous Functions)**.** *Functions If the functions f and g are continuous at $x = c$, then the following algebraic combinations are continuous at $x = c$.*

(1) *Sums: $f + g$.*

(2) *Differences: $f - g$.*

(3) *Constant multiples: $k \cdot f$, for any number k.*

(4) *Products:* $f \cdot g$.

(5) *Quotients:* f/g, *provided* $g(c) \neq 0$

(6) *Powers:* f^n, n *a positive integer.*

(7) *Roots:* $\sqrt[n]{f}$, *provided it is defined on an open interval containing* c, *where* n *is a positive integer.*

Using the above theorem, we the following important examples.

Example 2.4.3.

(a) *Every polynomial* $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ *is continuous because* $\lim_{x\to c} P(x) = P(c)$.

(b) *If* $P(x)$ *and* $Q(x)$ *are polynomials, then the rational function* $P(x)/Q(x)$ *is continuous wherever it is defined* $(Q(c) \neq 0)$.

Example 2.4.4. *The function* $f(x) = |x|$ *is continuous. If* $x > 0$, *we have* $f(x) = x$, *a polynomial. If* $x < 0$, *we have* $f(x) = -x$, *another polynomial. Finally, at the origin,* $\lim_{x\to 0} |x| = 0 = |0|$

The functions $y = \sin x$ and $y = \cos x$ are continuous at $x = 0$. Both functions are, in fact, continuous everywhere. It follows that all six trigonometric functions are then continuous wherever they are defined. For example, $y = \tan x$ is continuous on

$$\cdots \cup (-\pi/2, \pi/2) \cup (\pi/2, 3\pi/2) \cup \cdots$$

Inverse Functions and Continuity

The inverse function of any function continuous on an interval is continuous over its domain. This result is suggested by the observation that the graph of f^{-1}, being the reflection of the graph of f across the line $y = x$, cannot have any breaks in it when the graph of f has no breaks. A rigorous proof that f^{-1} is continuous whenever f is continuous on an interval is given in more advanced texts. It follows that the inverse trigonometric functions are all continuous over their domains.

We defined the exponential function $y = a^x$ in Chapter 1 informally by its graph. Recall that the graph was obtained from the graph of $y = a^x$ for x a rational number by "filling in the holes" at the irrational points x, so the function $y = a^x$ was defined to be continuous over the entire real line. The inverse function $y = \log_a x$ is also continuous. In particular, the natural exponential function $y = e^x$ and the natural logarithm function $y = \ln x$ are both continuous over their domains.

Composites All composites of continuous functions are continuous. The idea is that if $f(x)$ is continuous at $x = c$ and $g(x)$ is continuous at $x = f(c)$, then $g \circ f$ is continuous at $x = c$ (Figure 2.12). In this case, the limit as $x \to c$ is $g(f(c))$.

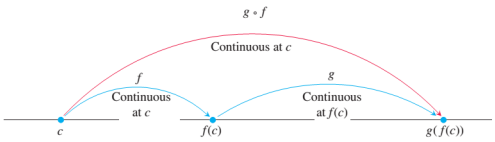


Figure 2.12

Theorem 2.8 (Composite of Continuous Functions)**.** *If f is continuous at c and g is continuous at $f(c)$, then the composite $g \circ f$ is continuous at c.*

Intuitively, Theorem 2.8 is reasonable because if x is close to c, then $f(x)$ is close to $f(c)$, and since g is continuous at $f(c)$, it follows that $g(f(x))$ is close to $g(f(c))$.

The continuity of composites holds for any finite number of functions. The only requirement is that each function be continuous where it is applied.

Example 2.4.5. *Show that the following functions are continuous on their natural domains.*

(a) $y = \sqrt{x^2 - 2x - 5}$.

Solution: *The square root function is continuous on $[0, \infty)$ because it is a root of the continuous identity function $f(x) = x$ (Part 7 , Theorem 2.7). The given function is then the composite of the polynomial $f(x) = x^2 - 2x - 5$ with the square root function $g(t) = \sqrt{t}$, and is continuous on its natural domain.*

(b) $y = \frac{x^{2/3}}{1+x^4}$.

Solution: *The numerator is the cube root of the identity function squared; the denominator is an everywhere-positive polynomial. Therefore, the quotient is continuous.*

(c) $y = \left|\frac{x-2}{x^2-2}\right|$.

Solution: *The quotient $(x - 2)/(x^2 - 2)$ is continuous for all $x \neq \pm\sqrt{2}$, and the function is the composition of this quotient with the continuous absolute value function.*

(d) $y = \left|\frac{x \sin x}{x^2+2}\right|$.

Solution: *Because the sine function is everywhere-continuous (Exercise 70), the numerator term $x \sin x$ is the product of continuous functions, and the denominator term $x^2 + 2$ is an everywhere-positive polynomial. The given function is the*

composite of a quotient of continuous functions with the continuous absolute value function.

Theorem 2.8 is actually a consequence of a more general result, which we now state and prove.

Theorem 2.9 (Limits of Continuous Functions)**.** *If g is continuous at the point b and $\lim_{x\to c} f(x) = b$, then*

$$\lim_{x\to c} g(f(x)) = g(b) = g\left(\lim_{x\to c} f(x)\right)$$

Proof. Let $\epsilon > 0$ be given. Since g is continuous at b, there exists a number $\delta_1 > 0$ such that

$$|g(y) - g(b)| < \epsilon \quad \text{whenever} \quad 0 < |y - b| < \delta_1$$

Since $\lim_{x\to c} f(x) = b$, there exists a $\delta > 0$ such that

$$|f(x) - b| < \delta_1 \quad \text{whenever} \quad 0 < |x - c| < \delta$$

If we let $y = f(x)$, we then have that

$$|y - b| < \delta_1 \quad \text{whenever} \quad 0 < |x - c| < \delta$$

which implies from the first statement that $|g(y) - g(b)| = |g(f(x)) - g(b)| < \epsilon$ whenever $0 < |x - c| < \delta$. From the definition of limit, this proves that $\lim_{x\to c} g(f(x)) = g(b)$. □

Example 2.4.6. *As an application of Theorem 2.9, we have the following calculations.*

(a)

$$\begin{aligned}\lim_{x\to\pi/2} \cos\left(2x + \sin\left(\frac{3\pi}{2} + x\right)\right) &= \cos \lim_{x\to\pi/2}\left(2x + \sin\left(\frac{3\pi}{2} + x\right)\right) \\ &= \cos(\pi + \sin 2\pi) = \cos\pi = -1.\end{aligned}$$

(b)

$$\begin{aligned}\lim_{x\to 1} \sin^{-1}\left(\frac{1-x}{1-x^2}\right) &= \sin^{-1}\left(\lim_{x\to 1}\frac{1-x}{1-x^2}\right) && \textit{Arcsine is continuous} \\ &= \sin^{-1}\left(\lim_{x\to 1}\frac{1}{1+x}\right) && \textit{Cancel the factor } (1-x) \\ &= \sin^{-1}\frac{1}{2} = \frac{\pi}{6}.\end{aligned}$$

(c) *Since the exponential is continuous, we have*

$$\begin{aligned}\lim_{x\to 0} \sqrt{x+1}e^{\tan x} &= \lim_{x\to 0}\sqrt{x+1}\cdot \exp\left(\lim_{x\to 0}\tan x\right) \\ &= 1\cdot e^0 = 1.\end{aligned}$$

2.4.1 Intermediate value property

Functions that are continuous on intervals have properties that make them particularly useful in mathematics and its applications. One of these is the Intermediate Value Property. A function is said to have the Intermediate Value Property if whenever it takes on two values, it also takes on all the values in between.

We end this section by the following theorem.

Theorem 2.10 (Intermediate Value Theorem for Continuous Functions)**.** *If f is a continuous function on a closed interval $[a, b]$, and if y_0 is any value between $f(a)$ and $f(b)$, then $y_0 = f(c)$ for some c in $[a, b]$.*

Theorem 2.10 says that continuous functions over finite closed intervals have the Intermediate Value Property. Geometrically, the Intermediate Value Theorem says that any horizontal line $y = y_0$ crossing the y-axis between the numbers $f(a)$ and $f(b)$ will cross the curve $y = f(x)$ at least once over the interval $[a, b]$.

2.5 Limits involving infinity; Asymptotes of graphs

Definition 2.11. (a) *We say that $f(x)$ has the limit L as x approaches infinity and write*

$$\lim_{x\to\infty} f(x) = L$$

if, for every number $\epsilon > 0$, there exists a corresponding number M such that for all x

$$x > M \quad \Rightarrow \quad |f(x) - L| < \epsilon$$

(b) *We say that $f(x)$ has the limit L as x approaches minus infinity and write*

$$\lim_{x\to-\infty} f(x) = L$$

if, for every number $\epsilon > 0$, there exists a corresponding number N such that for all x

$$x < N \quad \Rightarrow \quad |f(x) - L| < \epsilon$$

Example 2.5.1. *Show that*

(a) $\lim_{x\to\infty} \frac{1}{x} = 0$.

(b) $\lim_{x\to-\infty} \frac{1}{x} = 0$.

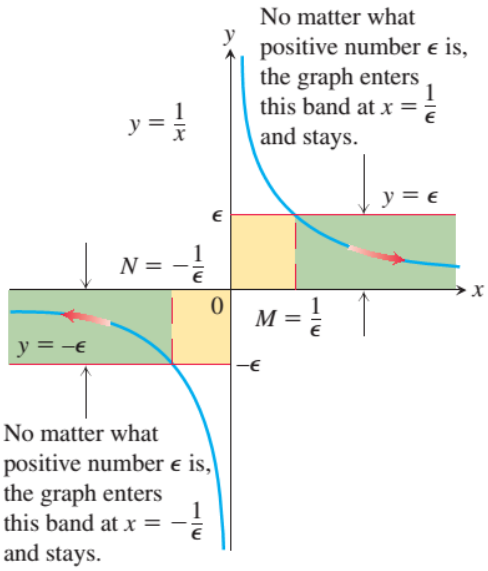


Figure 2.13

Solution: (a) Let $\epsilon > 0$ be given. We must find a number M such that for all x

$$x > M \quad \Rightarrow \quad \left|\frac{1}{x} - 0\right| = \left|\frac{1}{x}\right| < \epsilon$$

The implication will hold if $M = 1/\epsilon$ or any larger positive number (Figure 2.13). This proves $\lim_{x\to\infty}(1/x) = 0$.

Solution: (b) Let $\epsilon > 0$ be given. We must find a number N such that for all x

$$x < N \quad \Rightarrow \quad \left|\frac{1}{x} - 0\right| = \left|\frac{1}{x}\right| < \epsilon$$

The implication will hold if $N = -1/\epsilon$ or any number less than $-1/\epsilon$ (Figure 2.50). This proves $\lim_{x\to-\infty}(1/x) = 0$.

Limits at infinity have properties similar to those of finite limits.

Theorem 2.12. *All the Limit Laws in Theorem 1 are true when we replace* $\lim_{x\to c}$ *by* $\lim_{x\to\infty}$ *or* $\lim_{x\to-\infty}$ *. That is, the variable* x *may approach a finite number* c *or* $\pm\infty$.

2.5.1 Limits at Infinity of Rational Functions

To determine the limit of a rational function as $x \to \pm\infty$, we first divide the numerator and denominator by the highest power of x in the denominator. The result then depends on the degrees of the polynomials involved.

The following examples illustrate what happens when the degree of the numerator is less than or equal to the degree of the denominator.

Example 2.5.2.

(a)

$$\lim_{x\to\infty}\frac{5x^2+8x-3}{3x^2+2}=\lim_{x\to\infty}\frac{5+(8/x)-(3/x^2)}{3+(2/x^2)}$$
$$=\frac{5+0-0}{3+0}=\frac{5}{3}$$

where we divide numerator and denominator by x^2.

(b)

$$\lim_{x\to-\infty}\frac{11x+2}{2x^3-1}=\lim_{x\to-\infty}\frac{(11/x^2)+(2/x^3)}{2-(1/x^3)}$$
$$=\frac{0+0}{2-0}=0$$

where we divide numerator and denominator by x^3.

2.5.2 Horizontal Asymptotes

If the distance between the graph of a function and some fixed line approaches zero as a point on the graph moves increasingly far from the origin, we say that the graph approaches the line asymptotically and that the line is an asymptote of the graph.

Looking at $f(x)=1/x$ (see Figure 2.14), we observe that the x-axis is an asymptote of the curve on the right because

$$\lim_{x\to\infty}\frac{1}{x}=0$$

and on the left because

$$\lim_{x\to-\infty}\frac{1}{x}=0$$

We say that the x-axis is a horizontal asymptote of the graph of $f(x)=1/x$.

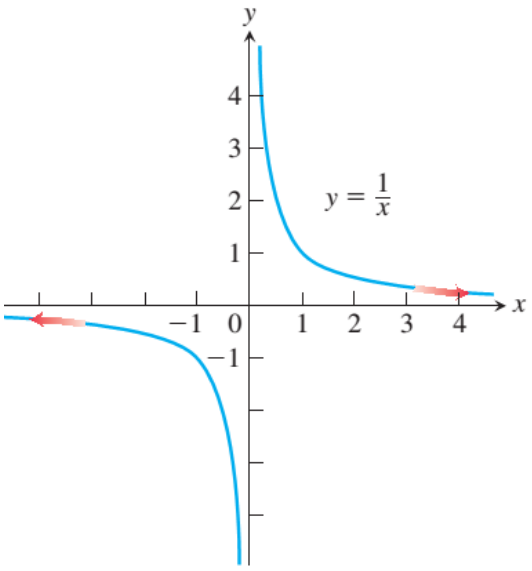


Figure 2.14

Definition 2.13. *A line $y = b$ is a horizontal asymptote of the graph of a function $y = f(x)$ if either*

$$\lim_{x\to\infty} f(x) = b \quad \text{or} \quad \lim_{x\to-\infty} f(x) = b.$$

The graph of the function

$$f(x) = \frac{5x^2 + 8x - 3}{3x^2 + 2}$$

has the line $y = 5/3$ as a horizontal asymptote on both the right and the left because

$$\lim_{x\to\infty} f(x) = \frac{5}{3} \quad \text{and} \quad \lim_{x\to-\infty} f(x) = \frac{5}{3}.$$

Example 2.5.3. *Find the horizontal asymptotes of the graph of*

$$f(x) = \frac{x^3 - 2}{|x|^3 + 1}$$

Solution: We calculate the limits as $x \to \pm\infty$.
For $x \geq 0$:

$$\lim_{x\to\infty} \frac{x^3 - 2}{|x|^3 + 1} = \lim_{x\to\infty} \frac{x^3 - 2}{x^3 + 1} = \lim_{x\to\infty} \frac{1 - (2/x^3)}{1 + (1/x^3)} = 1.$$

For $x < 0$:

$$\lim_{x\to-\infty} \frac{x^3 - 2}{|x|^3 + 1} = \lim_{x\to-\infty} \frac{x^3 - 2}{(-x)^3 + 1} = \lim_{x\to-\infty} \frac{1 - (2/x^3)}{-1 + (1/x^3)} = -1.$$

The horizontal asymptotes are $y = -1$ and $y = 1$. The graph is displayed in Figure 2.15. Notice that the graph crosses the horizontal asymptote $y = -1$ for a positive value of x.

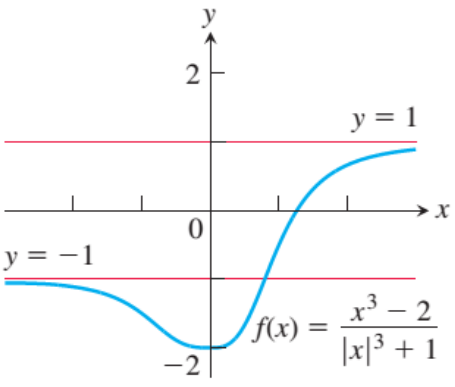


Figure 2.15

The Sandwich Theorem also holds for limits as $x \to \pm\infty$. You must be sure, though, that the function whose limit you are trying to find stays between the bounding functions at very large values of x in magnitude consistent with whether $x \to \infty$ or $x \to -\infty$.

Example 2.5.4. *Using the Sandwich Theorem, find the horizontal asymptote of the curve*

$$y = 2 + \frac{\sin x}{x}.$$

Solution: We are interested in the behavior as $x \to \pm\infty$. Since

$$0 \le \left|\frac{\sin x}{x}\right| \le \left|\frac{1}{x}\right|$$

and $\lim_{x\to\pm\infty} |1/x| = 0$, we have $\lim_{x\to\pm\infty} (\sin x)/x = 0$ by the Sandwich Theorem. Hence,

$$\lim_{x\to\pm\infty} \left(2 + \frac{\sin x}{x}\right) = 2 + 0 = 2$$

and the line $y = 2$ is a horizontal asymptote of the curve on both left and right.

2.5.3 Oblique Asymptotes

If the degree of the numerator of a rational function is 1 greater than the degree of the denominator, the graph has an oblique or slant line asymptote. We find an equation for the asymptote by dividing numerator by denominator to express f as a linear function plus a remainder that goes to zero as $x \to \pm\infty$.

Example 2.5.5. *Find the oblique asymptote of the graph of* $f(x) = \frac{x^2-3}{2x-4}$, *in Figure 2.16*

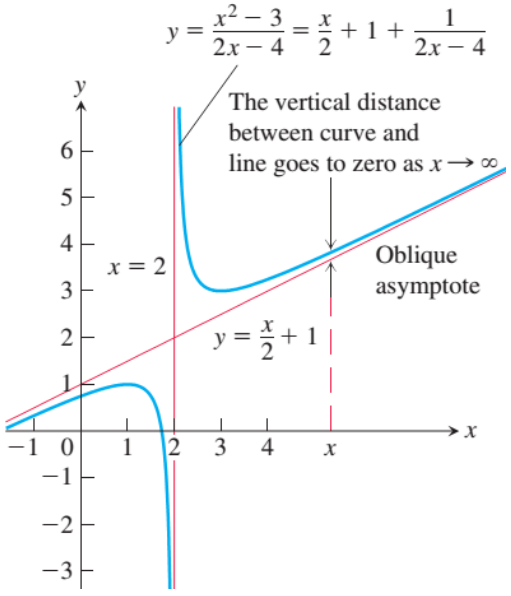


Figure 2.16

Solution: We are interested in the behavior as $x \to \pm\infty$. We divide $x^2 - 3$ by $2x - 4$ as follows:

$$\begin{array}{r} \qquad\qquad \frac{1}{2}x + 1. \\ \hline 2x - 4\big) \quad x^2 \qquad\quad - 3 \\ \qquad\quad -x^2 + 2x \\ \hline \qquad\qquad\quad 2x - 3 \\ \qquad\qquad -2x + 4 \\ \hline \qquad\qquad\qquad\quad 1 \end{array}$$

This tells us that

$$f(x) = \frac{x^2 - 3}{2x - 4} = (\underbrace{\frac{x}{2} + 1}_{\text{linear } g(x)}) + (\underbrace{\frac{1}{2x - 4}}_{\text{remainder}})$$

As $x \to \pm\infty$, the remainder, whose magnitude gives the vertical distance between the graphs of f and g, goes to zero, making the slanted line

$$g(x) = \frac{x}{2} + 1$$

an asymptote of the graph of f (Figure 2.16). The line $y = g(x)$ is an asymptote both to the right and to the left. The next subsection will confirm that the function $f(x)$ grows arbitrarily large in absolute value as $x \to 2$ (where the denominator is zero), as shown in the graph.

2.6 Infinite Limits

Let us look again at the function $(x) = \frac{1}{x}$. As $x \to 0^+$, the values of grow without bound, eventually reaching and surpassing every positive real number. That is, given any positive real number B, however large, the values of become larger still (Figure 2.17).

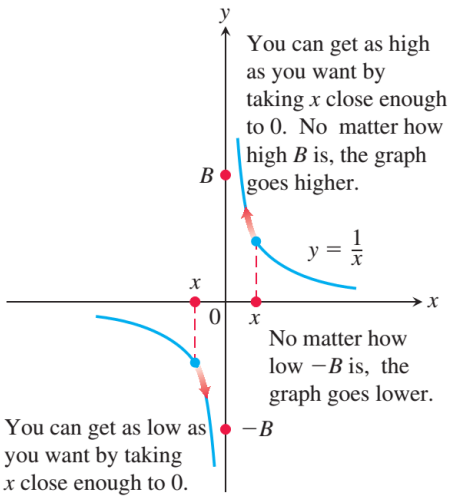


Figure 2.17

Thus, f has no limit as $x \to 0^+$.It is nevertheless convenient to describe the behavior of f by saying that $f(x)$ approaches ∞ as $x \to 0^+$. We write

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \frac{1}{x} = \infty$$

In writing this equation, we are not saying that the limit exists. Nor are we saying that there is a real number ∞, for there is no such number. Rather, we are saying that $\lim_{x\to 0^+}(1/x)$ does not exist because $1/x$ becomes arbitrarily large and positive as $x \to 0^+$.

As $x \to 0^-$, the values of $f(x) = 1/x$ become arbitrarily large and negative. Given any negative real number $-B$, the values of f eventually lie below $-B$. (See Figure 2.17.) We write

$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} \frac{1}{x} = -\infty$$

Again, we are not saying that the limit exists and equals the number $-\infty$. There *is* no real number $-\infty$. We are describing the behavior of a function whose limit as $x \to 0^-$does not exist because its values become arbitrarily large and negative.

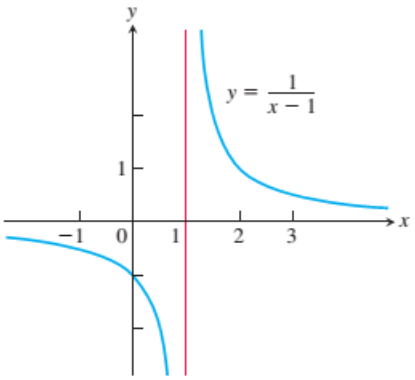


Figure 2.18

Example 2.6.1. *Find* $\lim_{x\to 1^+} \frac{1}{x-1}$ *and* $\lim_{x\to 1^-} \frac{1}{x-1}$.

Geometric Solution: The graph of $y = 1/(x-1)$ is the graph of $y = 1/x$ shifted 1 unit to the right (Figure 2.18). Therefore, $y = 1/(x-1)$ behaves near 1 exactly the way $y = 1/x$ behaves near 0:

$$\lim_{x\to 1^+} \frac{1}{x-1} = \infty \quad \text{and} \quad \lim_{x\to 1^-} \frac{1}{x-1} = -\infty.$$

Analytic Solution: Think about the number $x-1$ and its reciprocal. As $x \to 1^+$, we have $(x-1) \to 0^+$and $1/(x-1) \to \infty$. As $x \to 1^-$, we have $(x-1) \to 0^-$and $1/(x-1) \to -\infty$.

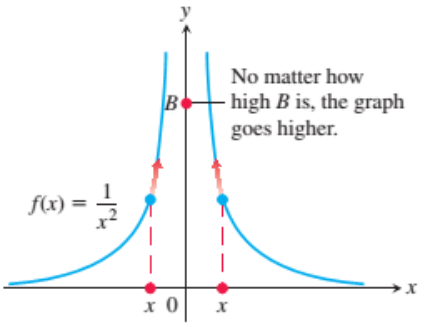


Figure 2.19

Example 2.6.2. *Discuss the behavior of*

$$f(x) = \frac{1}{x^2} \quad \text{as} \quad x \to 0.$$

Solution: As x approaches zero from either side, the values of $1/x^2$ are positive and become arbitrarily large (Figure 2.19). This means that

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{1}{x^2} = \infty$$

The function $y = 1/x$ shows no consistent behavior as $x \to 0$. We have $1/x \to \infty$ if $x \to 0^+$, but $1/x \to -\infty$ if $x \to 0^-$. All we can say about $\lim_{x\to 0}(1/x)$ is that it does not exist. The function $y = 1/x^2$ is different. Its values approach infinity as x approaches zero from either side, so we can say that $\lim_{x\to 0}(1/x^2) = \infty$.

More Examples: These examples illustrate that rational functions can behave in various ways near zeros of the denominator.

(a) $\lim_{x\to 2} \frac{(x-2)^2}{x^2-4} = \lim_{x\to 2} \frac{(x-2)^2}{(x-2)(x+2)} = \lim_{x\to 2} \frac{x-2}{x+2} = 0$

(b) $\lim_{x\to 2} \frac{(x-2)^2}{x^2-4} = \lim_{x\to 2} \frac{(x-2)^2}{(x-2)(x+2)} = \lim_{x\to 2} \frac{x-2}{x+2} = 0$

(c) $\lim_{x\to 2^+} \frac{x-3}{x^2-4} = \lim_{x\to 2^+} \frac{x-3}{(x-2)(x+2)} = -\infty$. The values are negative for $x > 2$, x near 2.

(d) $\lim_{x\to 2^-} \frac{x-3}{x^2-4} = \lim_{x\to 2^-} \frac{x-3}{(x-2)(x+2)} = \infty$. The values are positive for $x < 2$, x near 2.

(e) $\lim_{x\to 2} \frac{x-3}{x^2-4} = \lim_{x\to 2} \frac{x-3}{(x-2)(x+2)}$ does not exist. See parts (c) and (d).

(f) $\lim_{x\to 2} \frac{2-x}{(x-2)^3} = \lim_{x\to 2} \frac{-(x-2)}{(x-2)^3} = \lim_{x\to 2} \frac{-1}{(x-2)^2} = -\infty$.

In parts (a) and (b) the effect of the zero in the denominator at $x = 2$ is canceled because the numerator is zero there also. Thus a finite limit exists. This is not true in part (f), where cancellation still leaves a zero factor in the denominator

Example 2.6.3. *Find* $\lim_{x\to-\infty} \frac{2x^5-6x^4+1}{3x^2+x-7}$.

Solution: We are asked to find the limit of a rational function as $x \to -\infty$, so we divide the numerator and denominator by x^2, the highest power of x in the denominator:

$$\begin{aligned}\lim_{x\to-\infty} \frac{2x^5-6x^4+1}{3x^2+x-7} &= \lim_{x\to-\infty} \frac{2x^3-6x^2+x^{-2}}{3+x^{-1}-7x^{-2}} \\ &= \lim_{x\to-\infty} \frac{2x^2(x-3)+x^{-2}}{3+x^{-1}-7x^{-2}} \\ &= -\infty, \quad x^{-n} \to 0, x-3 \to -\infty\end{aligned}$$

because the numerator tends to $-\infty$ while the denominator approaches 3 as $x \to -\infty$.

2.6.1 Vertical Asymptotes

Notice that the distance between a point on the graph of $f(x) = 1/x$ and the y-axis approaches zero as the point moves vertically along the graph and away from the origin (Figure 2.20). The function $f(x) = 1/x$ is unbounded as x approaches 0 because

$$\lim_{x\to 0^+} \frac{1}{x} = \infty \quad \text{and} \quad \lim_{x\to 0^-} \frac{1}{x} = -\infty$$

We say that the line $x = 0$ (the y-axis) is a vertical asymptote of the graph of $f(x) = 1/x$ Observe that the denominator is zero at $x = 0$ and the function is undefined there.

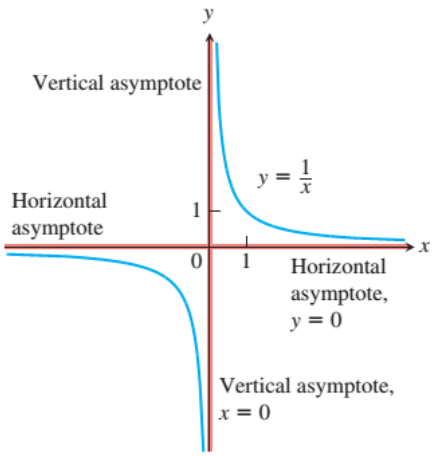


Figure 2.20

Definition 2.14. *A line $x = a$ is a vertical asymptote of the graph of a function $y = f(x)$ if either*

$$\lim_{x \to a^+} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \to a^-} f(x) = \pm\infty.$$

Example 2.6.4. *Find the horizontal and vertical asymptotes of the curve*

$$y = \frac{x+3}{x+2}.$$

Solution: We are interested in the behavior as $x \to \pm\infty$ and the behavior as $x \to -2$, where the denominator is zero.

The asymptotes are quickly revealed if we recast the rational function as a polynomial with a remainder, by dividing $(x+2)$ into $(x+3)$:

$$\begin{array}{r} 1. \\ \hline x+2 \big) \quad x+3 \\ \underline{-\,x-2} \\ 1 \end{array}$$

As $x \to \pm\infty$, the curve approaches the horizontal asymptote $y = 1$; as $x \to -2$, the curve approaches the vertical asymptote $x = -2$. We see that the curve in question is the graph of $f(x) = 1/x$ shifted 1 unit up and 2 units left (Figure 2.21). The asymptotes, instead of being the coordinate axes, are now the lines $y = 1$ and $x = -2$.

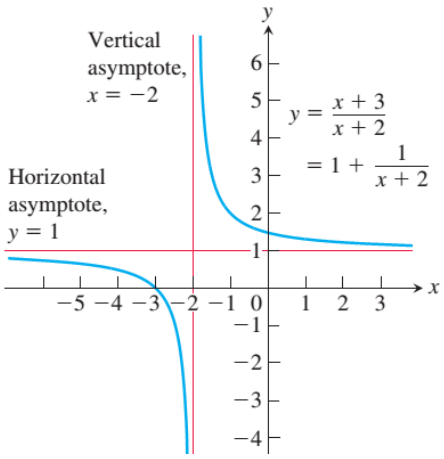


Figure 2.21

Example 2.6.5. *Find the horizontal and vertical asymptotes of the graph of*

$$f(x) = -\frac{8}{x^2 - 4}.$$

Solution: We are interested in the behavior as $x \to \pm\infty$ and as $x \to \pm 2$, where the denominator is zero. Notice that f is an even function of x, so its graph is symmetric with respect to the y-axis.

(a) The behavior as $x \to \pm\infty$. Since $\lim_{x\to\infty} f(x) = 0$, the line $y = 0$ is a horizontal asymptote of the graph to the right. By symmetry it is an asymptote to the left as well (Figure 2.22). Notice that the curve approaches the x-axis from only the negative side (or from below). Also, $f(0) = 2$.

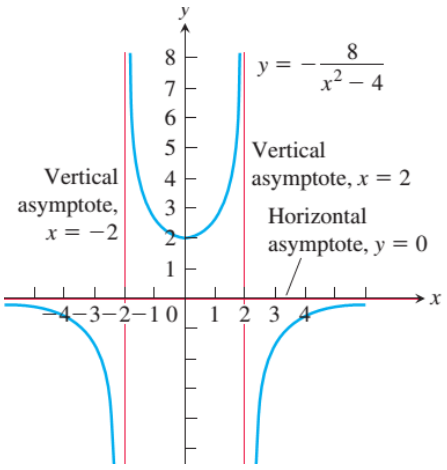


Figure 2.22

(b) The behavior as $x \to \pm 2$. Since

$$\lim_{x\to 2^+} f(x) = -\infty \quad \text{and} \quad \lim_{x\to 2^-} f(x) = \infty$$

the line $x = 2$ is a vertical asymptote both from the right and from the left. By symmetry, the line $x = -2$ is also a vertical asymptote. There are no other asymptotes because f has a finite limit at all other points.

Example 2.6.6. *The graph of the natural logarithm function has the y-axis (the line $x = 0$) as a vertical asymptote.*

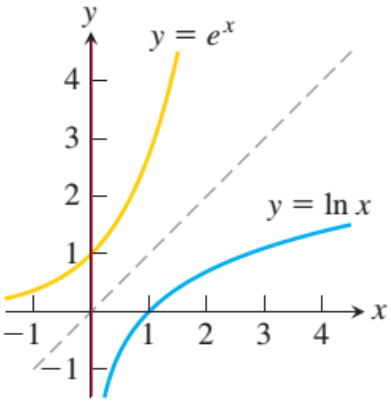


Figure 2.23

We see this from the graph sketched in Figure 2.23 (which is the reflection of the graph of the natural exponential function across the line $y = x$) and the fact that the x-axis is a horizontal asymptote of $y = e^x$. Thus,

$$\lim_{x \to 0^+} \ln x = -\infty$$

The same result is true for $y = \log_a x$ whenever $a > 1$.

Example 2.6.7. *The curves*

$$y = \sec x = \frac{1}{\cos x} \quad \text{and} \quad y = \tan x = \frac{\sin x}{\cos x}$$

both have vertical asymptotes at odd-integer multiples of $\pi/2$, where $\cos x = 0$ (Figure 2.24).

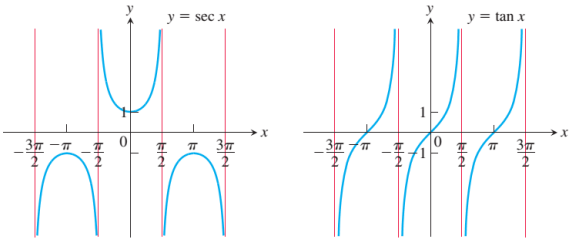


Figure 2.24

2.7 Exercises

Limits

1. Find the following limits:

   (a) $\lim_{x\to\pi} \sin(x - \sin x)$.

   (b) $\lim_{t\to 0} \sin\left(\frac{\pi}{2}\cos(\tan t)\right)$.

   (c) $\lim_{y\to 1} \sec\left(y\sec^2 y - \tan^2 y - 1\right)$.

   (d) $\lim_{x\to 0} \tan\left(\frac{\pi}{4}\cos\left(\sin x^{1/3}\right)\right)$.

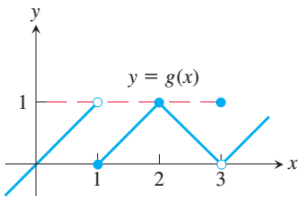
   (e) $\lim_{t\to 0} \cos\left(\frac{\pi}{\sqrt{19-3\sec 2t}}\right)$.

   (f) $\lim_{x\to\pi/6} \sqrt{\csc^2 x + 5\sqrt{3}\tan x}$.

   (g) $\lim_{x\to 0^+} \sin\left(\frac{\pi}{2}e^{\sqrt{x}}\right)$.

   (h) $\lim_{x\to 1} \cos^{-1}(\ln\sqrt{x})$.

2. For the function $g(x)$ graphed here, find the following limits or explain why they do not exist.



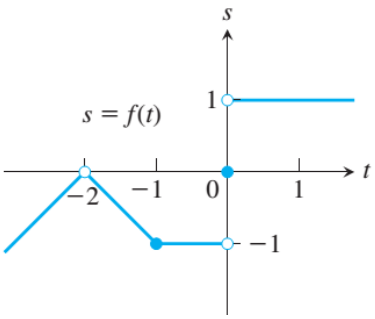
   (a) $\lim_{x\to 1} g(x)$

   (b) $\lim_{x\to 2} g(x)$

   (c) $\lim_{x\to 3} g(x)$

   (d) $\lim_{x\to 2.5} g(x)$.

3. For the function $f(t)$ graphed here, find the following limits or explain why they do not exist.



   (a) $\lim_{t\to -2} f(t)$

   (b) $\lim_{t\to -1} f(t)$

   (c) $\lim_{t\to 0} f(t)$

   (d) $\lim_{t\to -0.5} f(t)$.

4. Calculate the following limits:

(a) $\lim_{x\to 2} \frac{2x+5}{11-x^3}$

(b) $\lim_{s\to 2/3}(8-3s)(2s-1)$.

(c) $\lim_{x\to -1/2} 4x(3x+4)^2$

(d) $\lim_{y\to 2} \frac{y+2}{y^2+5y+6}$.

(e) $\lim_{y\to -3}(5-y)^{4/3}$

(f) $\lim_{z\to 4} \sqrt{z^2-10}$.

(g) $\lim_{h\to 0} \frac{3}{\sqrt{3h+1}+1}$

(h) $\lim_{h\to 0} \frac{\sqrt{5h+4}-2}{h}$.

(i) $\lim_{x\to 5} \frac{x-5}{x^2-25}$

(j) $\lim_{x\to -3} \frac{x+3}{x^2+4x+3}$.

(k) $\lim_{x\to -5} \frac{x^2+3x-10}{x+5}$

(l) $\lim_{x\to 2} \frac{x^2-7x+10}{x-2}$.

(m) $\lim_{t\to 1} \frac{t^2+t-2}{t^2-1}$

(n) $\lim_{t\to -1} \frac{t^2+3t+2}{t^2-t-2}$.

(o) $\lim_{x\to -2} \frac{-2x-4}{x^3+2x^2}$

(p) $\lim_{y\to 0} \frac{5y^3+8y^2}{3y^4-16y^2}$.

(q) $\lim_{u\to 1} \frac{u^4-1}{u^3-1}$

(r) $\lim_{v\to 2} \frac{v^3-8}{v^4-16}$.

(s) $\lim_{x\to 9} \frac{\sqrt{x}-3}{x-9}$

(t) $\lim_{x\to 4} \frac{4x-x^2}{2-\sqrt{x}}$.

(u) $\lim_{x\to 1} \frac{x-1}{\sqrt{x+3}-2}$

(v) $\lim_{x\to -1} \frac{\sqrt{x^2+8}-3}{x+1}$.

(w) $\lim_{x\to 2} \frac{\sqrt{x^2+12}-4}{x-2}$

(x) $\lim_{x\to -2} \frac{x+2}{\sqrt{x^2+5}-3}$.

(y) $\lim_{x\to -3} \frac{2-\sqrt{x^2-5}}{x+3}$

(z) $\lim_{x\to 4} \frac{4-x}{5-\sqrt{x^2+9}}$.

5. Calculate the following limits:

(a) $\lim_{x\to -0.5^-} \sqrt{\frac{x+2}{x+1}}$

(b) $\lim_{x\to 1^+} \sqrt{\frac{x-1}{x+2}}$.

(c) $\lim_{x\to -2^+} \left(\frac{x}{x+1}\right)\left(\frac{2x+5}{x^2+x}\right)$

(d) $\lim_{x\to 1^-} \left(\frac{1}{x+1}\right)\left(\frac{x+6}{x}\right)\left(\frac{3-x}{7}\right)$.

(e) $\lim_{h\to 0^+} \frac{\sqrt{h^2+4h+5}-\sqrt{5}}{h}$

(f) $\lim_{x\to 0} \frac{x^2-x+\sin x}{2x}$.

(g) $\lim_{x\to 0} \frac{x+x\cos x}{\sin x\cos x}$

(h) $\lim_{\theta\to 0} \frac{1-\cos\theta}{\sin 2\theta}$.

(i) $\lim_{x\to 0} \frac{x-x\cos x}{\sin^2 3x}$

(j) $\lim_{t\to 0} \frac{\sin(1-\cos t)}{1-\cos t}$.

(k) $\lim_{h\to 0} \frac{\sin(\sin h)}{\sin h}$

(l) $\lim_{\theta\to 0} \frac{\sin\theta}{\sin 2\theta}$.

(m) $\lim_{x\to 0} \frac{\sin 5x}{\sin 4x}$

(n) $\lim_{\theta\to 0} \theta\cos\theta$.

(o) $\lim_{\theta\to 0} \sin\theta\cot 2\theta$

(p) $\lim_{x\to 0} \frac{\tan 3x}{\sin 8x}$.

6. Later we will prove that for any number $x, \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = e^x$. Use this fact to find the following limits:

(a) $\lim_{x\to 0}(1-x)^{\frac{1}{x}}$

(b) $\lim_{x\to\infty}\left(\frac{x^2+1}{x^2-1}\right)^{x^2}$.

(c) $\lim_{x\to 1} x^{\frac{1}{1-x}}$

(d) $\lim_{x\to 0}(1-3x)^{\frac{1}{x}}$.

(e) $\lim_{x\to\infty}\left(1+\frac{2}{x}\right)^{-x}$.

(f) $\lim_{x\to\infty}(1+\tan x)^{\cot x}$

(g) $\lim_{x\to\infty}\left(\frac{x+1}{x-1}\right)^x$.

Continuity

1. At what points are the following functions continuous?

   (a) $y = \frac{1}{x-2} - 3x$

   (b) $y = \frac{1}{(x+2)^2} + 4.$

   (c) $y = \frac{x+1}{x^2-4x+3}$

   (d) $y = \frac{1}{|x|+1} - \frac{x^2}{2}.$

   (e) $y = |x-1| + \sin x$

   (f) $y = \frac{x+3}{x^2-3x-10}.$

   (g) $y = \frac{\cos x}{x}$

   (h) $y = \frac{x+2}{\cos x}.$

   (i) $y = \csc 2x$

   (j) $y = \tan \frac{\pi x}{2}.$

   (k) $y = \frac{x \tan x}{x^2+1}$

   (l) $y = \frac{\sqrt{x^4+1}}{1+\sin^2 x}.$

   (m) $y = \sqrt{2x+3}$

   (n) $y = \sqrt[4]{3x-1}.$

   (o) $y = (2x-1)^{1/3}$

   (p) $y = (2-x)^{1/5}.$

   (q) $g(x) = \begin{cases} \frac{x^2-x-6}{x-3}, & x \neq 3 \\ 5, & x = 3 \end{cases}$

   (r) $f(x) = \begin{cases} \frac{x^3-8}{x^2-4}, & x \neq 2, x \neq -2 \\ 3, & x = 2 \\ 4, & x = -2 \end{cases}$.

2. For what value of a is

$$f(x) = \begin{cases} x^2 - 1, & x < 3 \\ 2ax, & x \geq 3 \end{cases}$$

   continuous at every x?

3. For what value of b is

$$g(x) = \begin{cases} x, & x < -2 \\ bx^2, & x \geq -2 \end{cases}$$

   continuous at every x?

4. For what values of a is

$$f(x) = \begin{cases} a^2x - 2a, & x \geq 2 \\ 12, & x < 2 \end{cases}$$

   continuous at every x?

Limits Involving Infinity; Asymptotes of Graphs

1. Find the following limits:

(a) $\lim_{x\to-\infty}\left(\frac{1-x^3}{x^2+7x}\right)^5$

(b) $\lim_{x\to\infty}\sqrt{\frac{x^2-5x}{x^3+x-2}}$.

(c) $\lim_{x\to\infty}\frac{2\sqrt{x}+x^{-1}}{3x-7}$

(d) $\lim_{x\to\infty}\frac{2+\sqrt{x}}{2-\sqrt{x}}$.

(e) $\lim_{x\to-\infty}\frac{\sqrt[3]{x}-\sqrt[5]{x}}{\sqrt[3]{x}+\sqrt[5]{x}}$

(f) $\lim_{x\to\infty}\frac{x^{-1}+x^{-4}}{x^{-2}-x^{-3}}$.

(g) $\lim_{x\to\infty}\frac{2x^{5/3}-x^{1/3}+7}{x^{8/5}+3x+\sqrt{x}}$

(h) $\lim_{x\to-\infty}\frac{\sqrt[3]{x}-5x+3}{2x+x^{2/3}-4}$.

(i) $\lim_{x\to\infty}\frac{\sqrt{x^2+1}}{x+1}$

(j) $\lim_{x\to-\infty}\frac{\sqrt{x^2+1}}{x+1}$.

(k) $\lim_{x\to\infty}\frac{x-3}{\sqrt{4x^2+25}}$

(l) $\lim_{x\to-\infty}\frac{4-3x^3}{\sqrt{x^6+9}}$.

(m) $\lim_{x\to\infty}\frac{2x+3}{5x+7}$

(n) $\lim_{x\to-\infty}\frac{7x^3}{x^3-3x^2+6x}$.

(o) $\lim_{x\to\infty}\frac{3x^7+5x^2-1}{6x^3-7x+3}$

(p) $\lim_{x\to-\infty}\frac{3x+7}{x^2-2}$.

2. Find the limits:

(a) $\lim_{x\to0^+}\frac{1}{3x}$

(b) $\lim_{x\to0^-}\frac{5}{2x}$.

(c) $\lim_{x\to2^-}\frac{3}{x-2}$

(d) $\lim_{x\to3^+}\frac{1}{x-3}$.

(e) $\lim_{x\to-8^+}\frac{2x}{x+8}$

(f) $\lim_{x\to-5^-}\frac{3x}{2x+10}$.

(g) $\lim_{x\to7}\frac{4}{(x-7)^2}$

(h) $\lim_{x\to0}\frac{-1}{x^2(x+1)}$.

(i) $\lim_{x\to(\pi/2)^-}\tan x$

(j) $\lim_{x\to(-\pi/2)^+}\sec x$.

(k) $\lim_{\theta\to0^-}(1+\csc\theta)$

(l) $\lim_{\theta\to0}(2-\cot\theta)$.

3. Use limits to determine the equations for all vertical asymptotes.

(a) $y=\frac{x^2+4}{x-3}$.

(b) $f(x)=\frac{x^2-x-2}{x^2-2x+1}$.

(c) $y=\frac{x^2+x-6}{x^2+2x-8}$.

4. Use limits to determine the equations for all horizontal asymptotes.

(a) $y=\frac{1-x^2}{x^2+1}$

(b) $f(x)=\frac{\sqrt{x}+4}{\sqrt{x}+4}$

(c) $g(x)=\frac{\sqrt{x^2+4}}{x}$

(d) $y=\sqrt{\frac{x^2+9}{9x^2+1}}$

Chapter 3

Differentiation

Differential calculus is the study of the derivative, and differentiation is the process of computing derivatives. What is a derivative? There are three equally important answers:A derivative is a rate of change, it is the slope of a tangent line, and (more formally), it is the limit of a difference quotient, as we will explain shortly. In this chapter, we explore all three facets of the derivative and develop the basic rules of differentiation. When you master these techniques, you will possess one of the most useful and flexible tools that mathematics has to offer.

3.1 Definition of the Derivative

We begin with two questions: What is the precise definition of a tangent line? And how can we compute its slope?

The secant line through distinct points $P = (a, f(a))$ and $Q = (x, f(x))$ on the graph of a function f has slope [Figure 3.1]

$$\frac{\Delta f}{\Delta x} = \frac{f(x) - f(a)}{x - a}$$

where

$$\Delta f = f(x) - f(a) \quad \text{and} \quad \Delta x = x - a$$

The expression $\dfrac{f(x) - f(a)}{x - a}$ is called the difference quotient. Based on this intuition, we define the derivative $f'(a)$ (which is read " f prime of a ") at a point $x = a$ as the limit

$$\underbrace{f'(a)}_{\substack{\text{Slope of the} \\ \text{tangent line}}} = \underbrace{\lim_{x \to a} \frac{f(x) - f(a)}{x - a}}_{\text{Limit of slopes of secant lines}}$$

There is another way of writing the difference quotient using a new variable h :

$$h = x - a$$

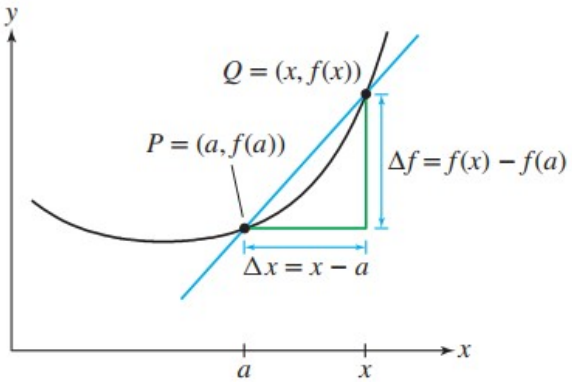


FIGURE 3.1

We have $x = a + h$ and, for $x \neq a$ (Figure 3.2),

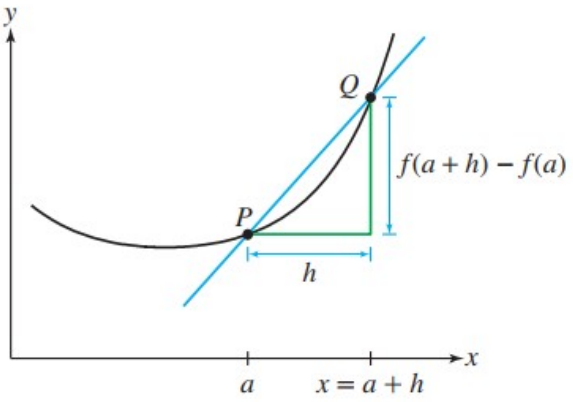


FIGURE 3.2

$$\frac{f(x) - f(a)}{x - a} = \frac{f(a + h) - f(a)}{h}.$$

The variable h approaches 0 as $x \to a$, so we can rewrite the derivative as

$$f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}.$$

Each way of writing the derivative is useful. The version using h is often more convenient in computations.

Definition 3.1. *The Derivative The derivative of f at a point a is the limit of the difference quotients (if it exists):*

$$f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}.$$

When the limit exists, we say that f is differentiable at a. An equivalent definition of the derivative at a point a is

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

We now investigate the derivative as a function derived from f by considering the limit at each point x in the domain of f

Definition 3.2. *The derivative of the function $f(x)$ with respect to the variable x is the function f' whose value at x is*

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

provided the limit exists.

Example 3.1.1. Differentiate

$$f(x) = \frac{x}{x-1}.$$

Solution: We use the definition of derivative, which requires us to calculate $f(x+h)$ and then subtract $f(x)$ to obtain the numerator in the difference quotient. We have

$$f(x) = \frac{x}{x-1} \quad \text{and} \quad f(x+h) = \frac{(x+h)}{(x+h)-1}.$$

Then

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \to 0} \frac{\frac{x+h}{x+h-1} - \frac{x}{x-1}}{h} \\ &= \lim_{h \to 0} \frac{1}{h} \frac{(x+h)(x-1) - x(x+h-1)}{(x+h-1)(x-1)} \\ &= \lim_{h \to 0} \frac{1}{h} \frac{-h}{(x+h-1)(x-1)} \\ &= \lim_{h \to 0} \frac{-1}{(x+h-1)(x-1)} = \frac{-1}{(x-1)^2}. \end{aligned}$$

Example 3.1.2. Find the derivative of$f(x) = \sqrt{x}$ for $x > 0$.
Solution:

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \end{aligned}$$

$$= \lim_{h\to 0} \frac{\sqrt{x+h}-\sqrt{x}}{h}\frac{\sqrt{x+h}+\sqrt{x}}{\sqrt{x+h}+\sqrt{x}}$$
$$= \lim_{h\to 0} \frac{x+h-x}{h(\sqrt{x+h}+\sqrt{x})}$$
$$= \lim_{h\to 0} \frac{1}{(\sqrt{x+h}+\sqrt{x})} = \frac{1}{2\sqrt{x}}.$$

Example 3.1.3. Compute $f'(3)$, where $f(x) = x^2 - 8x$.
Solution: Since

$$\begin{aligned} f'(3) &= \lim_{h\to 0} \frac{f(3+h)-f(3)}{h} \\ &= \lim_{h\to 0} \frac{((3+h)^2 - 8(3+h)) - (3^2 - 8(3))}{h} \\ &= \lim_{h\to 0} \frac{((9+6h+h^2) - (24+8h)) - (9-24)}{h} \\ &= \lim_{h\to 0} \frac{h^2-2h}{h} \\ &= \lim_{h\to 0} (h-2) = -2. \end{aligned}$$

3.2 Differentiation Rules

This section introduces several rules that allow us to differentiate constant functions, power functions, polynomials, rational functions, and certain combinations of them, simply and directly, without having to take limits each time.

3.2.1 Powers, Multiples, Sums, and Differences

A simple rule of differentiation is that the derivative of every constant function is zero. If f has the constant value $f(x) = c$, then

$$\frac{df}{dx} = \frac{d}{dx}(c) = 0.$$

Proof. We apply the definition of the derivative to $f(x) = c$, the function whose outputs have the constant value c . At every value of x, we find that

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\ &= \lim_{h\to 0} \frac{c-c}{h} = \lim_{h\to 0} 0 = 0. \end{aligned}$$

□

If n is any real number, then

$$\frac{d}{dx}x^n = nx^{n-1}$$

for all x where the powers x^n and x^{n-1} are defined.

Example 3.2.1. Differentiate the following powers of x.

(a) x^3 (b) $x^{2/3}$ (c) $x^{\sqrt{2}}$ (d) $\dfrac{1}{x^4}$ (e) $x^{-4/3}$ (f) $\sqrt{x^{2+\pi}}$.

Solution:

(a) $\dfrac{d}{dx}\left(x^3\right) = 3x^{3-1} = 3x^2.$

(b) $\dfrac{d}{dx}\left(x^{2/3}\right) = \dfrac{2}{3}x^{(2/3)-1} = \dfrac{2}{3}x^{-1/3}.$

(c) $\dfrac{d}{dx}\left(x^{\sqrt{2}}\right) = \sqrt{2}x^{\sqrt{2}-1}.$

(d) $\dfrac{d}{dx}\left(\frac{1}{x^4}\right) = \dfrac{d}{dx}\left(x^{-4}\right) = -4x^{-4-1} = -4x^{-5} = -\dfrac{4}{x^5}.$

(e) $\dfrac{d}{dx}\left(x^{-4/3}\right) = -\dfrac{4}{3}x^{-(4/3)-1} = -\dfrac{4}{3}x^{-7/3}.$

(f) $\dfrac{d}{dx}\left(\sqrt{x^{2+\pi}}\right) = \dfrac{d}{dx}\left(x^{1+(\pi/2)}\right) = \left(1+\dfrac{\pi}{2}\right)x^{1+(\pi/2)-1} = \dfrac{1}{2}(2+\pi)\sqrt{x^{\pi}}.$

The next rule says that when a differentiable function is multiplied by a constant, its derivative is multiplied by the same constant.

If u and v are differentiable functions of x, then their sum $u+v$ is differentiable at every point where u and v are both differentiable. At such points,

$$\frac{d}{dx}(u+v) = \frac{du}{dx} + \frac{dv}{dx}.$$

Proof. We apply the definition of the derivative to $f(x) = u(x) + v(x)$:

$$\begin{aligned}\frac{d}{dx}[u(x)+v(x)] &= \lim_{h\to 0}\frac{[u(x+h)+v(x+h)]-[u(x)+v(x)]}{h}\\ &= \lim_{h\to 0}\left[\frac{u(x+h)-u(x)}{h}+\frac{v(x+h)-v(x)}{h}\right]\\ &= \lim_{h\to 0}\frac{u(x+h)-u(x)}{h}+\lim_{h\to 0}\frac{v(x+h)-v(x)}{h} = \frac{du}{dx}+\frac{dv}{dx}.\end{aligned}$$

□

Example 3.2.2. Find the derivative of the polynomial

$$y = x^3 + \frac{4}{3}x^2 - 5x + 1.$$

Solution:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}x^3 + \frac{d}{dx}\left(\frac{4}{3}x^2\right) - \frac{d}{dx}(5x) + \frac{d}{dx}(1) \\ &= 3x^2 + \frac{4}{3}\cdot 2x - 5 + 0 \\ &= 3x^2 + \frac{8}{3}x - 5.\end{aligned}$$

3.2.2 Products and Quotients

While the derivative of the sum of two functions is the sum of their derivatives, the derivative of the product of two functions is not the product of their derivatives. For instance,

$$\frac{d}{dx}(x\cdot x) = \frac{d}{dx}\left(x^2\right) = 2x, \qquad \text{while} \qquad \frac{d}{dx}(x)\cdot\frac{d}{dx}(x) = 1\cdot 1 = 1$$

The derivative of a product of two functions is the sum of two products, as we now explain.

If u and v are differentiable at x, then so is their product uv, and

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

Proof of the Derivative Product Rule.

$$\frac{d}{dx}(uv) = \lim_{h\to 0}\frac{u(x+h)v(x+h) - u(x)v(x)}{h}$$

To change this fraction into an equivalent one that contains difference quotients for the derivatives of u and v, we subtract and add $u(x+h)v(x)$ in the numerator:

$$\begin{aligned}\frac{d}{dx}(uv) &= \lim_{h\to 0}\frac{u(x+h)v(x+h) - u(x+h)v(x) + u(x+h)v(x) - u(x)v(x)}{h} \\ &= \lim_{h\to 0}\left[u(x+h)\frac{v(x+h)-v(x)}{h} + v(x)\frac{u(x+h)-u(x)}{h}\right] \\ &= \lim_{h\to 0}u(x+h)\cdot\lim_{h\to 0}\frac{v(x+h)-v(x)}{h} + v(x)\cdot\lim_{h\to 0}\frac{u(x+h)-u(x)}{h}\end{aligned}$$

As h approaches zero, $u(x+h)$ approaches $u(x)$ because u, being differentiable at x, is continuous at x. The two fractions approach the values of dv/dx at x and du/dx at x.

Therefore,

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}.$$

□

Example 3.2.3. Find the derivative of

$$y = \left(x^2+1\right)\left(x^3+3\right).$$

Solution: From the Product Rule with $u = x^2+1$ and $v = x^3+3$, we find

$$\begin{aligned}\frac{d}{dx}\left[\left(x^2+1\right)\left(x^3+3\right)\right] &= \left(x^2+1\right)\left(3x^2\right) + \left(x^3+3\right)(2x)\\ &= 3x^4 + 3x^2 + 2x^4 + 6x\\ &= 5x^4 + 3x^2 + 6x.\end{aligned}$$

The derivative of the quotient of two functions is given by the Quotient Rule.

If u and v are differentiable at x and if $v(x) \neq 0$, then the quotient u/v is differentiable at x, and

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

In function notation,

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{g^2(x)}$$

Proof of the Derivative Quotient Rule.

$$\begin{aligned}\frac{d}{dx}\left(\frac{u}{v}\right) &= \lim_{h\to 0}\frac{\frac{u(x+h)}{v(x+h)} - \frac{u(x)}{v(x)}}{h}\\ &= \lim_{h\to 0}\frac{v(x)u(x+h) - u(x)v(x+h)}{hv(x+h)v(x)}.\end{aligned}$$

To change the last fraction into an equivalent one that contains the difference quotients for the derivatives of u and v, we subtract and add $v(x)u(x)$ in the numerator. We then get

$$\begin{aligned}\frac{d}{dx}\left(\frac{u}{v}\right) &= \lim_{h\to 0}\frac{v(x)u(x+h) - v(x)u(x) + v(x)u(x) - u(x)v(x+h)}{hv(x+h)v(x)}\\ &= \lim_{h\to 0}\frac{v(x)\frac{u(x+h)-u(x)}{h} - u(x)\frac{v(x+h)-v(x)}{h}}{v(x+h)v(x)}\\ &= \frac{v(x)\lim_{h\to 0}\frac{u(x+h)-u(x)}{h} - u(x)\lim_{h\to 0}\frac{v(x+h)-v(x)}{h}}{\lim_{h\to 0} v(x+h)v(x)}\\ &= \frac{v(x)\frac{du}{dx} - u(x)\frac{dv}{dx}}{v^2(x)}.\end{aligned}$$

□

Example 3.2.4. Find the derivative of

$$y = \frac{(x-1)(x^2-2x)}{x^4}.$$

Solution: Using the Quotient Rule here will result in a complicated expression with many terms. Instead, use some algebra to simplify the expression. First expand the numerator and divide by x^4 :

$$y = \frac{(x-1)(x^2-2x)}{x^4} = \frac{x^3-3x^2+2x}{x^4} = x^{-1} - 3x^{-2} + 2x^{-3}$$

Then use the Sum and Power Rules:

$$\begin{aligned}\frac{dy}{dx} &= -x^{-2} - 3(-2)x^{-3} + 2(-3)x^{-4} \\ &= -\frac{1}{x^2} + \frac{6}{x^3} - \frac{6}{x^4}.\end{aligned}$$

3.3 Derivatives of Trigonometric Functions

Many phenomena of nature are approximately periodic (electromagnetic fields, heart rhythms, tides, weather). The derivatives of sines and cosines play a key role in describing periodic changes. This section shows how to differentiate the six basic trigonometric functions.

To calculate the derivative of $f(x) = sinx$, since

$$\sin(x+h) = \sin x \cos h + \cos x \sin h.$$

Then

$$\begin{aligned}f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{\sin(x+h)-\sin x}{h} \\ &= \lim_{h\to 0} \frac{(\sin x\cos h + \cos x \sin h) - \sin x}{h} \\ &= \lim_{h\to 0} \frac{\sin x(\cos h - 1) + \cos x \sin h}{h} \\ &= \lim_{h\to 0} \left(\sin x \cdot \frac{\cos h - 1}{h}\right) + \lim_{h\to 0}\left(\cos x \cdot \frac{\sin h}{h}\right) \\ &= \sin x \cdot \lim_{h\to 0} \frac{\cos h - 1}{h} + \cos x \cdot \lim_{h\to 0} \frac{\sin h}{h} \\ &= \sin x \cdot 0 + \cos x \cdot 1 = \cos x.\end{aligned}$$

Example 3.3.1. Differentiate

$$y = x^2 \sin x.$$

Solution: Using the product rule's formula, we have

$$\begin{aligned}\frac{dy}{dx} &= x^2\frac{d}{dx}(\sin x) + \sin x\frac{d}{dx}\left(x^2\right)\\ &= x^2\cos x + 2x\sin x.\end{aligned}$$

Example 3.3.2. Differentiate

$$y = \frac{\sin x}{x}.$$

Solution: Using the quotient rule's formula, we have

$$\begin{aligned}\frac{dy}{dx} &= \frac{x\cdot\frac{d}{dx}(\sin x) - \sin x\cdot 1}{x^2}\\ &= \frac{x\cos x - \sin x}{x^2}.\end{aligned}$$

To find the derivative of $f(x) = \cos(x)$, since

$$\cos(x+h) = \cos x\cos h - \sin x\sin h$$

we can compute the limit of the difference quotient:

$$\begin{aligned}\frac{d}{dx}(\cos x) &= \lim_{h\to 0}\frac{\cos(x+h)-\cos x}{h}\\ &= \lim_{h\to 0}\frac{(\cos x\cos h - \sin x\sin h)-\cos x}{h}\\ &= \lim_{h\to 0}\frac{\cos x(\cos h - 1) - \sin x\sin h}{h}\\ &= \lim_{h\to 0}\cos x\cdot\frac{\cos h - 1}{h} - \lim_{h\to 0}\sin x\cdot\frac{\sin h}{h}\\ &= \cos x\cdot\lim_{h\to 0}\frac{\cos h - 1}{h} - \sin x\cdot\lim_{h\to 0}\frac{\sin h}{h}\\ &= \cos x\cdot 0 - \sin x\cdot 1 = -\sin x.\end{aligned}$$

Example 3.3.3. Differentiate

$$y = \sin x\cos x.$$

Solution:

$$\begin{aligned}\frac{dy}{dx} &= \sin x\frac{d}{dx}(\cos x) + \cos x\frac{d}{dx}(\sin x)\\ &= \sin x(-\sin x) + \cos x(\cos x)\\ &= \cos^2 x - \sin^2 x.\end{aligned}$$

Example 3.3.4. Differentiate

$$y = \frac{\cos x}{1 - \sin x}.$$

Solution:

$$\begin{aligned}\frac{dy}{dx} &= \frac{(1-\sin x)\frac{d}{dx}(\cos x) - \cos x \frac{d}{dx}(1-\sin x)}{(1-\sin x)^2} \\ &= \frac{(1-\sin x)(-\sin x) - \cos x(0-\cos x)}{(1-\sin x)^2} \\ &= \frac{1-\sin x}{(1-\sin x)^2} \\ &= \frac{1}{1-\sin x}.\end{aligned}$$

Example 3.3.5. Verify the formula

$$\frac{d}{dx}(\tan x) = \sec^2 x.$$

Solution: Use the quotient rule and the identity $\cos^2 x + \sin^2 x = 1$:

$$\begin{aligned}\frac{d}{dx}(\tan x) &= \left(\frac{\sin x}{\cos x}\right)' \\ &= \frac{\cos x \cdot (\sin x)' - \sin x \cdot (\cos x)'}{\cos^2 x} \\ &= \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.\end{aligned}$$

Example 3.3.6. Verify the formula

$$\frac{d}{dx}(\sec x) = \sec x \tan x.$$

Solution: Since

$$\sec x = \frac{1}{\cos x}.$$

Then

$$\begin{aligned}\frac{d}{dx}(\sec x) &= \frac{d}{dx}\frac{1}{\cos x} \\ &= \frac{\frac{d}{dx}(1)\cdot(\cos x) - \frac{d}{dx}\cos x \cdot (1)}{(\cos x)^2} \\ &= \frac{\sin x}{(\cos x)(\cos x)}\end{aligned}$$

$$= \left(\frac{\sin x}{\cos x}\right)\left(\frac{1}{\cos x}\right)$$
$$= \tan x \sec x.$$

Example 3.3.7. Differentiate

$$f(x) = \frac{\sec x}{1 + \tan x}.$$

Solution: The quotient rule gives

$$\begin{aligned} f'(x) &= \frac{(1+\tan x)\frac{d}{dx}(\sec x) - \sec x \frac{d}{dx}(1+\tan x)}{(1+\tan x)^2} \\ &= \frac{(1+\tan x)\sec x \tan x - \sec x \cdot \sec^2 x}{(1+\tan x)^2} \\ &= \frac{\sec x\,(\tan x + \tan^2 x - \sec^2 x)}{(1+\tan x)^2} \\ &= \frac{\sec x(\tan x - 1)}{(1+\tan x)^2}. \end{aligned}$$

We collect all the differentiation formulas for trigonometric functions in the following table. Remember that they are valid only when x is measured in radians.

Derivatives of Trigonometric Functions

$$\begin{aligned} \frac{d}{dx}(\sin x) &= \cos x & \frac{d}{dx}(\csc x) &= -\csc x \cot x \\ \frac{d}{dx}(\cos x) &= -\sin x & \frac{d}{dx}(\sec x) &= \sec x \tan x \\ \frac{d}{dx}(\tan x) &= \sec^2 x & \frac{d}{dx}(\cot x) &= -\csc^2 x. \end{aligned}$$

3.4 The Chain Rule

The Chain Rule is used to differentiate composite functions such as $y = \cos(x^3)$ and $y = \sqrt{x^4+1}$

Recall that a composite function is obtained by "plugging" one function into another. The composite of f and g, denoted $f \circ g$, is defined by

$$(f \circ g)(x) = f(g(x))$$

For convenience, we call f the outside function and g the inside function. Often, we write the composite function as $f(u)$, where $u = g(x)$. For example, $y = \cos(x^3)$ is the function $y = \cos u$, where $u = x^3$.

Chain Rule: If f and g are differentiable, then the composite function $(f \circ g)(x) = f(g(x))$ is differentiable and

$$(f(g(x)))' = f'(g(x))g'(x)$$

Example 3.4.1. Calculate the derivative of

$$y = \cos\left(x^3\right).$$

Solution: As noted above, $y = \cos(x^3)$ is a composite $f(g(x))$, where

$$\begin{aligned} f(u) &= \cos u, \qquad u = g(x) = x^3 \\ f'(u) &= -\sin u, \quad g'(x) = 3x^2 \end{aligned}$$

Since $u = x^3$, $f'(g(x)) = f'(u) = f'(x^3) = -\sin(x^3)$. So, by the Chain Rule

$$\begin{aligned} \frac{d}{dx}\cos\left(x^3\right) &= \underbrace{-\sin\left(x^3\right)}_{f'(g(x))}\underbrace{\left(3x^2\right)}_{g'(x)} \\ &= -3x^2\sin\left(x^3\right) \end{aligned}$$

Example 3.4.2. Calculate the derivative of $y = \sqrt{x^4+1}$.
Solution: The function $y = \sqrt{x^4+1}$ is a composite $f(g(x))$, where

$$\begin{aligned} f(u) &= u^{1/2}, \qquad u = g(x) = x^4 + 1 \\ f'(u) &= \frac{1}{2}u^{-1/2}, \qquad g'(x) = 4x^3 \end{aligned}$$

Note that $f'(g(x)) = \frac{1}{2}\left(x^4+1\right)^{-1/2}$, so by the Chain Rule,

$$\begin{aligned} \frac{d}{dx}\sqrt{x^4+1} &= \underbrace{\frac{1}{2}\left(x^4+1\right)^{-1/2}}_{f'(g(x))}\underbrace{\left(4x^3\right)}_{g'(x)} \\ &= \frac{4x^3}{2\sqrt{x^4+1}} \end{aligned}$$

Example 3.4.3. Calculate $\frac{dy}{dx}$ for

$$y = \tan\left(\frac{x}{x+1}\right).$$

Solution: The outside function is $f(u) = \tan u$. Because $f'(u) = \sec^2 u$, the Chain Rule gives us $\frac{d}{dx}\tan\left(\frac{x}{x+1}\right) = \sec^2\left(\frac{x}{x+1}\right) \underbrace{\frac{d}{dx}\left(\frac{x}{x+1}\right)}_{\substack{\text{Derivative of} \\ \text{inside function}}}$

Now, by the Quotient Rule,

$$\frac{d}{dx}\left(\frac{x}{x+1}\right) = \frac{(x+1)\frac{d}{dx}x - x\frac{d}{dx}(x+1)}{(x+1)^2} = \frac{1}{(x+1)^2}.$$

We obtain

$$\begin{aligned}\frac{d}{dx}\tan\left(\frac{x}{x+1}\right) &= \sec^2\left(\frac{x}{x+1}\right)\frac{1}{(x+1)^2} \\ &= \frac{\sec^2\left(\frac{x}{x+1}\right)}{(x+1)^2}.\end{aligned}$$

It is instructive to write the Chain Rule in Leibniz notation. Let

$$y = f(u) = f(g(x))$$

Then, by the Chain Rule,

$$\frac{dy}{dx} = f'(u)g'(x) = \frac{df}{du}\frac{du}{dx}$$

or

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}.$$

We now discuss some important special cases of the Chain Rule.

General Power and Exponential Rules: If g is differentiable, then

$$\frac{d}{dx}(g(x))^n = n(g(x))^{n-1}g'(x)$$

for any number n.

Example 3.4.4. Find the derivatives of

(a) $y = (x^2 + 7x + 2)^{-1/3}$.

(b) $y = \sec^4 t$.

Solution: Apply $\frac{d}{dx}g(x)^n = ng(x)^{n-1}g'(x)$.

(a)

$$\begin{aligned}\frac{d}{dx}\left(x^2+7x+2\right)^{-1/3} &= -\frac{1}{3}\left(x^2+7x+2\right)^{-4/3}\frac{d}{dx}\left(x^2+7x+2\right)\\ &= -\frac{1}{3}\left(x^2+7x+2\right)^{-4/3}(2x+7)\end{aligned}$$

(b)

$$\begin{aligned}\frac{d}{dt}\sec^4 t &= 4\sec^3 t\frac{d}{dt}\sec t\\ &= 4\sec^3 t(\sec t\tan t)\\ &= 4\sec^4 t\tan t\end{aligned}$$

Example 3.4.5. Calculate

$$\frac{d}{dx}\sqrt{1+\sqrt{x^2+1}}.$$

Solution: First apply the Chain Rule with inside function $u = 1+\sqrt{x^2+1}$:

$$\frac{d}{dx}\left(1+\left(x^2+1\right)^{1/2}\right)^{1/2} = \frac{1}{2}\left(1+\left(x^2+1\right)^{1/2}\right)^{-1/2}\frac{d}{dx}\left(1+\left(x^2+1\right)^{1/2}\right).$$

Then apply the Chain Rule again to the remaining derivative:

$$\begin{aligned}\frac{d}{dx}\left(1+\left(x^2+1\right)^{1/2}\right)^{1/2} &= \frac{1}{2}\left(1+\left(x^2+1\right)^{1/2}\right)^{-1/2}\left(\frac{1}{2}\left(x^2+1\right)^{-1/2}(2x)\right)\\ &= \frac{1}{2}x\left(x^2+1\right)^{-1/2}\left(1+\left(x^2+1\right)^{1/2}\right)^{-1/2}.\end{aligned}$$

Example 3.4.6. Find $f'(x)$ if

$$f(x) = \frac{1}{\sqrt[3]{x^2+x+1}}.$$

Solution: First rewrite $f : f(x) = \left(x^2+x+1\right)^{-1/3}$ Thus

$$\begin{aligned}f'(x) &= -\frac{1}{3}\left(x^2+x+1\right)^{-4/3}\frac{d}{dx}\left(x^2+x+1\right)\\ &= -\frac{1}{3}\left(x^2+x+1\right)^{-4/3}(2x+1).\end{aligned}$$

Example 3.4.7. Find the derivative of the function

$$g(t) = \left(\frac{t-2}{2t+1}\right)^9.$$

Solution: Combining the Power Rule, Chain Rule, and Quotient Rule, we get

$$\begin{aligned} g'(t) &= 9\left(\frac{t-2}{2t+1}\right)^8 \frac{d}{dt}\left(\frac{t-2}{2t+1}\right) \\ &= 9\left(\frac{t-2}{2t+1}\right)^8 \frac{(2t+1)\cdot 1 - 2(t-2)}{(2t+1)^2} = \frac{45(t-2)^8}{(2t+1)^{10}}. \end{aligned}$$

Example 3.4.8. Differentiate

$$y = (2x+1)^5\left(x^3 - x + 1\right)^4.$$

Solution: In this example we must use the Product Rule before using the Chain Rule:

$$\begin{aligned} \frac{dy}{dx} &= (2x+1)^5 \frac{d}{dx}\left(x^3 - x + 1\right)^4 + \left(x^3 - x + 1\right)^4 \frac{d}{dx}(2x+1)^5 \\ &= (2x+1)^5 \cdot 4\left(x^3 - x + 1\right)^3 \frac{d}{dx}\left(x^3 - x + 1\right) \\ &\quad + \left(x^3 - x + 1\right)^4 \cdot 5(2x+1)^4 \frac{d}{dx}(2x+1) \\ &= 4(2x+1)^5\left(x^3 - x + 1\right)^3\left(3x^2 - 1\right) + 5\left(x^3 - x + 1\right)^4 (2x+1)^4 \cdot 2 \\ &= 2(2x+1)^4\left(x^3 - x + 1\right)^3\left(17x^3 + 6x^2 - 9x + 3\right). \end{aligned}$$

3.5 Derivatives of Inverse Trigonometric Functions

In fact, if f is any one-to-one differentiable function, it can be proved that its inverse function f^{-1} is also differentiable, except where its tangents are vertical. This is plausible because the graph of a differentiable function has no corner or kink and so if we reflect it about $y = x$, the graph of its inverse function also has no corner or kink.

Recall the definition of the arcsine function:

$$y = \sin^{-1} x \quad \text{means} \quad \sin y = x \quad \text{and} \quad -\frac{\pi}{2} \leqslant y \leqslant \frac{\pi}{2}.$$

Differentiating $\sin y = x$ implicitly with respect to x, we obtain

$$\cos y \frac{dy}{dx} = 1 \quad \text{or} \quad \frac{dy}{dx} = \frac{1}{\cos y}.$$

Now $\cos y \geqslant 0$, since $-\pi/2 \leqslant y \leqslant \pi/2$, so

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}.$$

Therefore, $\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}$. Hence, $\frac{d}{dx}\left(\sin^{-1} x\right) = \frac{1}{\sqrt{1-x^2}}$.

The formula for the derivative of the arctangent function is derived in a similar way. If $y = \tan^{-1} x$, then $\tan y = x$. Differentiating this latter equation implicitly with respect to x, we have

$$\sec^2 y \frac{dy}{dx} = 1 \Longrightarrow \frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}.$$

That is, $\frac{d}{dx}\left(\tan^{-1} x\right) = \frac{1}{1+x^2}$.

The inverse trigonometric functions that occur most frequently are the ones that we have just discussed. The derivatives of the remaining four are given in the following table. The proofs of the formulas are left as exercises.

Derivatives of Inverse Trigonometric Functions

$$\begin{aligned} \frac{d}{dx}\left(\sin^{-1} x\right) &= \frac{1}{\sqrt{1-x^2}} & \frac{d}{dx}\left(\csc^{-1} x\right) &= -\frac{1}{x\sqrt{x^2-1}} \\ \frac{d}{dx}\left(\cos^{-1} x\right) &= -\frac{1}{\sqrt{1-x^2}} & \frac{d}{dx}\left(\sec^{-1} x\right) &= \frac{1}{x\sqrt{x^2-1}} \\ \frac{d}{dx}\left(\tan^{-1} x\right) &= \frac{1}{1+x^2} & \frac{d}{dx}\left(\cot^{-1} x\right) &= -\frac{1}{1+x^2} \end{aligned}$$

Example 3.5.1. Differentiate

(a) $y = \frac{1}{\sin^{-1} x}$.

(b) $f(x) = x \tan^{-1} \sqrt{x}$.

Solution:

(a)

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}\left(\sin^{-1} x\right)^{-1} = -\left(\sin^{-1} x\right)^{-2} \frac{d}{dx}\left(\sin^{-1} x\right) \\ &= -\frac{1}{\left(\sin^{-1} x\right)^2 \sqrt{1-x^2}} \end{aligned}$$

(b)

$$\begin{aligned} f'(x) &= x\frac{1}{1+(\sqrt{x})^2}\left(\frac{1}{2}x^{-1/2}\right) + \arctan\sqrt{x} \\ &= \frac{\sqrt{x}}{2(1+x)} + \arctan\sqrt{x} \end{aligned}$$

Example 3.5.2. Differentiate $y = \cot^{-1}\left(\frac{1}{x^2}\right)$.

Solution:

$$\begin{aligned} y' &= \left(\cot^{-1}\frac{1}{x^2}\right)' = -\frac{1}{1+\left(\frac{1}{x^2}\right)^2}\cdot\left(\frac{1}{x^2}\right)' \\ &= -\frac{1}{1+\frac{1}{x^4}}\cdot\left(-2x^{-3}\right) = \frac{2x^4}{\left(x^4+1\right)x^3} \\ &= \frac{2x}{1+x^4}. \end{aligned}$$

Example 3.5.3. Find the derivative of each of the following function:

$$f(x) = \cos^{-1}\left(x^3+x\right)$$

$$\begin{aligned} f'(x) &= \frac{d}{dx}\left[\cos^{-1}\left(x^3+x\right)\right] = -\frac{1}{\sqrt{1-\left(x^3+x\right)^2}}\cdot\left(3x^2+1\right) \\ &= -\frac{3x^2+1}{\sqrt{1-\left(x^3+x\right)^2}}. \end{aligned}$$

3.6 Derivative of the Natural Exponential Function

Thus the exponential function $f(x) = e^x$ has the property that it is its own derivative. The geometrical significance of this fact is that the slope of a tangent line to the curve $y = e^x$ is equal to the y-coordinate of the point .

$$\frac{d}{dx}\left(e^x\right) = e^x$$

Example 3.6.1. Differentiate the function $y = e^{\tan x}$.
Solution: To use the Chain Rule, we let $u = \tan x$. Then we have $y = e^u$, so

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = e^u\frac{du}{dx} = e^{\tan x}\sec^2 x.$$

In general , we have

$$\frac{d}{dx}\left(e^u\right) = e^u\frac{du}{dx}.$$

Example 3.6.2. Find y' if

$$y = e^{-4x}\sin 5x$$

Solution: Using Product Rule, we have

$$\begin{aligned} y' &= e^{-4x}(\cos 5x)(5) + (\sin 5x)e^{-4x}(-4) \\ &= e^{-4x}(5\cos 5x - 4\sin 5x). \end{aligned}$$

3.7 Derivative Of Logarithmic Functions

We begin by establishing that $f(x) = \ln x$ is differentiable for $x > 0$ by using the derivative definition to find its derivative. To obtain this derivative, we need the fact that $\ln x$ is continuous for $x > 0$. Since e^x is continuous, we know that $\ln x$ is continuous for $x > 0$. We will also need the limit

$$\lim_{v\to 0}(1+v)^{1/v} = e.$$

Using the definition of the derivative, we obtain

$$\begin{aligned} \frac{d}{dx}[\ln x] &= \lim_{h\to 0}\frac{\ln(x+h)-\ln x}{h} = \lim_{h\to 0}\frac{1}{h}\ln\left(\frac{x+h}{x}\right) = \lim_{h\to 0}\frac{1}{h}\ln\left(1+\frac{h}{x}\right) \\ &= \lim_{v\to 0}\frac{1}{vx}\ln(1+v) = \frac{1}{x}\lim_{v\to 0}\frac{1}{v}\ln(1+v) = \frac{1}{x}\lim_{v\to 0}\ln(1+v)^{1/v} \\ &= \frac{1}{x}\ln\left[\lim_{v\to 0}(1+v)^{1/v}\right] = \frac{1}{x}\ln e = \frac{1}{x}. \end{aligned}$$

Thus,

$$\frac{d}{dx}[\ln x] = \frac{1}{x}, \quad x > 0$$

A derivative formula for the general logarithmic function $\log_b x$ can be obtained the following from

$$\frac{d}{dx}\left[\log_b x\right] = \frac{d}{dx}\left[\frac{\ln x}{\ln b}\right] = \frac{1}{\ln b}\frac{d}{dx}[\ln x].$$

It follows from this that

$$\frac{d}{dx}\left[\log_b x\right] = \frac{1}{x\ln b}, \quad x > 0.$$

Example 3.7.1. Find $\frac{d}{dx}\left[\ln\left(x^2+1\right)\right]$.
Solution: Let $u = x^2 + 1$ we obtain

$$\begin{aligned} \frac{d}{dx}\left[\ln\left(x^2+1\right)\right] &= \frac{1}{x^2+1}\cdot\frac{d}{dx}\left[x^2+1\right] = \frac{1}{x^2+1}\cdot 2x \\ &= \frac{2x}{x^2+1}. \end{aligned}$$

Example 3.7.2. Find

$$\frac{d}{dx}\left[\ln\left(\frac{x^2\sin x}{\sqrt{1+x}}\right)\right].$$

Solution:

$$\begin{aligned}\frac{d}{dx}\left[\ln\left(\frac{x^2\sin x}{\sqrt{1+x}}\right)\right] &= \frac{d}{dx}\left[2\ln x+\ln(\sin x)-\frac{1}{2}\ln(1+x)\right]\\ &= \frac{2}{x}+\frac{\cos x}{\sin x}-\frac{1}{2(1+x)}\\ &= \frac{2}{x}+\cot x-\frac{1}{2+2x}.\end{aligned}$$

We now consider a technique called **logarithmic differentiation** that is useful for differentiating functions that are composed of products, quotients, and powers.

Example 3.7.3. *Find the derivative of*

$$y=\frac{x^2\sqrt[3]{7x-14}}{(1+x^2)^4}.$$

Solution: Take the natural logarithm of both sides and then use its properties, we can write

$$\ln y = 2\ln x+\frac{1}{3}\ln(7x-14)-4\ln\left(1+x^2\right).$$

Differentiating both sides with respect to x yields

$$\frac{1}{y}\frac{dy}{dx}=\frac{2}{x}+\frac{7/3}{7x-14}-\frac{8x}{1+x^2}.$$

Thus, on solving for dy/dx we obtain

$$\frac{dy}{dx}=\frac{x^2\sqrt[3]{7x-14}}{(1+x^2)^4}\left[\frac{2}{x}+\frac{1}{3x-6}-\frac{8x}{1+x^2}\right].$$

Example 3.7.4. Find

$$\frac{d}{dx}\ln\frac{x+1}{\sqrt{x-2}}.$$

Solution:

$$\begin{aligned}\frac{d}{dx}\ln\frac{x+1}{\sqrt{x+2}} &= \frac{d}{dx}\left[\ln(x+1)-\frac{1}{2}\ln(x-2)\right]\\ &= \frac{1}{x+1}-\frac{1}{2}\left(\frac{1}{x-2}\right).\end{aligned}$$

Steps in Logarithmic Differentiation

1. Take natural logarithms of both sides of an equation $y=f(x)$ and use the properties of logarithms to simplify.
2. Differentiate implicitly with respect to x.
3. Solve the resulting equation for y'.

Example 3.7.5. Differentiate

$$y = x^{\sqrt{x}}.$$

Solution: Since both the base and the exponent are variable, we use logarithmic differentiation:

$$\begin{aligned}
\ln y &= \ln x^{\sqrt{x}} = \sqrt{x}\ln x \\
\frac{y'}{y} &= \sqrt{x}\cdot\frac{1}{x} + (\ln x)\frac{1}{2\sqrt{x}} \\
y' &= y\left(\frac{1}{\sqrt{x}} + \frac{\ln x}{2\sqrt{x}}\right) = x^{\sqrt{x}}\left(\frac{2+\ln x}{2\sqrt{x}}\right).
\end{aligned}$$

Example 3.7.6. Differentiate

$$y = x^x + (\sin x)^{\cos x}.$$

Solution: Consider

$$y = y_1 + y_2$$

where $y_1 = x^x, \quad y_2 = (\sin x)^{\cos x}$ Since

$$\begin{aligned}
\ln y_1 &= \ln x^x = x\ln x \\
\Rightarrow \frac{1}{y_1}y_1' &= x\frac{1}{x} + \ln(x) \\
\Rightarrow y_1' &= x^x[1+\ln(x)]
\end{aligned}$$

and

$$\begin{aligned}
\ln y_2 &= \ln(\sin x)^{\cos x} = (\cos x)\ln(\sin x) \\
\Rightarrow \frac{1}{y_2}y_2' &= \frac{\cos^2 x}{\sin x} - \sin x\ln(\sin x) \\
\Rightarrow y_2' &= (\sin x)^{\cos x}[\cot x\csc x - \sin x\ln(\sin x)]
\end{aligned}$$

Then

$$\begin{aligned}
y' &= y_1' + y_2' \\
&= x^x[1+\ln(x)] + (\sin x)^{\cos x}[\cot x\csc x - \sin x\ln(\sin x)]
\end{aligned}$$

3.8 Derivatives of Hyperbolic Functions

The derivatives of the hyperbolic functions are easily computed. For example,

$$\frac{d}{dx}(\sinh x) = \frac{d}{dx}\left(\frac{e^x - e^{-x}}{2}\right) = \frac{e^x + e^{-x}}{2} = \cosh x$$

We list the differentiation formulas for the hyperbolic functions as in the following table . The remaining proofs are left as exercises. Note the analogy with the differentiation formulas for trigonometric functions, but beware that the signs are different in some cases.

Derivatives of Hyperbolic Functions

$$\begin{aligned} \frac{d}{dx}(\sinh x) &= \cosh x & \frac{d}{dx}(\operatorname{csch} x) &= -\operatorname{csch} x \coth x \\ \frac{d}{dx}(\cosh x) &= \sinh x & \frac{d}{dx}(\operatorname{sech} x) &= -\operatorname{sech} x \tanh x \\ \frac{d}{dx}(\tanh x) &= \operatorname{sech}^2 x & \frac{d}{dx}(\coth x) &= -\operatorname{csch}^2 x \end{aligned}$$

Example 3.8.1. Differentiate $y = \cosh\sqrt{x}$.

Solution:

$$\frac{dy}{dx} = \frac{d}{dx}(\cosh\sqrt{x}) = \sinh\sqrt{x} \cdot \frac{d}{dx}\sqrt{x} = \frac{\sinh\sqrt{x}}{2\sqrt{x}}.$$

Example 3.8.2. Differentiate $y = \cosh\sqrt{x}$.

Solution:

$$\frac{dy}{dx} = \frac{d}{dx}\left[\cosh\left(x^3\right)\right] = \sinh\left(x^3\right) \cdot \frac{d}{dx}\left[x^3\right] = 3x^2 \sinh\left(x^3\right).$$

Example 3.8.3. Differentiate $y = \ln(\tanh x)$.

Solution:

$$\frac{dy}{dx} = \frac{d}{dx}[\ln(\tanh x)] = \frac{1}{\tanh x} \cdot \frac{d}{dx}[\tanh x] = \frac{\operatorname{sech}^2 x}{\tanh x}.$$

3.9 Derivatives Of Inverse Hyperbolic Functions

Formulas for the derivatives of the inverse hyperbolic functions can be obtained as the following

$$\begin{aligned} \frac{d}{dx}\left[\sinh^{-1} x\right] &= \frac{d}{dx}\left[\ln\left(x + \sqrt{x^2+1}\right)\right] \\ &= \frac{1}{x + \sqrt{x^2+1}}\left(1 + \frac{x}{\sqrt{x^2+1}}\right) \\ &= \frac{\sqrt{x^2+1} + x}{\left(x + \sqrt{x^2+1}\right)\left(\sqrt{x^2+1}\right)} \\ &= \frac{1}{\sqrt{x^2+1}} \end{aligned}$$

The following table list the generalized derivative formulas and corresponding formulas for the inverse hyperbolic functions. Some of the proofs appear as exercises.

$$\begin{aligned}
\frac{d}{dx}\left(\sinh^{-1} u\right) &= \frac{1}{\sqrt{1+u^2}}\frac{du}{dx} \\
\frac{d}{dx}\left(\coth^{-1} u\right) &= \frac{1}{1-u^2}\frac{du}{dx}, \quad |u|>1 \\
\frac{d}{dx}\left(\cosh^{-1} u\right) &= \frac{1}{\sqrt{u^2-1}}\frac{du}{dx}, \quad u>1 \\
\frac{d}{dx}\left(\operatorname{sech}^{-1} u\right) &= -\frac{1}{u\sqrt{1-u^2}}\frac{du}{dx}, \quad 0<u<1 \\
\frac{d}{dx}\left(\tanh^{-1} u\right) &= \frac{1}{1-u^2}\frac{du}{dx}, \quad |u|<1 \\
\frac{d}{dx}\left(\operatorname{csch}^{-1} u\right) &= -\frac{1}{|u|\sqrt{1+u^2}}\frac{du}{dx}, \quad u\neq 0
\end{aligned}$$

Example 3.9.1. Verify

$$\frac{d}{dx}\coth x = -\operatorname{csch}^2 x.$$

Solution: By the Quotient Rule and the identity $\cosh^2 x - \sinh^2 x = 1$,

$$\begin{aligned}
\frac{d}{dx}\coth x &= \left(\frac{\cosh x}{\sinh x}\right)' \\
&= \frac{(\sinh x)(\cosh x)' - (\cosh x)(\sinh x)'}{\sinh^2 x} \\
&= \frac{\sinh^2 x - \cosh^2 x}{\sinh^2 x} = \frac{-1}{\sinh^2 x} = -\operatorname{csch}^2 x.
\end{aligned}$$

Example 3.9.2. Calculate:

(a) $\dfrac{d}{dx}\cosh\left(3x^2+1\right)$

(b) $\dfrac{d}{dx}\sinh x \tanh x.$

Solution:

(a) By the Chain Rule,

$$\frac{d}{dx}\cosh\left(3x^2+1\right) = 6x\sinh\left(3x^2+1\right).$$

(b) By the Product Rule,

$$\begin{aligned}\frac{d}{dx}(\sinh x \tanh x) &= \sinh x \operatorname{sech}^2 x + \tanh x \cosh x \\ &= \operatorname{sech} x \tanh x + \sinh x.\end{aligned}$$

Example 3.9.3. Verify the formula

$$\frac{d}{dx}\tanh^{-1} x = \frac{1}{1-x^2}.$$

Solution: Recall that if g is the inverse of f, then $g'(x) = 1/f'(g(x))$. Applying this to $f(x) = \tanh x$, and using the formula $(\tanh x)' = \operatorname{sech}^2 x$, we have

$$\frac{d}{dx}\tanh^{-1} x = \frac{1}{\operatorname{sech}^2\left(\tanh^{-1} x\right)}$$

Now let $t = \tanh^{-1} x$. Then

$$\begin{aligned}\cosh^2 t - \sinh^2 t &= 1 && \text{(basic identity)} \\ 1 - \tanh^2 t &= \operatorname{sech}^2 t && \text{(divide by } \cosh^2 t\text{)} \\ 1 - x^2 &= \operatorname{sech}^2\left(\tanh^{-1} x\right) && \text{(because } x = \tanh t\text{)}\end{aligned}$$

Hence

$$\frac{d}{dx}\tanh^{-1} x = \frac{1}{\operatorname{sech}^2\left(\tanh^{-1} x\right)} = \frac{1}{1-x^2}.$$

3.10 Implicit Differentiation

Most of the functions we have dealt with so far have been described by an equation of the form $y = f(x)$ that expresses y explicitly in terms of the variable x. We have learned rules for differentiating functions defined in this way. Another situation occurs when we encounter equations like

$$x^3 + y^3 - 9xy = 0, \quad y^2 - x = 0, \quad \text{or} \quad x^2 + y^2 - 25 = 0$$

These equations define an implicit relation between the variables x and y. In some cases we may be able to solve such an equation for y as an explicit function (or even several functions) of x. When we cannot put an equation $F(x, y) = 0$ in the form $y = f(x)$ to differentiate it in the usual way, we may still be able to find dy/dx by implicit differentiation. This section describes the technique.

Implicit Differentiation:

1. Differentiate both sides of the equation with respect to x, treating y as a differentiable function of x.

2. Collect the terms with dy/dx on one side of the equation and solve for dy/dx.

Example 3.10.1. Find dy/dx if $y^2 = x^2 + \sin xy$.
Solution: We differentiate the equation implicitly.

$$\begin{aligned} y^2 &= x^2 + \sin xy \\ \frac{d}{dx}\left(y^2\right) &= \frac{d}{dx}\left(x^2\right) + \frac{d}{dx}(\sin xy) \\ 2y\frac{dy}{dx} &= 2x + (\cos xy)\frac{d}{dx}(xy) \\ 2y\frac{dy}{dx} &= 2x + (\cos xy)\left(y + x\frac{dy}{dx}\right) \\ 2y\frac{dy}{dx} - (\cos xy)\left(x\frac{dy}{dx}\right) &= 2x + (\cos xy)y \\ (2y - x\cos xy)\frac{dy}{dx} &= 2x + y\cos xy \\ \frac{dy}{dx} &= \frac{2x + y\cos xy}{2y - x\cos xy}. \end{aligned}$$

Example 3.10.2. Find d^2y/dx^2 if $2x^3 - 3y^2 = 8$.
Solution: To start, we differentiate both sides of the equation with respect to x in order to find $y' = dy/dx$

$$\begin{aligned} \frac{d}{dx}\left(2x^3 - 3y^2\right) &= \frac{d}{dx}(8) \\ 6x^2 - 6yy' &= 0 \\ y' &= \frac{x^2}{y}, \qquad \text{when } y \neq 0. \end{aligned}$$

We now apply the Quotient Rule to find y''.

$$y'' = \frac{d}{dx}\left(\frac{x^2}{y}\right) = \frac{2xy - x^2y'}{y^2} = \frac{2x}{y} - \frac{x^2}{y^2}\cdot y'.$$

Finally, we substitute $y' = x^2/y$ to express y'' in terms of x and y.

$$y'' = \frac{2x}{y} - \frac{x^2}{y^2}\left(\frac{x^2}{y}\right) = \frac{2x}{y} - \frac{x^4}{y^3}, \qquad \text{when } y \neq 0.$$

3.11 Exercises

1. Compute $f'(a)$ using the limit definition of f at $x = a$

(a) $f(x) = x^2 - x, \quad a = 1$

(b) $f(x) = 5 - 3x, \quad a = 2$

(c) $f(x) = x^{-1}, \quad a = 4$

(d) $f(x) = x^3, \quad a = -2$

2. Compute dy/dx using the limit definition.

(a) $y = 4 - x^2$

(b) $y = \sqrt{2x+1}$

(c) $y = \frac{1}{2-x}$

(d) $y = \frac{1}{(x-1)^2}$

3. Compute the first derivative.

(a) $g(x) = x \sin(2^x)$

(b) $G(x) = 4^{c/x}$

(c) $F(t) = 3^{\cos 2t}$

(d) $y = (\cos x)^x$

(e) $y = (\sin x)^{\ln x}$

(f) $y = (\tan x)^{1/x}$

(g) $y = (\ln x)^{\cos x}$

(h) $y = (\tan^{-1} x)^2$

(i) $g(x) = \arccos \sqrt{x}$

(j) $y = \sin^{-1}(2x+1)$

(k) $R(t) = \operatorname{arcsin}(1/t)$

(l) $y = x \sin^{-1} x + \sqrt{1-x^2}$

(m) $y = \cos^{-1}\left(\sin^{-1} t\right)$

(n) $F(x) = x \sec^{-1}(x^3)$

(o) $y = \arctan \sqrt{\dfrac{1-x}{1+x}}$

4. Compute the first and the second derivatives.

(a) $y = 3x^5 - 7x^2 + 4$

(b) $y = 4x^{-3/2}$

(c) $y = t^{-7.3}$

(d) $y = 4x^2 - x^{-2}$

(e) $y = \frac{x+1}{x^2+1}$

(f) $y = \frac{3t-2}{4t-9}$

(g) $y = (x^4 - 9x)^6$

(h) $y = (3t^2 + 20t^{-3})^6$

(i) $y = (2 + 9x^2)^{3/2}$

(j) $y = (x+1)^3(x+4)^4$

(k) $y = \frac{z}{\sqrt{1-z}}$

(l) $y = \left(1 + \dfrac{1}{x}\right)^3$

(m) $y = \frac{x^4+\sqrt{x}}{x^2}$

(n) $y = \frac{1}{(1-x)\sqrt{2-x}}$

(o) $y = \sqrt{x + \sqrt{x + \sqrt{x}}}$

(p) $h(z) = \left(z + (z+1)^{1/2}\right)^{-3/2}$

(q) $y = \tan(t^{-3})$

(r) $y = 4\cos(2 - 3x)$

(s) $y = \sin(2x)\cos^2 x$

(t) $y = \sin\left(\frac{4}{\theta}\right)$

(u) $y = \frac{t}{1+\sec t}$

(v) $y = z \csc(9z+1)$

(w) $f(x) = x^5 + 5^x$

5. Use implicit differentiation to find dy/dx .

   (a) $x^2y + xy^2 = 6$

   (b) $x^3 + y^3 = 18xy$

   (c) $2xy + y^2 = x + y$

   (d) $x^3 - xy + y^3 = 1$

   (e) $x^2(x - y)^2 = x^2 - y^2$

   (f) $(3xy + 7)^2 = 6y$

   (g) $y^2 = \frac{x-1}{x+1}$

   (h) $x^3 = \frac{2x-y}{x+3y}$

   (i) $x = \sec y$

   (j) $xy = \cot(xy)$

   (k) $x + \tan(xy) = 0$

   (l) $x^4 + \sin y = x^3y^2$

   (m) $y \sin\left(\frac{1}{y}\right) = 1 - xy$

   (n) $x \cos(2x + 3y) = y \sin x$

6. Find y' if $e^{x/y} = x - y$.

7. Find an equation of the tangent line to the curve $xe^y + ye^x = 1$ at the point $(0, 1)$

8. Show that the function $y = e^x + e^{-x/2}$ satisfies the differential equation $2y'' - y' - y = 0$

9. Show that the function $y = Ae^{-x} + Bxe^{-x}$ satisfies the differential equation $y'' + 2y' + y = 0$

10. For what values of r does the function $y = e^{rx}$ satisfy the differential equation $y'' + 6y' + 8y = 0$?

11. Find the values of λ for which $y = e^{\lambda x}$ satisfies the equation $y + y' = y''$.

12. If $f(x) = e^{2x}$, find a formula for $f^{(n)}(x)$.

13. Find the thousandth derivative of $f(x) = xe^{-x}$.

14. Find y' if $\tan^{-1}(x^2y) = x + xy^2$.

15. Prove that $x^2y'' + xy' + y = 0$, where $y = \sin(\ln(x)) + \cos(\ln(x))$.

16. Prove that $x^4y'' + y = 0$, where $y = x \sin\left(\frac{1}{x}\right)$.

Chapter 4

Applications of Derivatives

In the previous chapter we focused almost exclusively on the computation of derivatives. In this chapter will focus on applications of derivatives. It is important to always remember that we didn't spend a whole chapter talking about computing derivatives just to be talking about them. There are many very important applications to derivatives.

The two main applications that we'll be looking at in this chapter are using derivatives to determine information about graphs of functions and optimization problems. These will not be the only applications however. We will be revisiting limits and taking a look at an application of derivatives that will allow us to compute limits that we haven't been able to compute previously.

4.1 Tangent and Normal Lines

The derivative of a function at a point is the slope of the tangent line at this point. The normal line is defined as the line that is perpendicular to the tangent line at the point of tangency. Therefore, we have:

1. The equation of the tangent line for the curve $y = f(x)$ at the point (x_1, y_1) is given by
$$\frac{y - y_1}{x - x_1} = m, \qquad \text{where } m = f'(x_1) := f'(x)|_{x=x_1}.$$

2. The equation of the normal line for the curve $y = f(x)$ at the point $x = a$ is given by
$$\frac{y - y_1}{x - x_1} = -\frac{1}{m}.$$

Example 4.1.1. Find the equation of the tangent and normal lines to the graph of $f(x) = \sqrt{x^2 + 3}$ at the point $(-1, 2)$.
Solution:

$$f(x) = \left(x^2 + 3\right)^{1/2} \Longrightarrow f'(x) = \frac{1}{2}\left(x^2 + 3\right)^{-1/2} \cdot (2x) = \frac{x}{\sqrt{x^2 + 3}}.$$

At the point $(-1, 2), f'(-1) = -1/2$ and the equation of the line is

$$y - y_1 = m\,(x - x_1) \Longrightarrow y - 2 = -\frac{1}{2}(x + 1) \Longrightarrow 2y - 4 = -x - 1 \Longrightarrow x + 2y = 3.$$

Similarly, the equation of the normal line at the point $(-1, 2)$ is

$$\frac{y - 2}{x + 1} = 2 \Longrightarrow y - 2 = 2(x + 1)y - 2x - 4 = 0.$$

Example 4.1.2. Find the equation of the tangent and normal lines to the graph of $y = x^2$ at the point $(-1, 1)$.
Solution: By differentiating, we have

$$f(x) = x^2 \Longrightarrow f'(x) = 2x \Longrightarrow m = f'(-1) = -2.$$

Hence, the equation of the tangent line at the point $(-1, 1)$ is

$$y - 1 = -2(x + 1) \Longrightarrow y + 2x + 1 = 0.$$

Similarly, the equation of the normal line at the point $(-1, 1)$ is

$$y - 1 = \frac{1}{2}(x + 1) \Longrightarrow 2y - x - 3 = 0.$$

4.2 Maclaurin and Taylor series

The Maclaurin series expansion for $y = f(x)$ (as a power in x) is defined by.

$$f(x) = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f''(0) + \frac{x^4}{4!}f^{(4)}(0) + \ldots. + \frac{x^n}{n!}f^{(n)}(0) + \ldots$$

Example 4.2.1. Find the Maclaurin series for $f(x) = e^x$, and hence for $f(x) = \sinh(x)$.
Solution: We have

$$\begin{aligned}
f(x) &= e^x & f(0) &= 1\\
f'(x) &= e^x & f'(0) &= 1\\
f''(x) &= e^x & f''(0) &= 1\\
f'''(x) &= e^x & f'''(0) &= 1\\
&\vdots\\
f^{(n)}(x) &= e^x & f^{(n)}(0) &= 1
\end{aligned}$$

Now, using the Maclaurin expansion, we get

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \ldots\ldots$$
$$e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \frac{x^5}{5!} + \ldots\ldots$$

Hence

$$\sinh x = \frac{e^x - e^{-x}}{2} = \frac{x}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \frac{x^9}{9!} + \ldots\ldots$$

Example 4.2.2. Find the Maclaurin series for $f(x) = \sin x$ and hence for$f(x) = \cos x$.
Solution: We know that the Maclaurin series expansion for $y = f(x)$ is defined by.Now, we have

$$\begin{aligned} f(x) &= \sin x & f(0) &= 0 \\ f'(x) &= \cos x & f'(0) &= 1 \\ f''(x) &= -\sin x & f''(0) &= 0 \\ f'''(x) &= -\cos x & f''(0) &= -1 \\ f^{(4)}(x) &= \sin x & f^{(4)}(0) &= 0 \\ f^{(5)}(x) &= \cos x & f^{(5)}(0) &= 1. \end{aligned}$$

Then

$$\begin{aligned} \sin x &= \frac{x}{1!} - \frac{x^3}{3!} + \frac{x^3}{5!} - \frac{x'}{7!}\ldots\ldots \\ \cos x &= \frac{d}{dx}[\sin x] \\ &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{5!} + \ldots\ldots \end{aligned}$$

Remark 4.1. *The Maclaurin expansions for* $\cosh x, \cos x, \ln(1+x), \ln(1-x), \tan^{-1} x$ *and more are readily found using similar application. However note that a power series expression for* $\ln x$ *is not possible since* $\ln 0$ *is not defined, and neither are derivatives of* $\ln x$ *at* $x = 0$.

Definition 4.2. *The Talyor series expansion for* $y = f(x)$ *at* $x = 0$ *(as a power in terms of* $(x - a)$ *is defined by:*

$$\begin{aligned} f(x) =& f(a) + \frac{x-a}{1!}f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f''(a) + \frac{(x-a)^4}{4!}f^{(4)}(a) \\ &+ \ldots.. + \frac{(x-a)^n}{n!}f^{(n)}(a) + \ldots \end{aligned}$$

OR

$$f(x+a) = f(a) + \frac{x}{1!}f'(a) + \frac{x^2}{2!}f''(a) + \frac{x^3}{3!}f''(a) + \frac{x^4}{4!}f^{(4)}(a) + \ldots. + \frac{x^n}{n!}f^{(n)}(a) + \ldots$$

This means that Maclaurin series expansion is the Taylor expansion at $x = 0$.

Example 4.2.3. Find the expansion of $\cos(a+x)$, and hence the expansion for $\cos(x)$.
Solution: We have

$$\begin{aligned} f(x) &= \cos x & f(a) &= \cos a \\ f'(x) &= -\sin x & f(a) &= -\sin a \\ f''(x) &= -\cos x & f''(a) &= -\cos a \\ f''(x) &= \sin x & f'''(a) &= \sin a \end{aligned}$$

The pattern now established

$$\cos(a+x) = \cos a - (\sin a)x - (\cos a)\frac{x^2}{2!} + (\sin a)\frac{x^3}{3!} + \ldots$$

Note that putting $a = 0$ in this expansion

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots.$$

Example 4.2.4. Find the expansion (the binomial expansion.) for $(a+x)^n$.
Solution: We have

$$\begin{aligned} f(x) &= x^n & f(a) &= a^n \\ f'(x) &= nx^{n-1} & f'(a) &= na^{n-1} \\ f''(x) &= n(n-1)x^{n-2} & f''(a) &= n(n-1)a^{n-2} \\ &\vdots \end{aligned}$$

Then

$$(a+x)^n = a^n + na^{n-1} + n(n-1)a^{n-2}\frac{x^2}{2!} + \ldots$$

readily recognizable as the binomial expansion.

Remark 4.3. *The series can be used for approximation. For example in the expansion for* $\cos(a+x)$*, a foregoing, putting* $a = \frac{\pi}{3}\,(60°)$ *and* $x = \frac{\pi}{180}\,(1°)$ *gives an approximate value for* $\cos 61°$*. The more terms taken, the better the approximation. So*

$$\cos 61° = \cos\frac{\pi}{3} - \left(\sin\frac{\pi}{3}\right)\left(\frac{\pi}{180}\right) - \left(\cos\frac{\pi}{3}\right)\frac{\left(\frac{\pi}{180}\right)^2}{2!} + \left(\sin\frac{\pi}{3}\right)\frac{\left(\frac{\pi}{180}\right)^3}{3!} + \ldots$$

Example 4.2.5. Find expansion for $\ln x$ in terms of powers of $(x-a)$.

Solution: One can be found using this first form of Taylor expansion. With

$$\begin{aligned} f(x) &= \ln x, & f(a) &= \ln a \\ f'(x) &= \frac{1}{x} & f'(a) &= \frac{1}{a} \\ f''(x) &= -\frac{1}{x^2} & f''(a) &= -\frac{1}{a^2} \\ f'''(x) &= \frac{2}{x^3} & f'''(a) &= \frac{2}{a^3} \end{aligned}$$

Then

$$\ln x = \ln a + \frac{1}{a}(x-a) - \frac{1}{a^2}\frac{(x-a)^2}{2!} + \frac{2}{a^3}\frac{(x-a)^3}{3!} + \dots$$

Remark 4.4. *Approximations when x is close to a such that $(x-a)$ is small, and consequently successive powers of $(x-a)$ [eg$(x-a)^2, (x-a)^3$] become smaller and smaller approximate values can be found.*

Example 4.2.6. In the a foregoing an approximate value of $\ln(1.01)$ can be found by letting $x = 1.01, a = 1$ and $(x-a) = 0.01$.
Solution :

$$\begin{aligned} \ln 1.01 &= \ln 1 + \frac{1}{1}(0.01) - \frac{1}{1^2}\frac{(0.01)^2}{2!} + \frac{2}{1^3}\frac{(0.01)^3}{3!} + \dots \\ &= 0 + 0.01 - 0.00005 + 0.000000333. \\ &= \underline{\underline{0.0099503}} \quad \text{Compare with calculator} \end{aligned}$$

Other approximations:

1. ***A*** linear approximation for $f(x)$ when x is close to a can be found by ignoring $(x-a)^2$ and higher powers, for then $f(x) \approx f(a) + (x-a)f'(a)$

2. ***A*** quadratic approximation for $f(x)$ when $\underline{x}$ is close to a can be found by ignoring $(x-a)^3$ and higher powers, for then $f(x) \approx f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots$

Example 4.2.7. Find the linear approximation for $\tan x$ when x is close to $\frac{\pi}{4}$.
Solution : Since

$$\begin{aligned} f(x) &= \tan x & f\left(\frac{\pi}{4}\right) &= \tan\frac{\pi}{4} = 1 \\ f'(x) &= \sec^2 x & f'\left(\frac{\pi}{4}\right) &= \sec^2\frac{\pi}{4} = 2. \end{aligned}$$

Then

$$f(x) \approx 1 + \left(x - \frac{\pi}{4}\right)2 \Longrightarrow f(x) \approx 1 - \frac{\pi}{2} + 2x.$$

4.3 Mean Value Theorem

We know that constant functions have zero derivatives, but could there be a more complicated function whose derivative is always zero? If two functions have identical derivatives over an interval, how are the functions related? We answer these and other questions in this chapter by applying the Mean Value Theorem. First we introduce a special case, known as Rolle's Theorem, which is used to prove the Mean Value Theorem.

Theorem 4.5 (Rolle's Theorem). *Suppose that $y = f(x)$ is continuous over the closed interval $[a, b]$ and differentiable at every point of its interior (a, b). If $f(a) = f(b)$, then there is at least one number c in (a, b) at which $f'(c) = 0$.*

Example 4.3.1. Show that the equation $x^3 + 3x + 1 = 0$ has exactly one real solution.
Solution: We define the continuous function

$$f(x) = x^3 + 3x + 1$$

Since $f(-1) = -3$ and $f(0) = 1$, the Intermediate Value Theorem tells us that the graph of f crosses the x-axis somewhere in the open interval $(-1, 0)$. Now, if there were even two points $x = a$ and $x = b$ where $f(x)$ was zero, Rolle's Theorem would guarantee the existence of a point $x = c$ in between them where f' was zero. However, the derivative

$$f'(x) = 3x^2 + 3$$

is never zero (because it is always positive). Therefore, f has no more than one zero.

Theorem 4.6 (The Mean Value Theorem). *Suppose $y = f(x)$ is continuous over a closed interval $[a, b]$ and differentiable on the interval's interior (a, b). Then there is at least one point c in (a, b) at which*

$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

Example 4.3.2. Let $f(x) = x^4 - x^3 + 1$. Show that there exists a number $c \in (0, 1)$ such that $f'(c) = 0$.
Solution: $f(x)$ is continuous on $[0, 1]$, differentiable on $(0, 1)$. Also $f(0) = 1 = f(1)$. So there exists $c \in (0, 1)$ such that $f'(c) = 0$ In fact

$$f'(c) = 4c^3 - 3c^2 = 0$$

So we can take

$$c = \frac{3}{4}$$

Example 4.3.3. Show that

$$|\sin x - \sin y| \leq |x - y|.$$

Solution: When $x = y$, the result is trivial. Suppose $x > y$. By the mean value theorem, there exists $c \in (y, x)$ such that

$$\frac{\sin x - \sin y}{x - y} = f'(c) = \cos c$$
$$\sin x - \sin y = \cos c(x - y)$$

Therefore

$$|\sin x - \sin y| = |\cos c||x - y| \leq |x - y|.$$

4.4 Extreme Values of Functions

The derivative of a function may be used to determine whether the function is increasing or decreasing on any intervals in its domain.

Definition 4.7. *Let $y = f(x)$ be a differentiable function. Then, we have*

1. *A point c is called critical point of $f(x)$ if $f'(c) = 0$.*
2. *The function is said to be increasing on I if $f'(x) > 0$ at each point in an interval I.*
3. *The function is said to be decreasing on I if $f'(x) < 0$ at each point in an interval I.*

Remark 4.8. *We use the first derivative test Or the second derivative test to check that a critical point $x = c$ of a function $y = f(x)$ is relative maximum or relative minimum or neither nor (See below)*

First Derivative Test: Suppose that $x = c$ is a critical point of $f(x)$ then,

1. If $f'(x) > 0$ to the left of $x = c$ and $f'(x) < 0$ to the right of $x = c$ then $x = c$ is a relative maximum.
2. If $f'(x) < 0$ to the left of $x = c$ and $f'(x) > 0$ to the right of $x = c$ then $x = c$ is a relative minimum.
3. If $f'(x)$ is the same sign on both sides of $x = c$ then $x = c$ is neither a relative maximum nor a relative minimum.

Second Derivative Test: Suppose that $x = c$ is a critical point of $f'(c)$ such that $f'(c) = 0$ and that $f''(x)$ is continuous in a region around $x = c$. Then,

1. If $f'(c) < 0$ then $x = c$ is a relative maximum.
2. If $f'(c) > 0$ then $x = c$ is a relative minimum.
3. If $f'(c) = 0$ then $x = c$ can be a relative maximum, relative minimum or neither.

Example 4.4.1. Find the critical points of

$$f(x) = x^3 - 9x^2 + 24x - 10$$

Solution: The function f is differentiable everywhere , so the critical points are the solutions of $f'(x) = 0$

$$f'(x) = 3x^2 - 18x + 24 = 3\left(x^2 - 6x + 8\right) = 3(x-2)(x-4) = 0.$$

The critical points are the roots $c = 2$ and $c = 4$.

Example 4.4.2. Find the critical points of

$$h(t) = 10te^{3-1^2}.$$

Solution: Here's the derivative for this function.

$$\mathrm{h'(t)} = 10\mathrm{e}^{3-1^2} + 10t\mathrm{e}^{3-1^2}(-2\mathrm{t}) = 10\mathrm{e}^{3-1^2} - 20\mathrm{t}^2\mathrm{e}^{3-1^2}.$$

Now, this looks unpleasant, however with a little factoring we can clean things up a little as follows,

$$h'(t) = 10e^{3-1^2}\left(1 - 2t^2\right).$$

This function will exist everywhere and so no critical points will come from that. Determining where this is zero is easier than it looks. We know that exponentials are never zero and so the only way the derivative will be zero is if,

$$1 - 2t^2 = 0 \Longrightarrow 1 = 2t^2 \Longrightarrow \frac{1}{2} = t^2.$$

We will have two critical points for this function.

$$t = \pm\frac{1}{\sqrt{2}}.$$

Classifying of intervals and relative maximum (minimum) points

1. Find the domain of $f(x)$ and calculate the first derivative to finding all its critical points .

2. Test all intervals in the domain of the function to the left and to the right of these critical points to determine if the derivative ispositive or negative. (If $f'(x) > 0$, then f is increasing on the interval, and if $f'(x) < 0$, then f is decreasing on the interval). positive or negative. (If $f'(x) > 0$, then f is increasing on the interval, and if $f'(x) < 0$, then f is decreasing on the interval). 3) Check that a critical point $x = c$ of a function $y = f(x)$ is relative maximum or relative minimum or neither nor using the first derivative test (or the second derivative test) . Check that a critical point $x = c$ of a function $y = f(x)$ is relative maximum or relative minimum or neither nor using the first derivative test (or the second derivative test) .

Example 4.4.3. For $f(x) = x^4 - 8x^2$ determine all intervals where f is increasing or decreasing. **Solution:** The domain of $f(x)$ is all real numbers, and its derivatives given by

$$f'(x) = 4x^3 - 16x = 4x(x-2)(x+2) = 0$$

So its critical points occur at $x = -2, 0,$ and 2 . Testing all intervals to the left and right of these values for $f'(x) = 4x^3 - 16x$, you find that

$$\begin{aligned} f'(x) &< 0 \text{ on } (-\infty, -2) \\ f'(x) &> 0 \text{ on } (-2, 0) \\ f'(x) &< 0 \text{ on } (0, 2) \\ f'(x) &> 0 \text{ on } (2, +\infty) \end{aligned}$$

Hence, f is increasing on $(-2, 0)$ and $(2, +\infty)$ and decreasing on $(-\infty, -2)$ and $(0, 2)$.

Example 4.4.4. For $f(x) = \sin x + \cos x$ on $[0, 2\pi]$, determine all intervals where f is increasing or decreasing.

Solution: The domain of $f(x)$ is restricted to the closed interval $[0, 2\pi]$, and its critical points occur at $\pi/4$ and $5\pi/4$. Testing all intervals to the left and right of these values for $f'(x) = \cos x - \sin x$, you find that

$$\begin{aligned} f'(x) &> 0 \text{ on } \left[0, \frac{\pi}{4}\right) \\ f'(x) &< 0 \text{ on } \left(\frac{\pi}{4}, \frac{5\pi}{4}\right) \\ f'(x) &> 0 \text{ on } \left(\frac{5\pi}{4}, 2\pi\right] \end{aligned}$$

Hence, f is increasing on $[0, \pi/4](5\pi/4, 2\pi)$ and decreasing on $(\pi/4, 5\pi/4)$.

Example 4.4.5. Find d classify all the critical points of the following function. Give the intervals where the function is increasing and decreasing.

$$g(t) = t\sqrt[3]{t^2-4}.$$

Solution: First we'll need the derivative so we can get our hands on the critical points. Note as well that we'll do some simplification on the derivative to help us find the critical points.

$$\begin{aligned} g'(t) &= \left(t^2-4\right)^{\frac{1}{3}} + \frac{2}{3}t^2\left(t^2-4\right)^{-\frac{2}{3}} \\ &= \left(t^2-4\right)^{\frac{1}{3}} + \frac{2t^2}{3\left(t^2-4\right)^{\frac{2}{3}}} \\ &= \frac{3\left(t^2-4\right)+2t^2}{3\left(t^2-4\right)^{\frac{2}{3}}} = \frac{5t^2-12}{3\left(t^2-4\right)^{\frac{2}{3}}}. \end{aligned}$$

So, it looks like we'll have four critical points here. They are,

$$\begin{aligned} &t = \pm 2 \quad \text{The derivative doesn't exist here.} \\ &t = \pm\sqrt{\frac{12}{5}} = \pm 1.549 \quad \text{The derivative is zero here.} \end{aligned}$$

Finding the intervals of increasing and decreasing will also give the classification of the critical points so let's get those first. Here is a number line with the critical points graphed and test points. So, it looks like we've got the following intervals of increasing and decreasing.

- Increase : $-\infty < x < -\sqrt{\frac{12}{5}}$ and $\sqrt{\frac{12}{5}} < x < \infty$
- Decrease : $-\sqrt{\frac{12}{5}} < \mathrm{x} < \sqrt{\frac{12}{5}}$

From this it looks like $t = -2$ and $t = 2$ are neither relative minimum or relative maximums since the function is increasing on both side of them. On the other hand, $t = -\sqrt{\frac{1}{5}}$ is a relative maximum and $t = \sqrt{\frac{12}{5}}$ is a relative minimum.

4.5 Concavity and Points of Inflection:

We have seen previously that the sign of the derivative provides us with information about where a function (and its graph) is increasing, decreasing or stationary. We now look at the "direction of bending" of a graph, i.e. whether the graph is "concave up" or "concave down".

Definition 4.9. *Let $y = f(x)$ be a differentiable function. A graph of $y = f(x)$ is said to be*

1. *concave up at a point if the tangent line to the graph at that point lies below the graph in the vicinity of the point (where* $f''(x) > 0$*)*

2. *concave down at a point if the tangent line lies above the graph in the vicinity of the point (where* $f''(x) < 0$*).*

3. *A point where the concavity changes (from up to down or down to up) is called a point of inflection (POI) (where* $f''(x) = 0$ *). Note that the tangent line to a graph at a point of inflection must cross the graph at that point.*

Steps to determining the concavity of the function's curve at any point

1. Calculate the second derivative.

2. Substitute the value of x.

3. If $f''(x) > 0$, the graph is concave upward at that value of x.

4. If $f''(x) = 0$, the graph may have a point of inflection at that value of x. To check, consider the value of $f''(x)$ at values of x to either side of the point of interest.

5. If $f''(x) < 0$, the graph is concave downward at that value of x.

Example 4.5.1. Find the points of inflection and the intervals on which

$$f(x) = 3x^5 - 5x^4 + 1$$

is concave up and concave down.
Solution: The first derivative is $f'(x) = 15x^4 - 20x^3$ and

$$f''(x) = 60x^3 - 60x^2 = 60x^2(x - 1)$$

The zeros of $f''(x) = 60x^2(x - 1)$ are $x = 0$ and $x = 1$. They divide the x-axis into three intervals: $(-\infty, 0), (0, 1)$, and $(1, \infty)$. We determine the sign of $f''(x)$ and the concavity of f by computing "test values" within each interval (Figure 7):

Interval	Test Value	Sign of $f''(x)$	Behavior of $f(x)$
$(-\infty, 0)$	$f''(-1) = -120$	$-$	Concave down
$(0, 1)$	$f''\left(\frac{1}{2}\right) = -\frac{15}{2}$	$-$	Concave down
$(1, \infty)$	$f''(2) = 240$	$+$	Concave up

We can read off the points of inflection from this table:

- $c = 0$: no point of inflection, because $f''(x)$ does not change sign at 0.

- $c = 1$: point of inflection, because $f''(x)$ changes sign at 1 .

Example 4.5.2. Find the point of inflection for

$$f(x) = x^4$$

Solution: Consider $f(x) = x^4$. Solving $f''(x) = 12x^2 = 0$ yields $x = 0$. At values of $x < 0$, the second derivative is positive. At values of $x > 0$, the second derivative is positive. $(0, 0)$ is a local minimum (is not inflection).

4.6 L'Hospital's Rule

What happens when we try to evaluate $\lim_{x\to 1} \dfrac{\ln x}{x-1}$?

- Can't do the limits of the numerator and denominator separately. Why?
- No common terms to cancel.

The limit may or may not exist and is called an indeterminate form of type ∞/∞. There is also an indeterminate form of type $0/0$, for example, $\lim_{x\to 0} \frac{x}{x^2}$.
L'Hospital's Rule: Suppose that we have one of the following cases,

$$\lim_{x\to a} \frac{\mathrm{f(x)}}{\mathrm{g(x)}} = \frac{0}{0} \quad \text{OR} \quad \lim_{x\to 2} \frac{\mathrm{f(x)}}{\mathrm{g(x)}} = \frac{\pm\infty}{\pm\infty}$$

where a can be any real number, infinity or negative infinity. In these cases we have,

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}.$$

Example 4.6.1. We have:

1. $\lim_{x\to 0} \dfrac{\sin 5x}{3x} = \frac{0}{0}$. Hence, using L'Hospital's rule, we get

$$\lim_{x\to 0} \frac{\sin 5x}{3x} = \mathrm{Lim}_{x\to 0} \frac{5\cos 5x}{3} = \frac{5}{3}$$

2. $\lim_{x\to 0} \dfrac{\ln(1+x)}{x} = \frac{0}{0}$. Hence, using L'Hospital's rule, we obtain

$$\lim_{x\to 0} \frac{\ln(1+x)}{x} = \mathrm{Lim}_{x\to 0} \frac{1}{1+x} = 1.$$

3. Here we use L'Hospital's Rule for three times and get

$$\begin{aligned}\lim_{x\to 0}\frac{e^x-e^{-x}-2x}{x-\sin x} &= \lim_{x\to 0}\frac{e^x+e^{-x}-2}{1-\cos x} = \lim_{x\to 0}\frac{e^x-e^{-x}}{\sin x}\\ &= \lim_{x\to 0}\frac{e^x-e^{-x}}{\sin x} = \lim_{x\to 0}\frac{e^x+e^{-x}}{\cos x} = \frac{2}{1} = 2.\end{aligned}$$

4. $\lim_{x\to 0}\frac{\sqrt{1+\sin x}-\sqrt{1-\sin x}}{x} = \frac{1-1}{0} = \frac{0}{0}$. Hence, using L'Hospital's rule, we get

$$\lim_{x\to 0}\frac{\sqrt{1+\sin x}-\sqrt{1-\sin x}}{x} = \frac{0}{0}$$

Hence, using L'Hospital's rule, we get

$$\begin{aligned}\lim_{x\to 0}\frac{\sqrt{1+\sin x}-\sqrt{1-\sin x}}{x} &= \lim_{x\to 0}\frac{\frac{\cos x}{2\sqrt{1+\sin x}}+\frac{\cos x}{2\sqrt{1-\sin x}}}{1}\\ &= \frac{1}{2}+\frac{1}{2} = 1\end{aligned}$$

5.

$$\begin{aligned}\lim_{x\to 0}(\csc x-\cot x) &= \lim_{x\to 0}\left(\frac{1}{\sin x}-\frac{\cos x}{\sin x}\right)\\ &= \lim_{x\to 0}\left(\frac{1-\cos x}{\sin x}\right) = \frac{1-1}{0} = \frac{0}{0}\\ &= \lim_{x\to 0}\left(\frac{1-\cos x}{\sin x}\right) = \lim_{x\to 0}\left(\frac{\sin x}{\cos x}\right) = 0\end{aligned}$$

L'Hospital's Rule works great on the two indeterminate forms $0/0$ and $\pm\infty\pm\infty$. However, there are many more indeterminate forms $\{0\cdot\mp\infty, \infty-\infty, 0^0, \infty^0, 1^\infty\}$. Let's take a look at some of those and see how we deal with those kinds of indeterminate forms.

Example 4.6.2. Evaluate the following limit $\lim_{x\to 0^+}(x\ln x)$.
Solution: Since,

$$\lim_{x\to 0^+}(x\ln x) = 0\cdot(-\infty)$$

Now, in the limit, we get the indeterminate form $(0)(-\infty)$. L'Hospital's Rule won't work on products, it only works on quotients. However, we can turn this into a fraction if we rewrite things a little.

$$\lim_{x\to 0^-} x\ln x = \lim_{x\to 0^*}\frac{\ln x}{1/x}$$

The function is the same, just rewritten, and the limit is now in the form $-\infty/\infty$ and we can now use L'Hospital's Rule.

$$\lim_{x\to 0^+} \mathrm{x}\ln x = \lim_{x\to 1^\circ} \frac{\ln x}{1/x} = \lim_{x\to 0^-} \frac{1/x}{-1/x^2}$$

Now, this is a mess, but it cleans up nicely.

$$\lim_{x\to 0^\circ} x\ln x = \lim_{x\to 0^\circ} \frac{1/x}{-1/x^2} = \lim_{x\to 0^-} (-x) = 0$$

Example 4.6.3. $\lim_{x\to 0^+}(1+\sin 4x)^{\cot x}$.
Solution: Since $\lim_{x\to 0^+}(1+\sin 4x)^{\cot x} = 1^\infty$ (indeterminate power), putting

$$y = (1+\sin 4x)^{\cot x} \Rightarrow \lim_{x\to 0^+} y?$$

Taking the natural logarthim ln, we have

$$\ln y = \cot \ln(1+\sin 4x) = \frac{\ln(1+\sin 4x)}{\tan x}$$
$$\lim_{x\to 0^+} \ln y = \lim_{x\to 0^+} \frac{\ln(1+\sin 4x)}{\tan x} = \frac{0}{0}$$

Now, we can use L'Hospital rules

$$\begin{aligned}\ln \lim_{x\to 0^+} y &= \lim_{x\to 0^+} \frac{4\cos 4x/(1+\sin 4x)}{\sec^2 x} \\ &= \lim_{x\to 0^+} \frac{4\cos 4x}{\sec^2 x(1+\sin 4x)} \\ &= \lim_{x\to 0^+} \frac{4\cos 4x\cos^2 x}{(1+\sin 4x)} = 1\end{aligned}$$

Hence, we obtain $\lim_{x\to 0^+} y = e$.

Example 4.6.4. Prove that for any x, $\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = e^x$.
Solution: Let $y = \left(1+\frac{x}{n}\right)^n$. Then

$$\ln y = \ln\left(1+\frac{x}{n}\right)^n = n\ln\left(1+\frac{x}{n}\right).$$

By the application of l'Hôpital's Rule, in which we differentiate with respect to n:

$$\begin{aligned}\lim_{n\to\infty} n\ln\left(1+\frac{x}{n}\right) &= \lim_{n\to\infty} \frac{\ln(1+x/n)}{1/n} \\ &= \lim_{n\to\infty} \frac{\left(\frac{1}{1+x/n}\right)\cdot\left(-\frac{x}{n^2}\right)}{-1/n^2} = \lim_{n\to\infty} \frac{x}{1+x/n} = x.\end{aligned}$$

Using the fact that $\lim_{n\to\infty} \ln y = \ln \lim_{n\to\infty} y$, we have

$$\ln \lim_{n\to\infty} y = x \Longrightarrow \lim_{n\to\infty} y = e^x.$$

4.7 EXERCISES

1. Find the equation of the tangent and normal lines to following curves at the corresponding points:

   (a) $x^2 + y^2 - 4x + 6y - 3 = 0; \quad (2. - 7)$

   (b) $x^2 - y^2 = 9 \quad ; (3\sqrt{2}, 3)$

   (c) $y^2 = x - 1 \quad ; (5, 2)$

   (d) $x = 2y^2 - y + 1; (4, -1)$

   (e) $x = \ln t \quad ; \quad y = 2t^2 \quad ; \quad (0, 2)$

2. Find the equation of the tangent $y = x^3 - 6x + 2$ and parallel to the line $y = 6x - 2$

3. Prove the two tangent lines for $x^2 - 4y + 4 = 0$ at $(3/2, 0)$ are perpendicular.

4. Determine the increasing and decreasing intervals for the following functions:

   (a) $y = 2x^2 - 4x + 5$

   (b) $y = x^2 - 4x + 1$

   (c) $y = x^3 - 3x^2 + 5$

   (d) $y = 2x^3 - 3x^2 + 1$

   (e) $y = x^4 - 2x^2 + 1$

   (f) $y = x^4 - 4x^2 + 5$

5. Find the maximum and minimum extremes for the following functions:

   (a) $y = x^2 - 2x + 3$

   (b) $y = x^4 - 8x^2 + 2$

   (c) $y = -x^4 + 2x^3$

   (d) $y = x^3 - 9x^2 + 15x + 3$

6. Evaluate the following limits;

   (a) $\lim_{x\to 0} \dfrac{1 - \cos x}{x^2}$

   (b) $\lim_{x\to \pi/2} \dfrac{2\cos x}{2x - \pi}$

   (c) $\lim_{x\to 0} \dfrac{\cos^{-1}(1 + x)}{\ln(1 + x)}$

   (d) $\lim_{x\to 0} \dfrac{x - \tan x}{x - \sin x}$

   (e) $\lim_{x\to\infty} \dfrac{\ln x}{\sqrt[3]{x}}$

   (f) $\lim_{x\to 0} \dfrac{\tan x - x}{x^3}$

7. Determine the point(s) on $y = x^2 + 1$ that are closest to $(0, 2)$.

8. Determine the area of the largest rectangle that can be inscribed in a circle of radius 4.

9. For the following function find the inflection points and use the second derivative test, if possible, to classify the critical points. Also, determine the intervals of increase, decrease and the intervals of concave up, concave down and sketch the graph of the functions:

   (a) $y = x(6-x)^{\frac{2}{3}}$ (b) $y = x^3 + 3x^2 - 9x - 20$

10. Use the second derivative test to classify the critical points of the function $y = 3x^5 - 5x^3 + 3$

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